THE ANATOMY OF THE CONTINUUM

Desk Volume One

The formal statements, the complete proofs, and the problems belonging to the audio season of the same name.

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Issued chapter by chapter alongside the episodes.
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How to Use This Volume

This volume is the desk's half of a strict partnership. The episodes carry motivation, history, definitions in words, and the *strategy* of proofs; this volume carries precise notation, formal statements, complete proofs, and the problems. Neither half stands alone, and the intended order is fixed: listen first, then work. Nothing appears here that the corresponding episode has not motivated, and nothing is claimed aloud as proved unless the proof appears here — or the narration says plainly where the full argument lives, or that it lies beyond the course.

The episode digest. Each chapter opens with an unnumbered digest of its episode: the argument in one paragraph; then the episode retold beat by beat under its own section titles — each beat restating the claim, the idea of the argument, and the image that carried it, keyed to the formal items here; then the toolkit acquired, the debts, and the dates worth keeping. Enough to reconstruct the hour without replaying it. The episode still carries the motivation; the digest presumes a first listening.

The notation ledger. Each chapter opens with a table pairing the episode's spoken phrases with their written symbols. It is the bridge between ear and desk; cross it first.

Conventions. Two are set once and hold throughout. First, the natural numbers begin at one: $\mathbb{N} = \{1, 2, 3, \dots\}$; when zero is wanted, we write $\mathbb{N} \cup \{0\}$. Second, a fact may occasionally be used *on credit*: stated, employed, and not yet proved. Every such credit is flagged where it occurs, entered in the running ledger, and either paid in a named later chapter or declared honestly to be beyond the course, with references.

The problems. Three tiers, framed aloud at each episode's close. *Tier I, Calisthenics:* routine and definitional; designed to be gotten right, because confidence is a mathematical instrument. *Tier II, The Real Thing:* the substance; each problem teaches a technique or completes something the audio sketched. *Tier III, For the Ambitious* (starred): genuinely hard, or a door the course will not itself walk through.

Hints and solutions. Hints for Tiers II and III sit in their own section, spoiler-safe. Solutions are complete and terse for Tier I; provided for *selected* Tier II problems — those completing arguments sketched aloud; and withheld for Tier III, where hints are all the help on offer. The policy is deliberate: self-study requires closure on the essentials, but complete solution manuals rot discipline. Work first; hints second; solutions as audit, never as route.

Contents

Chapter 1

The Scandal of the Infinite

Companion to Episode One. The episode carried the story — Pythagoras to Weierstrass — and one complete proof. This chapter supplies the formal dress for that proof, the toolkit it drew on, the generalizations promised aloud, and the first problem sets.

Episode Digest

The argument of the hour. Mathematics was forced into rigour by three crises: the Greeks discovered that their concept of number was too small for the world and retreated into geometry for two millennia; the calculus of Newton and Leibniz conquered the sciences while resting on an increment that was nonzero and zero as convenience required; and Fourier's claim that *any* function is a sum of waves broke the old regime open, so that answering him created modern analysis. The rebuild runs Bolzano → Cauchy → Weierstrass → Dedekind → Cantor, and this season walks it.

The episode, beat by beat. **The Cult of Number.** The Pythagoreans found harmony governed by small whole-number ratios — halve a string for the octave, divide it two-to-three for the fifth — and concluded *all is number*: any two lengths share a common measuring unit, so every ratio of magnitudes is a fraction. Hippasus is said to have drowned for the counterexample; the drowning is legend (sources late, contradictory, improved in the telling; some scholars back the pentagon, not the square, as the first offender), but the theorem is not.

The Diagonal. In a unit square, the tilted square erected on the diagonal contains four of the half-square triangles where the original contains two: the diagonal's square is exactly 2. So the creed stakes everything on one question — is there a fraction whose square is 2? No instrument can settle it (a large denominator hides below any resolution); only proof can, and it must prove a negative over infinitely many candidates.

The First Impossible Thing. Suppose $\sqrt{2} = p/q$ in lowest terms. Squaring: $p^2 = 2q^2$, so p^2 is even, so p is even (odd numbers have odd squares — proved by contrapositive: $(2k+1)^2 = 2(2k^2+2k)+1$). Write $p = 2r$: then $q^2 = 2r^2$, so q is even too — and a lowest-terms fraction with both entries even is an absurdity (Theorem 1.6). Coda from the same armoury: multiply any finite list of primes and add one; the result's prime divisor is a stranger to the list — the number itself need not be prime (Theorem 1.7). The wonder beat: one finite argument dispatching infinitely many candidates at once — what no search or measurement could ever certify.

The Retreat into Geometry. Eudoxus salvaged rigour without new numbers: two ratios are *equal* when every fraction that undershoots one undershoots the other, and likewise overshooting — a magnitude known by the company it keeps among the fractions. (File that: it is Dedekind's idea, twenty-two centuries early.) His method of exhaustion corners curved areas between inscribed polygons and a double contradiction — neither more nor less — by which Archimedes proved the parabolic segment is exactly four-thirds its inscribed triangle; his private *Method* (weighing shapes on an idealized balance, then certifying by exhaustion) surfaced in 1906 as a palimpsest under a book of prayers.

Zeno, Taken Seriously. The Dichotomy and Achilles decompose motion into infinitely many stages completed in finite time; the schoolbook reply — the stages' durations sum as a geometric series — *presumes* the concept of limit, the very thing in question, so it relocates the paradox rather than resolving it (an answer that presumes the thing in question is an intention to answer). The Arrow: at any single instant the arrow occupies exactly its own space and does not move; velocity at an instant is distance zero over time zero — the derivative's question, twenty-two centuries early.

The Machine That Worked. Newton (fluxions, the plague years) and Leibniz (dx, in print 1684) built one machine in two costumes: slope-finding and area-finding, secretly inverse (the fundamental theorem, glimpsed, unproved). The crack, in four lines: to find how x^2 grows, nudge x by h; the growth is $2xh + h^2$; divide by h — so $h \neq 0$ — getting $2x + h$; then discard h — so $h = 0$. Right answer, scandalous procedure (Remark 1.11). Footnotes of the human colour: the Royal Society's priority commission found for its own president, in a report drafted by Newton himself; and L'Hôpital's rule was purchased, under retainer, from Johann Bernoulli.

The Bishop Files His Report. Berkeley's *Analyst* (1734): the increment is treated as something when divided by and nothing when discarded — fluxions are "the ghosts of departed quantities" — and yet the answers come out right, for which he proposed a mechanism: two errors, equal and opposite, cancelling. Every substantive objection was correct; the profession attacked the reviewer and kept shipping; the ticket stayed open a century.

The Age of Nerve. Euler solved Basel (1735) by treating the sine as an infinite polynomial and factoring it by its roots: out fell $\pi^2/6$ — unlicensed, correct, verified numerically, and legalized only later (by Weierstrass). Grandi's $1 - 1 + 1 - \cdots$ assigned the value $\frac{1}{2}$: answers without definitions are moods (modern summation theories do license the $\frac{1}{2}$ — but they are theories, with definitions). Abel, 1826: divergent series are the invention of the devil.

The Heat of the Matter. An iron rod, ends held at ice: heat flows from hot to cold, faster where the profile is steeper, and a point warms by net inflow — so crests fall and troughs fill. Sine arches decay *in place*, and a wave of n arches dies n^2 times faster (fine wiggles are hot gossip; broad features are the slow news). Fourier's claim: *any* initial profile is a sum of these waves, with a recipe — multiply by the wave, average: an integral. Four alarms: what is an infinite sum of functions; can smoothness sum to a jag; what is the integral of an arbitrary function; can two profiles share all their waves (uniqueness)? The invoice: Dirichlet 1829 (convergence for decent functions, jumps split midway — plus a deranged function his theorem cannot touch, reserved for Episode Six); Riemann 1854 (his integral, born as a lemma for trigonometric series); Cantor (uniqueness $\to$ sets of exceptional points $\to$ set theory, through the tradesman's entrance).

The Great Rebuild. Bolzano (Prague, 1817) proved the intermediate value theorem "purely analytically" and went unread. Cauchy (1821) put limits at the foundation — and proved, *his* word, that a convergent sum of continuous functions is continuous, which Fourier's jagged assemblies contradict in the street; where the proof leaks is Episode Seven's business. Weierstrass abolished motion: no flowing, no approaching — only tolerance and threshold, inequalities that hold or fail; arithmetic, standing still. Kronecker: God made the whole numbers; all else is the work of man — and his campaign against Cantor is Episode Three's sorrow. Dedekind and Cantor were left queued for the next two hours.

The Season Ahead. Monsters placed under contract: the deranged function (E6), a function continuous exactly at the irrationals (E6), the nowhere-differentiable coastline (E7, with Baire's verdict that the monsters are the majority), the integral's defeat and Volterra's hole in the fundamental theorem (E8) — and Season Two paraded by name, Vitali to Kolmogorov.

The toolkit acquired. Proof by contradiction (the shape of Theorem 1.6); the contrapositive and why it is legitimate (Lemma 1.1); induction, demonstrated twice (Propositions 1.2, 1.3 — Bernoulli is hired for the season); well-ordering, discharging the episode's two confessions (Lemmas 1.4, 1.5); quantifier order and *eventually* versus *infinitely often* (§1.2 — Episode Four's grammar, pre-installed); unique factorization on credit (Theorem 1.8) powering the general criterion $\sqrt{n} \in \mathbb{Q} \iff n$ a perfect square (Theorem 1.9, Corollary 1.10; Theodorus stopped at seventeen); the nudge in lawful symbols (Remark 1.11). Supplied solutions complete the audio's sketches: irrational arithmetic (II.1) and $\log_2 3 \notin \mathbb{Q}$ (II.2).

Debts opened. D1, what a number actually *is* (paid in Episode Two). D2, Zeno — the race due Episode Four, the Arrow Episode Seven. D3, Fourier's waves — instalments across both seasons; the largest account. D4, the monsters — as contracted above. D5, the exchange of limiting processes — memorandum; formally issued in Episode Four.

Dates worth keeping. 1665–66 Newton's plague years; 1684 and 1687 Leibniz and Newton in print; 1696 L'Hôpital's textbook; 1734 *The Analyst*; 1735 Basel; 1797 Lagrange's attempted reconstruction; 1807 and 1822 Fourier; 1817 Bolzano; 1821 Cauchy's *Course*; 1829 Dirichlet; 1854 Riemann's qualifying essay; 1856 Weierstrass to Berlin; 1858 and 1872 Dedekind; 1872 the Weierstrass monster; 1874 and 1891 Cantor.

1.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"for every" / "there exists"	$\forall$ / $\exists$	the quantifiers; §1.2
"the whole numbers"	$\mathbb{N} = \{1, 2, 3, \dots\}$	zero excluded, by standing convention
"the integers"	$\mathbb{Z}$	$\dots, -2, -1, 0, 1, 2, \dots$
"the rationals" / "fractions"	$\mathbb{Q}$	ratios p/q with $p \in \mathbb{Z}$, $q \in \mathbb{N}$
"the real numbers"	$\mathbb{R}$	*on credit until Chapter 2*
"p over q, in lowest terms"	p/q, $\gcd(p, q) = 1$	the phrase is licensed by Lemma 1.4
"root two"	$\sqrt{2}$	see the caution after Theorem 1.6
"x squared"	x^2	
"the nudge"	h	Newton wrote o; we write h, which is audible and is not mistakable for a zero
"the slope quotient"	$\dfrac{(x+h)^2 - x^2}{h}$	Remark 1.11
"a half, plus a quarter, plus an eighth, and so on, is one"	$\displaystyle\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$	*on credit*; paid in Chapter 4
"pi squared over six"	$\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$	*on credit*; proof beyond this course — see the chapter notes
"the open interval from a to b"	(a, b)	§1.4
"f of x"	$f(x)$	§1.4

1.2 The Grammar of Quantifiers

Modern analysis is written almost entirely with two words: *for every* ($\forall$) and *there exists* ($\exists$). Everything difficult about the subject's definitions lives in the *order* in which they arrive.

Compare the two claims about kitchenware from the episode. Let $F(p, \ell)$ abbreviate "lid ℓ fits pot p." Then

$$\underbrace{\forall p\ \exists \ell\ F(p, \ell)}_{\text{every pot has a lid}} \qquad \text{versus} \qquad \underbrace{\exists \ell\ \forall p\ F(p, \ell)}_{\text{one lid fits every pot}}\ .$$

In the first, the lid may be chosen *after* the pot is revealed, and may depend on it. In the second, a single lid is nominated in advance and must then survive every pot. The second claim implies the first; the converse fails, as any kitchen confirms. Whenever an existential follows a universal, the witness may depend on everything quantified before it. This one observation carries more of analysis than any theorem in this chapter.

Negation. Negation walks past each quantifier in turn, flipping it, and finally negates the interior:

$$\neg\,\forall x\, P(x) \iff \exists x\, \neg P(x), \qquad \neg\,\exists x\, P(x) \iff \forall x\, \neg P(x).$$

Applied to a nested statement:

$$\neg\,\big(\forall p\ \exists \ell\ F(p,\ell)\big) \iff \exists p\ \forall \ell\ \neg F(p,\ell) \quad \text{(“some pot fits no lid”).}$$

Note what negation does *not* do: it never reorders the variables.

Eventually and infinitely often. Two patterns recur constantly from Chapter 4 onward. Let $P(n)$ be a property of natural numbers.

$$\begin{aligned} \text{“}P \text{ holds } \textit{eventually}\text{”} &\quad \text{means} \quad \exists N \in \mathbb{N}\ \ \forall n \geq N :\ P(n); \\ \text{“}P \text{ holds } \textit{infinitely often}\text{”} &\quad \text{means} \quad \forall N \in \mathbb{N}\ \ \exists n \geq N :\ P(n). \end{aligned}$$

Eventually implies infinitely often; not conversely (Problem I.7). The two are dual under negation: P fails to hold eventually precisely when $\neg P$ holds infinitely often — flip the quantifiers and negate the interior, exactly as above.

1.3 The Proof Toolkit

Contradiction. To prove a statement S: assume $\neg S$, derive by valid steps a statement known to be false, and conclude S. The episode's set piece has this shape, with S the assertion that no rational has square two, and the false terminus a fraction in lowest terms with both entries even.

Contrapositive. The implication $P \Rightarrow Q$ and its contrapositive $\neg Q \Rightarrow \neg P$ are logically equivalent: each is false in exactly one situation, namely P true and Q false. One may always prove whichever is more convenient. The episode's servant-lemma is the standard illustration:

Lemma 1.1 (Parity of squares)**.** *If $n \in \mathbb{Z}$ is odd, then n^2 is odd. Equivalently (contrapositive): if n^2 is even, then n is even.*

Proof. If n is odd, then $n = 2k+1$ for some $k \in \mathbb{Z}$, and

$$n^2 = (2k+1)^2 = 4k^2 + 4k + 1 = 2\,(2k^2 + 2k) + 1,$$

which is twice an integer plus one: odd. □

Induction. If a property holds at 1, and holding at n forces holding at $n+1$, then it holds at every natural number. Two specimens, both of which will work for a living later in the season:

Proposition 1.2. *For every $n \in \mathbb{N}$,* $\quad 1 + 2 + \cdots + n = \dfrac{n(n+1)}{2}.$

Proof. At $n = 1$ both sides equal 1. Assuming the identity at n,

$$1 + 2 + \cdots + n + (n+1) = \frac{n(n+1)}{2} + (n+1) = (n+1)\Big(\frac{n}{2} + 1\Big) = \frac{(n+1)(n+2)}{2},$$

which is the identity at $n+1$. □

Proposition 1.3 (Bernoulli's inequality). *For every real $x \geq -1$ and every $n \in \mathbb{N}$,* $(1+x)^n \geq 1+nx$.

Proof. At $n = 1$ the two sides are equal. Assume the inequality at n. Since $1 + x \geq 0$, multiplying both sides of $(1+x)^n \geq 1+nx$ by $(1+x)$ preserves the inequality:

$$(1+x)^{n+1} \geq (1+nx)(1+x) = 1+(n+1)x+nx^2 \geq 1+(n+1)x,$$

the last step because $nx^2 \geq 0$. □

The hypothesis $x \geq -1$ is doing real work: it is what makes the multiplication legitimate. Bernoulli looks like nothing; it will be found carrying luggage in half the estimates of this season.

Well-ordering. *Every non-empty subset of* $\mathbb{N}$ *has a least element.* We take this as an axiom of the natural numbers; it is equivalent to the principle of induction (see the chapter notes). The episode confessed two uses of it, discharged now.

Lemma 1.4 (Lowest terms exist). *Every positive rational number r can be written $r = p/q$ with $p, q \in \mathbb{N}$ and* $\gcd(p, q) = 1$.

Proof. The set $D = \{\, n \in \mathbb{N} : r = m/n \text{ for some } m \in \mathbb{N} \,\}$ is non-empty, since r is a positive rational. By well-ordering, D has a least element q; write $r = p/q$. If some integer $d > 1$ divided both p and q, then $r = (p/d)\big/(q/d)$ would exhibit a denominator smaller than q in D, contradicting minimality. Hence $\gcd(p, q) = 1$. □

Lemma 1.5 (Prime divisors exist). *Every integer $n \geq 2$ has a prime divisor.*

Proof. The set of divisors of n that exceed 1 is non-empty (it contains n), so by well-ordering it has a least element d. If d were composite, say $d = ab$ with $1 < a < d$, then a would divide n and exceed 1, contradicting the minimality of d. Hence d is prime. □

1.4 Sets, Functions, and the Written Dialect

Sets. A set is specified either by listing, $\{1, 2, 3\}$, or by the *set-builder* form

$$\{\, x \in A : P(x) \,\} \qquad \text{— “the set of } x \text{ in } A \text{ such that } P(x) \text{ holds.”}$$

Set-builder notation is banned aloud and welcomed on paper; the episode described such sets in words, and this volume writes them. Membership is $x \in A$; containment is $B \subseteq A$; union and intersection are $A \cup B$ and $A \cap B$; the set with nothing in it is $\varnothing$.

Intervals. For $a < b$ real (on credit until Chapter 2):

$$(a, b) = \{x : a < x < b\}, \qquad [a, b] = \{x : a \leq x \leq b\},$$

with the half-open $[a, b)$, $(a, b]$ read off the brackets, and the unbounded (a, ∞), $(-\infty, b]$ defined likewise. The symbol ∞ here is punctuation, not a number.

Functions. A function $f\colon A \to B$ assigns to *each* element $x \in A$ exactly one element $f(x) \in B$. Nothing requires a formula: any unambiguous assignment qualifies — a dictum usually credited to Dirichlet, whose deranged example (Episode Six) will test everyone's sincerity about it. The terms *injective*, *surjective*, *bijective* are deferred to Chapter 3, where they earn their keep in the arithmetic of the infinite.

1.5 The Theorems of the Hour

Theorem 1.6 (Irrationality of $\sqrt{2}$)**.** *No rational number has square equal to* 2*.*

Proof. Suppose, for contradiction, that $r \in \mathbb{Q}$ satisfies $r^2 = 2$. Since $r^2 = (-r)^2$, we may take $r > 0$, and by Lemma 1.4 write $r = p/q$ with $p, q \in \mathbb{N}$, $\gcd(p,q) = 1$. Then

$$p^2 = 2q^2.$$

The right side is even, so p^2 is even, so p is even by Lemma 1.1; write $p = 2r'$ with $r' \in \mathbb{N}$. Substituting, $4r'^2 = 2q^2$, hence

$$q^2 = 2r'^2,$$

so q^2 is even, so q is even by Lemma 1.1 again. Then 2 divides both p and q, contradicting $\gcd(p,q) = 1$. No such r exists. □

Caution, in the spirit of the honesty clause. The theorem is purely negative: it says no *rational* squares to 2. That there exists a real number whose square is 2 — that the symbol $\sqrt{2}$ names an inhabitant of $\mathbb{R}$ at all — is a genuine further claim, and it is precisely the business of Chapter 2, where completeness manufactures the number. Tonight we know only that *if* the diagonal has a length, that length is no fraction.

Two other murders of the same conjecture. A swifter and stranger proof by infinite descent is Problem III.1. A one-line proof by unique factorization falls out of Theorem 1.9 below.

Theorem 1.7 (Euclid)**.** *There are infinitely many primes.*

Proof. Let $p_1, \ldots, p_k$ be any finite list of primes, and set

$$N = p_1 p_2 \cdots p_k + 1.$$

Then $N \geq 2$, so by Lemma 1.5 it has a prime divisor d. If d were one of the p_i, then d would divide both N and the product $p_1 \cdots p_k$, hence their difference 1 — impossible. So d is a prime not on the list. No finite list of primes is complete. □

Remark. N itself need not be prime — only its prime divisors are new. The smallest cautionary example: $2 \cdot 3 \cdot 5 \cdot 7 \cdot 11 \cdot 13 + 1 = 30031 = 59 \times 509$.

Theorem 1.8 (Fundamental theorem of arithmetic — stated on credit)**.** *Every integer $n \geq 2$ can be written as a product of primes, and the factorization is unique up to the order of the factors.*

Honesty clause. Existence of the factorization follows from Lemma 1.5 by induction; *uniqueness* is the substantive half, and this course does not prove it — the honest article requires Euclid's lemma and a page of number theory that belongs to another subject. References in the chapter notes. The theorem is hereby entered in the ledger as a permanent, flagged credit: used below and occasionally later, never proved here.

Theorem 1.9 (Rational roots are exact roots)**.** *For $n \in \mathbb{N}$: $\sqrt{n}$ is rational if and only if n is a perfect square.*

Proof. If $n = m^2$ with $m \in \mathbb{N}$, then $\sqrt{n} = m \in \mathbb{Q}$. Conversely, suppose $\sqrt{n} = p/q$ in lowest terms; then

$$p^2 = n\,q^2.$$

Fix any prime ℓ, and let a and b be the exponents of ℓ in the factorizations of p and q (Theorem 1.8 makes these well-defined), and c its exponent in n. Comparing the exponent of ℓ on the two sides of $p^2 = nq^2$, uniqueness of factorization gives

$$2a = c + 2b, \qquad \text{hence} \qquad c = 2(a - b),$$

an even number (and non-negative). Since every prime appears in n to an even power, n is a perfect square: take m with each prime exponent halved, and $n = m^2$. □

Corollary 1.10. $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$, $\sqrt{6}$, $\sqrt{7}$, $\sqrt{8}$, $\sqrt{10}$ *are irrational; in general $\sqrt{n} \notin \mathbb{Q}$ for every $n \in \mathbb{N}$ that is not a perfect square.*

Theodorus of Cyrene, in the generation after Hippasus, is reported (in Plato's *Theaetetus*) to have proved the cases through $n = 17$ one at a time — and there to have stopped, for reasons still argued about. Theorem 1.9 is the industrialized version.

Remark 1.11 (The nudge, in symbols)*.* The episode's forbidden computation, written honestly: for $h \neq 0$,

$$\frac{(x+h)^2 - x^2}{h} = \frac{2xh + h^2}{h} = 2x + h.$$

Every step above is lawful, *because* $h \neq 0$. The eighteenth century's crime was the next step: setting $h = 0$ in the result while having divided by h on the way to it. The lawful replacement — the *limit* of the quotient as h tends to 0, a number defined without h ever being zero — is exactly the machinery of Chapter 4, deployed on this quotient in Chapter 7. Until then the equation above is a fact about nonzero h, and nothing more.

1.6 Problems

Tier I — Calisthenics

I.1 All variables range over $\mathbb{N}$. (a) Write out the formal negation of $\forall m\, \exists n : \; n > m$. (b) Decide, with one-line justifications, whether the statement and its negation are true. (c) Decide whether $\exists n\, \forall m : \; n > m$ is true. Draw the moral about order.

I.2 Prove by contrapositive: if $3 \nmid n$ then $3 \nmid n^2$. (Two cases: $n = 3k + 1$ and $n = 3k + 2$.) Conclude that $3 \mid n^2$ implies $3 \mid n$.

I.3 Using Problem I.2, prove that $\sqrt{3}$ is irrational, by the method of Theorem 1.6.

I.4 Prove by induction: $1 + 3 + 5 + \cdots + (2n - 1) = n^2$.

I.5 Prove by induction: $2^n > n$ for every $n \in \mathbb{N}$.

I.6 Show that there is no smallest positive rational number.

I.7 Let $a_n = (-1)^n$ and $b_n = n$. (a) Express formally, and decide: "$a_n = 1$ infinitely often." (b) Express formally, and decide: "$a_n = 1$ eventually." (c) Show that $b_n > 100$ eventually, giving an explicit threshold N.

I.8 Show that $\mathbb{Q}$ is closed under addition and multiplication, and that the reciprocal of a nonzero rational is rational.

Tier II — The Real Thing

II.1 (Arithmetic of the irrationals; solution provided.) Prove: (a) if $r \in \mathbb{Q}$ and $x \notin \mathbb{Q}$, then $r + x \notin \mathbb{Q}$; (b) if $r \in \mathbb{Q}$, $r \neq 0$, and $x \notin \mathbb{Q}$, then $rx \notin \mathbb{Q}$. Then exhibit: (c) two irrationals whose sum is rational; (d) two irrationals whose product is rational. Moral: the irrationals are not closed under anything convenient.

II.2 (Solution provided.) Prove that $\log_2 3$ is irrational.

II.3 Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

II.4 (Quantifier games.) (a) Write Goldbach's conjecture — *every even integer greater than* 2 *is a sum of two primes* — as a formal quantified statement. (b) Negate it formally and simplify until no negation touches a quantifier. (c) Do the same for: *there are infinitely many primes p such that $p + 2$ is prime*. (d) With m, n ranging over $\mathbb{N}$, decide the truth of $\forall m\, \exists n : n^2 > m$ and of $\exists n\, \forall m : n^2 > m$.

II.5 (The one-colour herd.) Here is a proof by induction that all horses are the same colour. *Base case:* in any herd of one horse, all horses trivially share a colour. *Step:* assume every herd of k horses is monochrome, and take a herd of $k + 1$. Remove one horse: the remaining k are monochrome. Put it back and remove a different horse: again k horses, again monochrome. The two sub-herds overlap, so all $k + 1$ horses share one colour. *Conclusion:* all horses are the same colour. Locate the flaw — precisely.

Tier III — For the Ambitious ⋆

⋆ **III.1** (Descent.) Suppose $p, q \in \mathbb{N}$ satisfy $p^2 = 2q^2$. Set

$$p' = 2q - p, \qquad q' = p - q.$$

Show that $p', q' \in \mathbb{N}$, that $p'^2 = 2q'^2$, and that $q' < q$. Conclude, by well-ordering, that no such p, q exist — a second proof of Theorem 1.6, in which the hypothetical fraction manufactures a strictly smaller copy of itself forever.

⋆ **III.2** (Existence without exhibition.) Prove that there exist irrational numbers x and y such that x^y is rational. Your proof will, if you follow the intended route, be constitutionally unable to say *which* pair works. Sit with that sensation; it returns, at industrial scale, in Season Two.

1.7 Hints (Tiers II and III)

II.3 Suppose $s = \sqrt{2} + \sqrt{3}$ were rational. Compute s^2 and isolate the term $\sqrt{6}$. Then let Theorem 1.9 pronounce on 6.

II.4 (d) Who moves first? In the first statement n may be chosen after seeing m; in the second it may not.

II.5 Run the inductive step at the smallest value it claims to cover. How many horses are in the overlap of the two sub-herds when $k+1=2$?

III.1 For the identity, expand and use $p^2 = 2q^2$:

$$p'^2 - 2q'^2 = (2q-p)^2 - 2(p-q)^2 = 2q^2 - p^2 = 0.$$

For the bounds, show first that $p^2 = 2q^2$ forces $q < p < 2q$ (compare p^2 with q^2 and with $4q^2$); then read off $0 < q' < q$ and $p' \in \mathbb{N}$. Well-ordering does the mourning.

III.2 You need not decide anything. Consider the number $c = \sqrt{2}^{\sqrt{2}}$. Either c is rational — and then a pair is found — or it is irrational; in that case consider $c^{\sqrt{2}}$ and compute it. Afterwards, for the ambitious beyond ambition: the Gelfond–Schneider theorem settles which case actually holds; look it up only after solving.

1.8 Solutions

Tier I (complete, terse)

I.1 (a) $\exists m\, \forall n : \ n \le m$. (b) The statement is true: given m, take $n = m+1$. Its negation is therefore false. (c) False: given any candidate n, take $m = n$; then $n > n$ fails. Moral: swapping $\forall$ and $\exists$ changes the claim entirely.

I.2 If $n = 3k+1$: $n^2 = 9k^2 + 6k + 1 = 3(3k^2+2k)+1$, not divisible by 3. If $n = 3k+2$: $n^2 = 9k^2 + 12k + 4 = 3(3k^2+4k+1)+1$, not divisible by 3. Contrapositive of the conclusion: done.

I.3 Suppose $\sqrt{3} = p/q$ in lowest terms (Lemma 1.4). Then $p^2 = 3q^2$, so $3 \mid p^2$, so $3 \mid p$ by I.2; write $p = 3r$. Then $9r^2 = 3q^2$, so $q^2 = 3r^2$, so $3 \mid q$ likewise. Then 3 divides both p and q: contradiction.

I.4 Base: $1 = 1^2$. Step: $n^2 + (2(n+1)-1) = n^2 + 2n + 1 = (n+1)^2$.

I.5 Base: $2^1 = 2 > 1$. Step: $2^{n+1} = 2 \cdot 2^n > 2n \ge n+1$ for $n \ge 1$.

I.6 If $r > 0$ is rational, then $r/2$ is rational and $0 < r/2 < r$. No positive rational is smallest.

I.7 (a) $\forall N\, \exists n \ge N : \ a_n = 1$ — true: given N, take any even $n \ge N$. (b) $\exists N\, \forall n \ge N : \ a_n = 1$ — false: beyond any N there is an odd n, where $a_n = -1$. (c) $N = 101$ works: $n \ge 101$ gives $b_n = n > 100$.

I.8 $\dfrac{p}{q} + \dfrac{r}{s} = \dfrac{ps+qr}{qs}$ and $\dfrac{p}{q} \cdot \dfrac{r}{s} = \dfrac{pr}{qs}$, with nonzero denominators; and if $p \ne 0$, the reciprocal of p/q is q/p.

Tier II (selected)

II.1 (a) If $r + x = s$ with $r, s \in \mathbb{Q}$, then $x = s - r \in \mathbb{Q}$ by closure (I.8) — contradicting $x \notin \mathbb{Q}$. (b) If $rx = s \in \mathbb{Q}$ with $r \ne 0$, then $x = s/r \in \mathbb{Q}$ by closure — same contradiction. (c) $\sqrt{2}$ and

$-\sqrt{2}$ are both irrational (the second by part (b) with $r = -1$), and their sum is $0 \in \mathbb{Q}$. (d) $\sqrt{2} \cdot \sqrt{2} = 2 \in \mathbb{Q}$.

II.2 Note $\log_2 3 > 0$. Suppose $\log_2 3 = p/q$ with $p, q \in \mathbb{N}$. Exponentiating, $2^{p/q} = 3$, hence

$$2^p = 3^q.$$

The left side is even (as $p \geq 1$); the right side is odd (a product of odd factors). Even cannot equal odd; no such p, q exist. (For connoisseurs: this is also immediate from uniqueness of factorization, Theorem 1.8 — a power of 2 and a power of 3 share no prime.)

Problems II.3–II.5 are left with their hints, per standing policy; III.1 and III.2 receive hints only.

Notes and References

- **Abbott**, *Understanding Analysis*, 2nd ed. — the season's spiritual companion; its Chapter 1 parallels this one and is warmly recommended for parallel problems.
- **Hardy and Wright**, *An Introduction to the Theory of Numbers* — the honest article on Theorem 1.8 (unique factorization) and on Euclid's lemma.
- **Aigner and Ziegler**, *Proofs from THE BOOK* — several proofs of the irrationality of $\sqrt{2}$, the descent argument among them, and six proofs of Euclid's theorem.
- **Dunham**, *Euler: The Master of Us All* — the Basel problem and the age of nerve, told with affection and full computations.
- **Fourier**, *The Analytical Theory of Heat* (Freeman translation) — the primary source; the opening chapters remain readable and startling.
- **Berkeley**, *The Analyst* (1734) — in the public domain and easily found online; the finest hostile code review in history rewards a direct reading.
- **Plato**, *Meno* (the doubling of the square) and *Theaetetus* (Theodorus and the roots through 17).
- On well-ordering and induction: their equivalence is proved in any introduction to mathematical logic or discrete mathematics; Abbott's Chapter 1 exercises include it.

Chapter 2

What Is a Number, Actually?

Companion to Episode Two. The episode located a hole in the rationals from the inside, stated the axiom of completeness, and narrated Dedekind's construction and three dividend theorems. This chapter supplies the formal statements, the proofs in full — including the payment of Chapter 1's outstanding caution — and the problem sets.

Episode Digest

The argument of the hour. The rationals, for all their virtues, contain bounded climbing crowds whose frontier posts are unmanned — holes, demonstrable from inside the crowd. The axiom of completeness is the decree that in the real numbers every such post is occupied; Dedekind's cuts are the warehouse that proves the decree can be honoured; and three dividend theorems — Archimedean, density, nested intervals — are the axiom's first interest payments, each one false in the rationals.

The episode, beat by beat. **The Society of Fractions.** The reference letter before the demotion: the rationals are a *field* (the four operations never exile you, and behave as the schoolroom promised), an *ordered* field (comparison cooperates with arithmetic), *dense in themselves* (between any two, their average — so no fraction has a next-door neighbour; succession is abolished), and sufficient for every finite measurement ever taken. And yet: the diagonal.

The Anatomy of a Hole. How does one point at an absence using only members of the crowd? The *shortfall club* — every positive fraction whose square falls short of two — is non-empty and bounded above (nothing at or past 2 qualifies). Press the ceilings down: a rational ceiling cannot have square equal to 2 (Chapter 1) nor below 2 (it would be a member, and every member is overtopped by another member); and any rational ceiling with square above 2 admits a slightly smaller rational ceiling. So the members climb forever, the ceilings descend forever, and *no fraction stands at the frontier* (Proposition 2.5; the single fraction $q' = (2q+2)/(q+2)$ performs both halves — Problem II.1, solution supplied). Density makes it eerier, not milder: from inside $\mathbb{Q}$, the hole's neighbourhood is exactly as crowded as an honest number's. A hole in a dense crowd is a missing resident in a city with no vacant streets.

Bounds, and the Best of Them. Upper bound = ceiling; the *supremum* is the least ceiling, in two locked clauses: (i) no member exceeds it; (ii) dip beneath it by any margin, however slight, and some member overtakes you (Definitions 2.2, 2.3; Lemma 2.4 and the slack remark). The open interval $(0, 1)$ has no maximum but supremum 1: a frontier no one stands on — suprema require frontiers, not champions, which is the entire point. The marching set $1 - \frac{1}{n}$ has supremum 1: the collection *approaches* something, and the supremum names it without a verb — Weierstrass's abolition of motion, already in service. In the new vocabulary the hole compresses to one sentence: the shortfall club is bounded above with no supremum *in* $\mathbb{Q}$.

The Single Axiom. *Every non-empty set of reals bounded above has a least upper bound* (Axiom 2.6) — a specification, not an observation: the reals are commissioned as the ordered field with no unmanned frontiers. Remove it and the season's theorems die in $\mathbb{Q}$: the climbing decimals $1,\ 1.4,\ 1.41, \ldots$ are a bounded climb arriving nowhere. Solvency (does anything satisfy the specification?) is Dedekind's deed below; uniqueness (only one thing does, up to renaming) is on credit (Remark 2.8). Infima come free by reflection (Proposition 2.7).

The Man Who Dated His Idea. Dedekind: Gauss's last doctoral student, the least theatrical of the great mathematicians. Zurich, 1858, his first calculus course: at the theorem that a bounded climbing quantity settles toward a value, he found he possessed no proof — nobody did; one gestured at the picture. He asked what the line's gaplessness *says* when gesturing is forbidden, found the answer on 24 November 1858 (a Wednesday), put it in a drawer for fourteen years, and published thirty pages in 1872 when rival constructions forced his hand.

The Audacity. A point does something banal: it cuts the line in two. Invert it — keep the cut, hide the knife: the line's continuity is precisely the claim that *every* cut lands on somebody. Run the knife through the rationals at the shortfall club: left camp, the club and everything below; right camp, the rest; a lawful, complete severing — performed at nobody. Dedekind's move: *then the cut shall be the number*. That pair of camps *is* $\sqrt{2}$ (Definition 2.9); rationals embed as the cuts they perform; the ownerless cuts are exactly the missing numbers, hired wholesale. Numbers as free creations of the human mind: when reality declines to supply an object your theorems require, precision itself serves as a foundry. (Why not decimals instead? Addition carries right-to-left and an unending decimal has no rightmost digit; and base ten is a biographical accident of our hands. Cuts carry only order.) The audit: order is containment of left camps; addition is memberwise (Proposition 2.11); multiplication is a swamp, tasted in Problem III.1 and cited to the professionals. The decisive inspection: the supremum of any bounded family of cuts is the *union of their left camps* — a cut of cuts is a cut (Theorem 2.10) — so one round of hole-filling terminates forever; no second storey needed. As of 1872 the complete ordered field is not a wish but a warehouse.

The Rival Firm. Cantor (1872, with Méray's priority from 1869 and Heine systematizing): a real number as a *huddling sequence* of rationals — terms eventually crowding within any tolerance of one another; where no destination exists, the journey is appointed the destination. Same audacity, different material; same product, by the uniqueness certificate. Dedekind's cuts are native to order; Cantor's sequences carry only distances — and so his is the construction that *travels*, to spaces of functions where Season Two banks. The founders met by chance at Interlaken that very summer; the correspondence that followed will carry next episode's proof across Europe in an envelope. Hilbert's third stance: write the axioms, visit the warehouse once, and thereafter take existence as read — which is how this course now proceeds.

The End of the Ghosts. The Archimedean property: some whole number exceeds any given real (Theorem 2.12). Proof, entire: were $\mathbb{N}$ bounded, completeness would grant $s = \sup \mathbb{N}$ (*the bell, first ring*); dipping by one, some $n > s - 1$; then $n + 1 > s$ — a ceiling with a whole number above it. Corollary: below any positive tolerance sits some $\frac{1}{n}$ — so $\mathbb{R}$ harbours *no infinitesimals*:

Berkeley's ghosts have no forwarding address here. This is a genuine theorem, not a triviality: ordered fields with lawful ghosts exist (Problem III.2's jurisdiction of rational functions, ordered by eventual dominance), and by this proof read in reverse, none of them is complete. Honest refinement: Robinson (1960s) rehoused the infinitesimals rigorously — elsewhere.

The Picket Fence. Between any two reals lies a fraction (Theorem 2.15): choose mesh $\frac{1}{n}$ finer than the gap (Archimedes); erect the fence of all multiples of $\frac{1}{n}$; the *first post past the left edge* (secured by well-ordering, not by gesture) cannot have crossed the whole gap in a one-mesh stride, so it stands inside. Corollary by the sliding trick: shift the gap down by $\sqrt{2}$, plant a fraction, slide back — rational plus irrational is irrational (Chapter 1, II.1, now load-bearing), so the irrationals are dense too (Corollary 2.16). The two societies permeate each other everywhere — equal partners, surely? Next episode weighs them; the scale does not balance. And you have trusted the fence all your life: decimals are fences with meshes of one tenth, one hundredth, one thousandth.

The Russian Dolls. Nested closed bounded intervals always share a survivor (Theorem 2.17): the left endpoints climb, ceilinged by every right endpoint; their supremum (*the bell, second ring*) is wedged at or above every beginning and at or below every end — inside every doll. Watch it die twice (Remark 2.18): open dolls $(0, \frac{1}{n})$ leak the survivor out through the disowned boundary; unbounded dolls $[n, \infty)$ let the quarry escape rightward forever. Widths shrinking beneath every tolerance leave exactly one prisoner — the trap configuration. "Closed and bounded" surfaces here for the first time (enthroned in Episode Five); the dolls are contracted to Episode Three as the murder weapon in Cantor's 1874 proof, and to Episode Four's bisection hunt. (Third ring of the bell: on paper, in Theorem 2.13, where the axiom manufactures $\sqrt{2}$ itself and pays Chapter 1's caution — Corollary 2.14.)

The toolkit acquired. The supremum and its two clauses, with the approximation lemma and the slack remark (Definitions 2.2, 2.3; Lemma 2.4); the axiom and its free mirror (Axiom 2.6, Proposition 2.7); cuts, their order, their addition, their completeness by pooling (Definition 2.9, Proposition 2.11, Theorem 2.10); the Archimedean property in both forms (Theorem 2.12); the existence of $\sqrt{2}$ by two-sided squeeze on a supremum (Theorem 2.13); density both ways (Theorem 2.15, Corollary 2.16); the nested interval theorem with its fine print and unique-survivor form (Theorem 2.17, Remark 2.18). Supplied solutions complete the audio's sketches: the hole (II.1) and the sum rule for suprema (II.2). ε works on paper throughout; its audio ceremony is reserved for Episode Four.

Debts. D1 — **paid**: a number is a cut of the rationals; the season's first stamped account. D7 — **opened**: the axiom of completeness to be caught in every major theorem; the bell's running count is three. All other accounts standing.

Dates worth keeping. 24 November 1858, Zurich — the idea, dated by its owner; 1869 Méray in print; 1872 — Dedekind and Cantor both publish, and meet by chance at Interlaken; the 1960s — Robinson's rigorous rehousing of the infinitesimals, one jurisdiction over.

2.1 Notation Ledger

Spoken in the episode	Written here	Remarks
“bounded above” / “an upper bound”	$\exists M\ \forall a \in A:\ a \le M$	Definition 2.2
“the supremum — the least upper bound”	$\sup A$	Definition 2.3
“the infimum — the greatest lower bound”	$\inf A$	Definition 2.3
“dip beneath it by any margin, however slight, and some member overtakes you”	$\forall \varepsilon > 0\ \exists a \in A:\ a > \sup A - \varepsilon$	Lemma 2.4
“the axiom of completeness”	every non-empty $A \subseteq \mathbb{R}$ bounded above has $\sup A \in \mathbb{R}$	Axiom 2.6
“the shortfall club”	$S = \{\, q \in \mathbb{Q} : q > 0,\ q^2 < 2 \,\}$	Proposition 2.5
“a cut” (the left camp travels alone)	$\alpha \subset \mathbb{Q}$ per Definition 2.9	§2.6
“the Archimedean property”	$\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}:\ n > x$	Theorem 2.12
“the picket fence”	the multiples k/n, $k \in \mathbb{Z}$	Theorem 2.15
“the Russian dolls”	$I_1 \supseteq I_2 \supseteq \cdots$, closed and bounded	Theorem 2.17
“closed” / “open” interval	$[a, b]$ / (a, b)	Chapter 1, §1.4

The Greek letter ε (epsilon) appears above and throughout this chapter as the standing name for an arbitrary positive margin. Its formal introduction in the audio, with the adversarial ceremony it deserves, is Episode Four’s affair; on paper it is already employed, doing clerical work.

2.2 The Rationals as an Ordered Field

The rationals $\mathbb{Q}$ satisfy the *field axioms*: addition and multiplication are associative and commutative, multiplication distributes over addition, there are neutral elements $0 \neq 1$, every element has an additive inverse, and every nonzero element has a multiplicative inverse. They further carry an order $<$ compatible with the arithmetic: for all a, b, c,

$$a < b \ \Rightarrow\ a + c < b + c, \qquad a < b \text{ and } c > 0 \ \Rightarrow\ ac < bc,$$

and any two elements are comparable. A system satisfying all of this is an *ordered field*; $\mathbb{Q}$ is one, and $\mathbb{R}$ is commissioned as one. Two consequences used silently below: squaring preserves order among positives ($0 < x < y \Rightarrow x^2 < y^2$, multiply twice), hence also its converse for positives ($0 < x, y$ and $x^2 < y^2 \Rightarrow x < y$).

Proposition 2.1 ($\mathbb{Q}$ is dense in itself)**.** *If $p, q \in \mathbb{Q}$ and $p < q$, then $p < \dfrac{p+q}{2} < q$, and the average is rational.*

Proof. Rationality: closure (Chapter 1, Problem I.8). Order: add p (respectively q) to both sides of $p < q$ and halve. $\square$

2.3 Bounds, Suprema, Infima

Definition 2.2. Let $A \subseteq \mathbb{R}$ be non-empty. A number M is an *upper bound* of A if $a \leq M$ for every $a \in A$; if an upper bound exists, A is *bounded above*. *Lower bound* and *bounded below* are the mirror notions; A is *bounded* if both hold.

Definition 2.3. A number s is the *supremum* (least upper bound) of A, written $s = \sup A$, if

(i) s is an upper bound of A, and

(ii) no number smaller than s is an upper bound of A.

The *infimum* $\inf A$ (greatest lower bound) is defined by the mirror clauses. If $\sup A \in A$ it is the *maximum* of A; a supremum need not belong to A (examples below), which is the entire point of the concept.

Lemma 2.4 (Approximation property)**.** *Let s be an upper bound of A. Then $s = \sup A$ if and only if for every $\varepsilon > 0$ there exists $a \in A$ with $a > s - \varepsilon$.*

Proof. Clause (ii) of Definition 2.3 says: no $s - \varepsilon$ with $\varepsilon > 0$ is an upper bound; and "$s - \varepsilon$ is not an upper bound" says exactly that some $a \in A$ exceeds it. The two statements are word-for-word translations of one another across the quantifier grammar of Chapter 1. □

A slack remark, used constantly: if $u \geq v - \varepsilon$ for *every* $\varepsilon > 0$, then $u \geq v$ (otherwise take $\varepsilon = v - u > 0$). Approximation arguments end with this line so often that we state it once here and thereafter use it without ceremony.

Calibrating examples. $\sup[0,1] = 1 = \max[0,1]$. $\sup(0,1) = 1 \notin (0,1)$: no maximum exists (for $x \in (0,1)$ the average $\frac{x+1}{2}$ is a larger member), yet the frontier is definite. $\sup\{\, 1 - \frac{1}{n} : n \in \mathbb{N} \,\} = 1$, not attained (Problem I.6); $\inf\{\, \frac{1}{n} : n \in \mathbb{N} \,\} = 0$, not attained (Problem I.5; the proof needs Theorem 2.12).

2.4 The Hole, Precisely

Proposition 2.5 (The shortfall club has no rational supremum)**.** *Let $S = \{\, q \in \mathbb{Q} : q > 0,\ q^2 < 2 \,\}$. Then S is non-empty and bounded above, and* no rational number is a least upper bound of S*: every member of S is exceeded by another member, and every rational upper bound of S lies strictly above some smaller rational upper bound.*

Proof. Problem II.1, with full solution supplied — it completes the squeeze narrated in the episode, and the single fraction that performs both halves of it deserves to be met at the desk. □

2.5 The Axiom of Completeness

Axiom 2.6 (Completeness; the least-upper-bound property)**.** *Every non-empty set of real numbers that is bounded above has a least upper bound in* $\mathbb{R}$.

Together with the ordered-field requirements of §2.2, this is the entire specification of $\mathbb{R}$: *the* complete ordered field. Proposition 2.5 is the certificate that $\mathbb{Q}$ fails the specification.

Proposition 2.7 (Infima come free)**.** *Every non-empty $A \subseteq \mathbb{R}$ bounded below has a greatest lower bound, namely* $\inf A = -\sup(-A)$.

Proof. Let $-A = \{ -a : a \in A \}$. A number m is a lower bound of A if and only if $-m$ is an upper bound of $-A$; so $-A$ is non-empty and bounded above, and Axiom 2.6 yields $s = \sup(-A)$. Then $-s$ is a lower bound of A (negate clause (i)), and any lower bound m of A supplies the upper bound $-m$ of $-A$, whence $-m \geq s$, that is, $m \leq -s$. So $-s$ is the greatest lower bound. □

Remark 2.8 (Uniqueness of $\mathbb{R}$ — on credit)*.* Any two complete ordered fields are isomorphic: identical in all but the names of their elements. The proof is honest bookkeeping of some length and is *not* given in this course; see the chapter notes. Consequence: the construction of §2.6 and the Cantor–Méray construction from Cauchy sequences deliver the same object, and no theorem below cares which warehouse its numbers shipped from.

2.6 Dedekind's Construction, in Outline

Following Dedekind (1872), we build an object meeting the specification out of $\mathbb{Q}$ alone. By standing convention the cut travels light: we carry only the left camp; the right camp is its complement.

Definition 2.9 (Cut)**.** A *cut* is a set $\alpha \subset \mathbb{Q}$ such that (i) $\alpha \neq \varnothing$ and $\alpha \neq \mathbb{Q}$; (ii) α is downward closed: if $q \in \alpha$ and $p < q$, then $p \in \alpha$; (iii) α has no greatest element. Order: $\alpha < \beta$ means $\alpha \subsetneq \beta$. Each rational r embeds as the cut $r^* = \{ q \in \mathbb{Q} : q < r \}$; the embedding preserves order and (once defined) arithmetic. Cuts *not* of the form r^* are precisely the new numbers: the set $\{ q \in \mathbb{Q} : q \leq 0 \} \cup S$, with S the shortfall club, is a cut of the new kind — Proposition 2.5 in cut clothing — and *is*, by definition, $\sqrt{2}$.

Theorem 2.10 (The cuts are complete)**.** *Every non-empty family of cuts that is bounded above (some cut contains every member of the family) has a least upper bound among the cuts.*

Proof. Let $\{\alpha_i\}_{i \in I}$ be such a family, bounded above by the cut γ, and set $\alpha = \bigcup_{i \in I} \alpha_i$. Then α is a cut: it is non-empty (it contains any one α_i's members) and proper ($\alpha \subseteq \gamma \neq \mathbb{Q}$); it is downward closed, being a union of downward-closed sets; and it has no greatest element, since any $q \in \alpha$ lies in some α_i, which contains some $q' > q$. By construction $\alpha_i \subseteq \alpha$ for every i, so α is an upper bound; and any upper bound δ contains every α_i, hence contains their union α. So $\alpha = \sup_i \alpha_i$. □

Proposition 2.11 (Addition of cuts)**.** *For cuts α, β, the set $\alpha + \beta = \{ a + b : a \in \alpha,\ b \in \beta \}$ is a cut.*

Proof. Non-empty: clear. Proper: pick $a' \notin \alpha$ and $b' \notin \beta$; by downward closure, $a < a'$ and $b < b'$ for all $a \in \alpha$, $b \in \beta$, so $a + b < a' + b'$, and hence $a' + b' \notin \alpha + \beta$. Downward closed: if $q < a + b$ with $a \in \alpha$, $b \in \beta$, write $q = a'' + b$ where $a'' = q - b < a$; then $a'' \in \alpha$ and $q \in \alpha + \beta$. No greatest element: given $a + b$, pick $a' > a$ in α; then $a' + b$ is a larger member. □

Honesty clause. Additive inverses require a small dodge:

$$-\alpha = \{ q \in \mathbb{Q} : \text{there exists } r > q \text{ with } -r \notin \alpha \},$$

which sidesteps the boundary of rational cuts; and multiplication must be defined by cases on sign and verified case by case. This is the swamp: virtuous drudgery containing no idea. Problem III.1 has you taste it; the notes cite where it is performed in full, once, for all humanity.

2.7 Four Theorems from One Axiom

From here to the end of the season we stand on Axiom 2.6 and never again open the warehouse. Deployments of the axiom are marked ⊛ — the bell of the episode, in print.

Theorem 2.12 (Archimedean property)**.** (i) *For every $x \in \mathbb{R}$ there exists $n \in \mathbb{N}$ with $n > x$.* (ii) *For every real $\varepsilon > 0$ there exists $n \in \mathbb{N}$ with $\dfrac{1}{n} < \varepsilon$.*

Proof. (i) Suppose not: some x is an upper bound of $\mathbb{N}$. Then $\mathbb{N}$ is non-empty and bounded above, so $s = \sup \mathbb{N}$ exists (⊛). Since $s - 1 < s$, clause (ii) of Definition 2.3 gives $n \in \mathbb{N}$ with $n > s - 1$; then $n + 1 \in \mathbb{N}$ and $n + 1 > s$, contradicting clause (i). (ii) Given $\varepsilon > 0$, apply (i) to $x = 1/\varepsilon$ to obtain $n > 1/\varepsilon$; then $1/n < \varepsilon$. □

Remark. Equivalently: $\mathbb{R}$ contains no infinitesimals — no $\iota > 0$ with $\iota < 1/n$ for every n. Ordered fields containing such elements exist (Problem III.2); by the argument above read in reverse, none of them is complete.

Theorem 2.13 (Existence of $\sqrt{2}$)**.** *There is a unique real $s > 0$ with $s^2 = 2$.*

Proof. Let $A = \{\, x \in \mathbb{R} : x > 0,\ x^2 < 2 \,\}$. Then $1 \in A$, and 2 is an upper bound: if $x \geq 2$ then $x^2 \geq 4 > 2$, so $x \notin A$. By Axiom 2.6, $s = \sup A$ exists (⊛), and $s \geq 1 > 0$. We rule out both alternatives to $s^2 = 2$.

Suppose $s^2 < 2$. For any $n \in \mathbb{N}$,

$$\left(s + \tfrac{1}{n}\right)^2 = s^2 + \tfrac{2s}{n} + \tfrac{1}{n^2} \ \leq\ s^2 + \tfrac{2s+1}{n},$$

using $\frac{1}{n^2} \leq \frac{1}{n}$. By Theorem 2.12(ii) choose n with $\frac{1}{n} < \dfrac{2 - s^2}{2s + 1}$; then $\left(s + \frac{1}{n}\right)^2 < 2$, so $s + \frac{1}{n} \in A$, contradicting clause (i) of the supremum.

Suppose $s^2 > 2$. For any $n \in \mathbb{N}$,

$$\left(s - \tfrac{1}{n}\right)^2 = s^2 - \tfrac{2s}{n} + \tfrac{1}{n^2} \ >\ s^2 - \tfrac{2s}{n}.$$

Choose n with $\frac{1}{n} < \dfrac{s^2 - 2}{2s}$ and also $n \geq 2$; then $\left(s - \frac{1}{n}\right)^2 > 2$, and $s - \frac{1}{n} \geq 1 - \frac{1}{2} > 0$. For any $x \in A$: $x^2 < 2 < \left(s - \frac{1}{n}\right)^2$ with both sides positive, so $x < s - \frac{1}{n}$ (squaring reflects order among positives, §2.2). Thus $s - \frac{1}{n}$ is an upper bound of A smaller than s, contradicting clause (ii).

Hence $s^2 = 2$. Uniqueness: if $0 < s < t$ then $s^2 < t^2$, so two distinct positive numbers cannot both square to 2. □

Corollary 2.14 (Irrational numbers exist)**.** $\sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$; *in particular* $\mathbb{R} \neq \mathbb{Q}$.

Proof. Existence: Theorem 2.13. Irrationality: Theorem 1.6 of Chapter 1. *This discharges the caution recorded after that theorem: the diagonal now has a length, and the length is a number.* □

Theorem 2.15 ($\mathbb{Q}$ is dense in $\mathbb{R}$)**.** *For all real $a < b$ there exists $r \in \mathbb{Q}$ with $a < r < b$.*

Proof. By Theorem 2.12(ii) choose $n \in \mathbb{N}$ with $\frac{1}{n} < b - a$. We claim some integer m satisfies $m - 1 \le na < m$; granting the claim,

$$a < \frac{m}{n} = \frac{m-1}{n} + \frac{1}{n} \le a + \frac{1}{n} < a + (b - a) = b,$$

so $r = m/n$ serves. For the claim — the first picket past the edge — use Theorem 2.12(i) to pick $p \in \mathbb{N}$ with $p > -na$, so that $na + p > 0$; by (i) again, the set $T = \{ j \in \mathbb{N} : j > na + p \}$ is non-empty, and by well-ordering (Chapter 1) it has a least element j_0. Set $m = j_0 - p \in \mathbb{Z}$. Then $m > na$. And $m - 1 \le na$: if $j_0 \ge 2$, then $j_0 - 1 \in \mathbb{N}$ lies outside T by minimality, so $j_0 - 1 \le na + p$, that is, $m - 1 \le na$; if $j_0 = 1$, then $m - 1 = -p < na$ directly, since $p > -na$. □

Corollary 2.16 (The irrationals are dense in $\mathbb{R}$)**.** *For all real $a < b$ there exists $x \in \mathbb{R} \setminus \mathbb{Q}$ with $a < x < b$.*

Proof. Apply Theorem 2.15 to $a - \sqrt{2} < b - \sqrt{2}$: some rational r satisfies $a - \sqrt{2} < r < b - \sqrt{2}$. Then $x = r + \sqrt{2}$ lies in (a, b) and is irrational — rational plus irrational is irrational, by Chapter 1, Problem II.1(a), now load-bearing. □

Theorem 2.17 (Nested interval theorem)**.** *Let $I_n = [a_n, b_n]$ be closed bounded intervals with $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. Then $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$.*

Proof. Nesting gives $a_1 \le a_2 \le \cdots$ and $b_1 \ge b_2 \ge \cdots$. Cross-check: $a_n \le b_m$ for all n, m — if $n \ge m$, then $a_n \le b_n \le b_m$; if $n < m$, then $a_n \le a_m \le b_m$. So $A = \{ a_n : n \in \mathbb{N} \}$ is non-empty and bounded above by every b_m; let $s = \sup A$ (⊛). Clause (i): $a_n \le s$ for all n. Clause (ii), applied against each upper bound b_m: $s \le b_m$ for all m. Hence $s \in [a_n, b_n]$ for every n. □

Remark 2.18 (Fine print, and the unique survivor). Both hypotheses are load-bearing. *Closed:* the open intervals $(0, \frac{1}{n})$ are nested with empty intersection — 0 is excluded by each, and any $t > 0$ is expelled once $\frac{1}{n} < t$, by Theorem 2.12(ii). *Bounded:* the closed rays $[n, \infty)$ are nested with empty intersection — any t is passed by some n, by Theorem 2.12(i). Finally, if in Theorem 2.17 the widths shrink beneath every tolerance — for every $\varepsilon > 0$ some $b_n - a_n < \varepsilon$ — then the common point is unique: two survivors $s < t$ would force $b_n - a_n \ge t - s > 0$ for all n. This is the form in which the dolls serve as a trap: Chapter 3 (Cantor, 1874) and Chapter 4 (the bisection hunt) both hunt with it.

2.8 Problems

Tier I — Calisthenics

I.1 For each set, find sup and inf where they exist, and state whether each is attained: (a) $\{ \frac{1}{n} : n \in \mathbb{N} \}$; (b) $\{ x \in \mathbb{R} : x^2 < 2x \}$; (c) $(0, 2] \cup \{3\}$; (d) $\{ \frac{n}{n+1} : n \in \mathbb{N} \}$; (e) $\{ (-1)^n (1 + \frac{1}{n}) : n \in \mathbb{N} \}$.

I.2 Prove that a set has at most one supremum.

I.3 Prove: if $\varnothing \neq A \subseteq B$ and B is bounded above, then $\sup A \le \sup B$.

I.4 Prove: if an upper bound of A belongs to A, then it is the maximum of A and equals $\sup A$.

I.5 Prove $\inf\{\,\frac{1}{n} : n \in \mathbb{N}\,\} = 0$. (Theorem 2.12(ii) is the engine.)

I.6 Prove $\sup\{\,1 - \frac{1}{n} : n \in \mathbb{N}\,\} = 1$, giving for each $\varepsilon > 0$ an explicit threshold n that overtakes $1 - \varepsilon$.

I.7 Let $c > 0$ and $cA = \{\,ca : a \in A\,\}$, with A non-empty and bounded above. Prove $\sup(cA) = c \sup A$.

I.8 For non-empty A, B bounded above, prove $\sup(A \cup B) = \max\{\sup A, \sup B\}$.

Tier II — The Real Thing

II.1 (The hole, completed; solution provided.) Prove Proposition 2.5 in full. Follow the plan: (a) show any rational $q > 0$ with $q^2 > 2$ is an upper bound of S; (b) show a rational upper bound cannot have square less than 2, nor equal to 2; (c) for rational $q > 0$, set

$$q' = \frac{2q+2}{q+2}$$

and compute the signs of $q' - q$ and of ${q'}^2 - 2$ in terms of the sign of $q^2 - 2$; conclude both that S has no greatest member and that every rational upper bound of S sits above a smaller one.

II.2 (The sum rule; solution provided.) For non-empty A, B bounded above, with $A + B = \{\,a + b : a \in A,\ b \in B\,\}$, prove $\sup(A + B) = \sup A + \sup B$.

II.3 (Eudoxus, notarized.) Prove that every real x satisfies $x = \sup\{\,q \in \mathbb{Q} : q < x\,\}$: each number is recoverable from — because constituted by — its rational entourage.

II.4 (The dolls fail in $\mathbb{Q}$.) Exhibit closed intervals with *rational* endpoints, nested and with widths shrinking beneath every tolerance, whose intersection contains no rational number. Conclude that the nested interval theorem fails in $\mathbb{Q}$, and is therefore completeness in disguise rather than a free gift of order.

II.5 (Interleaving.) Prove: between any two distinct rationals lies an irrational, and between any two distinct irrationals lies a rational.

II.6 (The climbing escort.) Let $s = \sup A$. Prove that for every $n \in \mathbb{N}$ there exists $a_n \in A$ with $s - \frac{1}{n} < a_n \leq s$. (In Episode Four this becomes: the supremum is a limit of members.)

Tier III — For the Ambitious ⋆

⋆ **III.1** (The swamp, tasted.) For cuts α, β that are *positive* (each contains some rational > 0), define

$$\alpha \cdot \beta = \{\,q \in \mathbb{Q} : q \leq 0\,\} \cup \{\,ab : a \in \alpha,\ b \in \beta,\ a > 0,\ b > 0\,\}.$$

Verify that $\alpha \cdot \beta$ is a cut, noting exactly where positivity is used. Then attempt to write down the definition for arbitrary signs, in terms of $-\alpha$ and the positive case, and observe the case count. You may stop when the character-building begins; the notes cite where the full verification is performed.

⋆ **III.2** (The jurisdiction of the ghosts.) Let F be the set of rational functions $f = p/q$ (p, q real polynomials, $q \neq 0$), and declare $f > 0$ when the leading coefficients of p and q have the same sign — equivalently, when $f(t) > 0$ for all sufficiently large real t ("eventual dominance"). (a) Check the declaration is well-defined on equivalent fractions and makes F an ordered field (verify totality, and closure of positives under sum and product). (b) Show the element t satisfies $t > n$ for every $n \in \mathbb{N}$. (c) Show $1/t$ is positive yet smaller than $\frac{1}{n}$ for every n: a lawful infinitesimal. (d) Conclude F is not Archimedean, and hence — by the proof of Theorem 2.12 read in reverse — not complete. Berkeley's ghosts, housed one jurisdiction over; completeness is the eviction order.

2.9 Hints (Tiers II and III)

II.3 Clause (i) is the definition of an upper bound. For clause (ii), dip to $x - \varepsilon$ and let Theorem 2.15 plant a picket in $(x - \varepsilon,\ x)$.

II.4 Let a_n be the n-digit decimal truncation of $\sqrt{2}$ (a rational), and consider $I_n = [a_n,\ a_n + 10^{-n}]$. Check the nesting from the truncation property, and identify the unique survivor via Remark 2.18.

II.5 Corollary 2.16 and Theorem 2.15, each aimed at the appropriate pair.

II.6 Lemma 2.4 with $\varepsilon = \frac{1}{n}$.

III.1 Properness: take $a' \notin \alpha$, $b' \notin \beta$; every positive product ab satisfies $ab < a'b'$. Downward closure is where the adjoined tail $\{q \leq 0\}$ earns its keep. For signs: there are nine sign patterns for (α, β) and the definition must be assembled from $|\alpha| \cdot |\beta|$ with a sign rule; write out three cases and extrapolate the tedium.

III.2 For well-definedness: replacing p/q by pr/qr multiplies both leading coefficients by the same leading coefficient. For (b): $t - n$ has numerator with positive leading coefficient. For (d): if F were complete, the proof of Theorem 2.12 would run verbatim and evict your infinitesimal. Further reading in the notes.

2.10 Solutions

Tier I (complete, terse)

I.1 (a) $\sup = 1$, attained ($n = 1$); $\inf = 0$, not attained (Problem I.5). (b) $x^2 < 2x \iff x(x-2) < 0 \iff 0 < x < 2$: the set is $(0, 2)$; $\sup = 2$, $\inf = 0$, neither attained. (c) $\sup = 3$, attained; $\inf = 0$, not attained. (d) $\frac{n}{n+1} = 1 - \frac{1}{n+1}$: $\sup = 1$, not attained; $\inf = \frac{1}{2}$, attained ($n = 1$). (e) The even-indexed values $1 + \frac{1}{n}$ descend from $\frac{3}{2}$; the odd-indexed values $-(1 + \frac{1}{n})$ ascend from -2. $\sup = \frac{3}{2}$, attained ($n = 2$); $\inf = -2$, attained ($n = 1$).

I.2 If s and t are both suprema, each is an upper bound; s least gives $s \leq t$, and t least gives $t \leq s$; so $s = t$.

I.3 $\sup B$ is an upper bound of B, hence of $A \subseteq B$; and $\sup A$ is the *least* upper bound of A, so $\sup A \leq \sup B$.

I.4 Let $M \in A$ be an upper bound. It is the maximum by definition. Any upper bound u satisfies $u \geq M$ (since $M \in A$), so M is least: $M = \sup A$.

I.5 0 is a lower bound. If $m > 0$ were a lower bound, Theorem 2.12(ii) yields n with $\frac{1}{n} < m$, a member beneath m: contradiction. So every lower bound is ≤ 0, and $\inf = 0$; it is not attained since $\frac{1}{n} > 0$ always.

I.6 1 is an upper bound since $1 - \frac{1}{n} < 1$. Given $\varepsilon > 0$, choose $n > \frac{1}{\varepsilon}$ (Theorem 2.12(i) applied to $1/\varepsilon$); then $\frac{1}{n} < \varepsilon$ and $1 - \frac{1}{n} > 1 - \varepsilon$. By Lemma 2.4, $\sup = 1$; not attained.

I.7 For $a \in A$: $ca \leq c \sup A$, so $c \sup A$ is an upper bound of cA. If u is any upper bound of cA, then $a \leq u/c$ for all a, so $u/c \geq \sup A$, that is, $u \geq c \sup A$. Hence $\sup(cA) = c \sup A$.

I.8 Say $\sup A \geq \sup B$. Every element of $A \cup B$ is $\leq \sup A$, so $\sup A$ is an upper bound. Any upper bound of $A \cup B$ bounds A, hence is $\geq \sup A$. So $\sup(A \cup B) = \sup A = \max\{\sup A, \sup B\}$.

Tier II (selected)

II.1 Throughout, recall from §2.2 that for positives, $x < y \iff x^2 < y^2$. S is non-empty ($1 \in S$) and bounded above by 2.

(a) Let $q > 0$ with $q^2 > 2$. For any $x \in S$: $x^2 < 2 < q^2$, both positive, so $x < q$. Thus q is an upper bound.

(c) — computed first, since (b) uses it. For rational $q > 0$ and $q' = \dfrac{2q+2}{q+2}$ (rational, and positive since numerator and denominator are positive):

$$q' - q = \frac{2q + 2 - q(q+2)}{q+2} = \frac{2 - q^2}{q+2}, \qquad {q'}^2 - 2 = \frac{(2q+2)^2 - 2(q+2)^2}{(q+2)^2} = \frac{2\,(q^2-2)}{(q+2)^2}.$$

So ${q'}^2 - 2$ has the *same* sign as $q^2 - 2$, while $q' - q$ has the *opposite* sign.

(b) A positive rational upper bound u cannot have $u^2 = 2$ (Chapter 1, Theorem 1.6). If $u^2 < 2$, then $u \in S$, and by (c) the rational $u' > u$ also has ${u'}^2 < 2$, so $u' \in S$ exceeds u — contradicting that u is an upper bound. (The same computation shows S has no greatest member.) Hence every rational upper bound has square exceeding 2.

Conclusion. Let u be any rational upper bound; by (b), $u^2 > 2$; by (c), $u' < u$ with ${u'}^2 > 2$ and $u' > 0$; by (a), u' is an upper bound. Every rational upper bound sits strictly above another: no least one exists. ∎

II.2 Let $s = \sup A$, $t = \sup B$. For $a \in A, b \in B$: $a + b \leq s + t$, so $s + t$ is an upper bound of $A + B$. Given $\varepsilon > 0$, Lemma 2.4 provides $a > s - \frac{\varepsilon}{2}$ and $b > t - \frac{\varepsilon}{2}$, whence $a + b > s + t - \varepsilon$. So any upper bound u of $A + B$ satisfies $u > s + t - \varepsilon$ for every $\varepsilon > 0$, and the slack remark after Lemma 2.4 gives $u \geq s + t$. Hence $s + t$ is least: $\sup(A + B) = \sup A + \sup B$. ∎

Problems II.3–II.6 are left with their hints, per standing policy; III.1 and III.2 receive hints only.

Notes and References

- **Dedekind**, *Continuity and Irrational Numbers* (1872), in *Essays on the Theory of Numbers*, Beman translation (Dover; public domain) — thirty pages, and among the most consequential per page ever printed. The preface contains the dated idea and the deferred publication.
- **Abbott**, *Understanding Analysis*, 2nd ed., §§1.3–1.4 (suprema, completeness, and the dividend theorems, with parallel problems) and §8.6 (the cut construction carried out as a guided project).
- **Rudin**, *Principles of Mathematical Analysis*, Chapter 1 and its Appendix — the cut construction in full, multiplication swamp included: the once-for-all-humanity reference of §2.6.
- **Landau**, *Foundations of Analysis* — the entire arithmetic of the number systems built without one wasted word or one spared detail; bracing.
- **Tao**, *Analysis I* — the Cantor–Méray construction (Cauchy sequences) performed in full, the rival firm's own prospectus.
- **Ebbinghaus et al.**, *Numbers* — Chapter 2 for the uniqueness of the complete ordered field (Remark 2.8's credit), and much else besides.
- **Goldblatt**, *Lectures on the Hyperreals* — where Problem III.2's ghosts are given a rigorous afterlife (Robinson's non-standard analysis).

Chapter 3

Cantor, or the Sizes of Infinity

Companion to Episode Three. The episode weighed the infinite crowds: pairing as the meaning of size, the countable menagerie, both of Cantor's uncountability proofs, the halting-problem bridge, the census of numbers, the continuum hypothesis, and the birth of the dust. This chapter formalizes the lot, with one credit taken and flagged.

Episode Digest

The argument of the hour. Size is pairing, not counting; by that honest standard the rationals — dense as they are — can be marched single file, and the reals cannot. Uncountability is proved twice: by the Russian dolls (1874) and by the diagonal (1891), which then escapes mathematics and proves the halting problem undecidable. A census shows every number algebra can describe forms a countable film on the line: the transcendentals — the unnameable — predominate. Cantor's question of whether anything lies between the two sizes breaks loose from the axioms themselves; and the middle-thirds dust is born: uncountable, yet containing no interval at all.

The episode, beat by beat. **What Same Size Must Mean.** A shepherd matches sheep to pebbles: comparison precedes counting, and *same size* means a perfect pairing — a bijection, everyone partnered, none left over (Definitions 3.1, 3.2). Galileo (1638): the squares dance with all the whole numbers — a part equal to the whole; right observation, wrong conclusion. Dedekind inverts the scandal into the definition: *infinite* means "dances with a proper part of itself" — when mathematics cannot defeat a monster, it hires it and issues a uniform. Hilbert's hotel, checked in and out briskly: adding one guest, or countably many, moves nothing.

The Single File. Countable = dances with $\mathbb{N}$ = admits an exhaustive roster with every member at a finite position. The queue is ours to design: the integers yield to the from-the-middle zigzag ($0, 1, -1, 2, -2, \ldots$; Proposition 3.4). One planted caution: tonight's rosters are God's-eye lists, not necessarily printable — the gap where the saboteur will live.

The Zigzag. First shock: $\mathbb{Q}$ is countable (Theorem 3.7). Seat the fractions in an infinite grid (row = denominator), march the *finite* diagonals, skip duplicates — every fraction reached at a finite address. Density is no obstacle to being listed. The moral instrument: countably many countable crowds pool into one countable crowd (Theorem 3.8; Problem II.1, with its honest footnote on infinitely many acts of choice). In the listener's dialect: countable = enumerable, a lazy stream — and the theorem coming asserts a stream no generator can produce.

The First Murder, by Russian Dolls. December 1873, the letters to Dedekind; print 1874, under a camouflaged title. Given *any* alleged complete roster of $\mathbb{R}$: split a closed interval into three closed thirds — a single point can touch at most two — and pass to a third avoiding suspect one; inside it, evict suspect two; forever. Closed, bounded, nested, widths beneath every tolerance: the dolls (*the bell, fourth ring*) trap a survivor differing from every listed number (Theorem 3.10). The hunt is local — run it inside any interval, so every interval is uncountable, and the irrationals are uncountable (Corollary 3.11): Episode Two's scale crashes. The force is completeness itself; in $\mathbb{Q}$ the trap can close on a hole, as it must, since $\mathbb{Q}$ queues.

The Diagonal. 1891, the set piece. Strip to flip-records: endless heads-tails sequences, $\{0,1\}^{\mathbb{N}}$. Picture the adversary's roster as an infinite table; read one cell per row, straight down the diagonal, and *write the opposite*: $t_n = 1 - s_n^{(n)}$. The fugitive differs from row n at column n, for every n: no roster is complete (Theorem 3.12). Four moves — list, diagonal, flip, contradiction — using no completeness, no order, no arithmetic: it defeats every list at once, and the list defeats itself. Same blade, general form: no set surjects onto its power set ($D = \{a : a \notin f(a)\}$ is unclaimed — Theorem 3.13), so the sizes ladder upward without end.

The Same Blade on Your Own Hardware. Programs are finite texts, hence countable (Corollary 3.9). Suppose an oracle decides halting. Build the saboteur: on input n, ask the oracle about program n run on input n — the diagonal cell — and do the *opposite*. The saboteur is a program, so it has an index; feed it its own index, and it halts if and only if it does not (Remark 3.14; Turing 1936, with Gödel 1931 name-checked as the third of the family). The plague spreads to essentially every question about what programs *do*: the deep reason testing never became proving.

The Census of Numbers. Algebraic = root of an integer polynomial (Definition 3.15) — everything algebra can name. The algebraics are countable (Theorem 3.17: enumerate by height; the chapter's one credit — at most n roots for degree n). Uncountable minus countable is uncountable (Lemma 3.16), so transcendentals *predominate*, in every interval (Corollary 3.18) — yet the specimen cabinet is nearly bare: Liouville 1844 (a decimal with ones at the factorial positions, flattered by fractions beyond algebraic tolerance), Hermite 1873 (e), Lindemann 1882 (π — the circle unsquarable). Cantor exhibits none: existence without exhibition, at industrial scale. Sharper still: any fixed language's finite texts are countable, so almost every real is not merely unnamed but unnameable (Remark 3.19, with the logician's asterisk).

The Arithmetic of the Infinite, and the Question That Broke. The countable size is christened $\aleph_0$. Adding one, adding $\aleph_0$, even multiplying by $\aleph_0$ — all slide off (hotel; interleaving; the zigzag). What moves the needle is the power set: the subsets of $\mathbb{N}$ *are* the flip-records, and their crowd is the continuum's ($2^{\aleph_0} = |\mathbb{R}|$, Problem III.2). The continuum hypothesis — nothing strictly between $\aleph_0$ and the continuum — was Hilbert's problem number one (1900). Gödel 1940: consistent with the axioms. Cohen 1963, by forcing: so is its negation. The axioms are silent (Remark 3.24); the door is marked, and analysis lives happily on this side of it.

The Dust Is Born. Delete the open middle third of $[0,1]$; then of each survivor; forever (Definition 3.20). The endpoints survive and are countable — but an *itinerary* (left or right at every stage, an endless two-symbol record) steers a nest of closed thirds, widths $3^{-n} \leq 1/n$, onto exactly one grain (*the bell, fifth ring*); distinct itineraries part into disjoint children: the flip-records inject into the dust, which is therefore uncountable (Proposition 3.21) — almost no grain is an endpoint; the dust is mostly ghosts. Yet no interval survives: a stretch inside C must sit in one component of stage n, so its width undercuts 3^{-n} for every n — zero, by Chapter 2's slack remark (Proposition 3.22). Ternary costume: expansions avoiding the digit 1; self-similar, copies of itself all the way down (Remark 3.23). Contracts filed: Chapter 5, Volume Two, and one sealed question of extent — deliberately unasked tonight.

Paradise and Its Enemies. Kronecker: a serious finitist creed — and a disgraceful campaign: editorial obstruction, Berlin barred, reported epithets. The position was defensible; the conduct was not. Weierstrass, to his credit, backed the young work. Cantor's health broke in 1884 and periodically after; he died in 1918, in care — three sentences, no gawking. What wants remembering: Dedekind's steady friendship, and the verdict of history — set theory became the foundation of everything, Hilbert declared (1926) that no one shall expel us from Cantor's paradise, and Cantor himself had already written (1883) the season's creed at full generality: the essence of mathematics lies precisely in its freedom. Kronecker died in 1891 — the year of the diagonal.

The toolkit acquired. The grammar of pairings (Definitions 3.1–3.2, Proposition 3.3) and a drawer of explicit bijections (Proposition 3.4; Problems I.1–I.4); subsets of countable sets are countable (Lemma 3.5); the pairing function (Proposition 3.6); $\mathbb{Q}$ countable (Theorem 3.7); pooling with its choice footnote (Theorem 3.8, II.1); the 1874 dolls proof (Theorem 3.10 ⊛) and the diagonal (Theorem 3.12); Cantor's theorem (Theorem 3.13); the census (Theorem 3.17, II.2; Corollary 3.18); the dust's address system and interval-freeness (Propositions 3.21 ⊛, 3.22). Supplied solutions complete the audio's sketches: pooling (II.1) and the census (II.2).

Debts. D2 (Zeno: the race) moves to the top of the pile — due next episode. D7 (the bell): rung twice tonight — the 1874 dolls and the dust's addressing — running count **five**. **New:** the dust's three contracts (Episode Five; Season Two; one sealed question of extent). D6 (existence without exhibition) gains the census as a second entry; the family bill falls due in Season Two, Episode Three. The dolls' murder-weapon contract from Episode Two: executed.

Dates worth keeping. 1638 Galileo's final book; December 1873 the letters; 1874 the first proof, camouflaged; 1844 Liouville; 1873 Hermite; 1882 Lindemann; 1883 the dust, and "freedom"; 1884 the first collapse; 1891 the diagonal — and Kronecker's death; 1900 Hilbert's list; 1918 Cantor's death, in care; 1926 "paradise"; 1931 Gödel; 1936 Turing; 1940 and 1963 the two halves of independence.

3.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"a perfect dance" / "same size"	a bijection $f\colon A \to B$; $A \sim B$	Definitions 3.1–3.2
"countable" / "the single file"	$A \sim \mathbb{N}$ (or A finite)	Definition 3.2
"uncountable"	infinite and not $\sim \mathbb{N}$	Definition 3.2
"aleph nought"	$\aleph_0$, the cardinality of $\mathbb{N}$	§3.7
"the flip-records"	$\{0,1\}^{\mathbb{N}}$, all maps $\mathbb{N} \to \{0,1\}$	Theorem 3.12
"the crowd of subcollections"	the power set $\mathcal{P}(A)$	Theorem 3.13
"two to the aleph nought"	$2^{\aleph_0} = \lvert\mathcal{P}(\mathbb{N})\rvert = \lvert\mathbb{R}\rvert$	Problem III.2
"algebraic" / "transcendental"	root of a $\mathbb{Z}$-polynomial / not	Definition 3.15
"the dust"	the Cantor set $C = \bigcap_n C_n$	Definition 3.20
"the itinerary"	$\sigma \in \{L, R\}^{\mathbb{N}}$	Proposition 3.21

3.2 Pairings and Size

Definition 3.1. Let $f\colon A \to B$. Then f is *injective* if distinct inputs have distinct outputs ($f(a) = f(a') \Rightarrow a = a'$); *surjective* if every $b \in B$ is $f(a)$ for some a; *bijective* if both — a perfect pairing, under which each element of either set has exactly one partner in the other. (These terms were deferred from Chapter 1, §1.4; here they earn their keep.)

Definition 3.2. Sets A and B have the *same cardinality*, written $A \sim B$, if a bijection $A \to B$ exists. A set is *countably infinite* if it $\sim \mathbb{N}$; *countable* if finite or countably infinite; *uncountable* if infinite and not countable. To be countably infinite is exactly to admit an exhaustive roster $a_1, a_2, a_3, \ldots$ in which every element appears at exactly one finite position.

Proposition 3.3. *$\sim$ is an equivalence: $A \sim A$; if $A \sim B$ then $B \sim A$; if $A \sim B \sim C$ then $A \sim C$.*

Proof. The identity is a bijection; the inverse of a bijection is a bijection; the composition of bijections is a bijection. (Each claim is a one-line check against Definition 3.1.) □

Proposition 3.4. *$\mathbb{N} \sim \mathbb{Z}$, via $f(n) = n/2$ for n even, $f(n) = -(n-1)/2$ for n odd.*

Proof. Problem I.1 (a routine verification: $1, 2, 3, 4, 5, \cdots \mapsto 0, 1, -1, 2, -2, \ldots$; each integer is hit exactly once). □

Calibrating examples. $n \mapsto 2n$ pairs $\mathbb{N}$ with the evens; $n \mapsto n^2$ with the squares (Galileo's dance: a proper part matching the whole — for infinite sets, the membership card, not a paradox). Any two open intervals pair by a straight-line map (Problem I.3); and $(-1, 1) \sim \mathbb{R}$ via the purely algebraic bijection

$$g(x) = \frac{x}{1 - |x|}, \qquad g^{-1}(y) = \frac{y}{1 + |y|}$$

(Problem I.4) — so every open interval is, size-wise, the whole line.

3.3 The Countable Menagerie

Lemma 3.5. *Every subset of a countable set is countable. In particular, every infinite subset $S \subseteq \mathbb{N}$ satisfies $S \sim \mathbb{N}$.*

Proof. It suffices to prove the second statement (a subset of a countable set transports, along the roster, to a subset of $\mathbb{N}$). Let $S \subseteq \mathbb{N}$ be infinite. Define $s_1 = \min S$ (well-ordering, Chapter 1) and, recursively, $s_{k+1} = \min\{ s \in S : s > s_k \}$ — the sets minimized are non-empty because S is infinite. The map $k \mapsto s_k$ is injective (strictly increasing) and surjective onto S: any $s \in S$ has only finitely many predecessors in S, so $s = s_k$ where $k-1$ counts them. □

Proposition 3.6. $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$, *via the diagonal enumeration*

$$P(m,n) = \frac{(m+n-2)(m+n-1)}{2} + m.$$

Proof. P lists the pairs diagonal by diagonal: the pairs with $m+n = d+1$ form the d-th diagonal, of length d; the summand $\frac{(m+n-2)(m+n-1)}{2} = 1 + 2 + \cdots + (m+n-2)$ counts all seats on earlier diagonals (Proposition 1.2, Chapter 1), and m locates the seat within the current one. Distinct pairs get distinct positions, and every position is reached: bijective. □

Theorem 3.7 ($\mathbb{Q}$ is countable)**.** *The rationals are countably infinite.*

Proof. Each nonzero rational has a unique lowest-terms form p/q with $p \in \mathbb{Z} \setminus \{0\}$, $q \in \mathbb{N}$, $\gcd(|p|, q) = 1$ (Lemma 1.4, Chapter 1); send 0 to $(0,1)$. This injects $\mathbb{Q}$ into $\mathbb{Z} \times \mathbb{N}$, which is countable: $\mathbb{Z} \times \mathbb{N} \sim \mathbb{N} \times \mathbb{N} \sim \mathbb{N}$ by Propositions 3.4 and 3.6. So $\mathbb{Q}$ is (transporting along these bijections) an infinite subset of a countable set, hence countable by Lemma 3.5. □

Theorem 3.8 (Pooling: countable unions)**.** *If $A_1, A_2, A_3, \ldots$ are countable sets, then $\bigcup_n A_n$ is countable.*

Proof. Problem II.1, solution supplied — including an honest footnote on the infinitely many acts of choice involved. □

Corollary 3.9. *The set of all finite strings over a finite alphabet is countable (list by length, alphabetically within a length: finitely many per length, then pool). Hence: all computer programs form a countable set; so do all finite subsets of $\mathbb{N}$ (encode as strings). By contrast, the* infinite *strings are the subject of §3.5.*

3.4 The First Uncountability Proof (1874)

Theorem 3.10 (Cantor, 1874: $\mathbb{R}$ is uncountable)**.** *No sequence $x_1, x_2, x_3, \ldots$ of real numbers exhausts $\mathbb{R}$. Indeed, for any such sequence and any closed interval $[a,b]$ with $a < b$, there is a point of $[a,b]$ equal to no x_n.*

Proof. Set $I_0 = [a,b]$. Given the closed interval $I_{n-1} = [u,v]$ of width $w = v - u > 0$, split it into its three closed thirds

$$\left[u,\ u + \tfrac{w}{3}\right], \quad \left[u + \tfrac{w}{3},\ u + \tfrac{2w}{3}\right], \quad \left[u + \tfrac{2w}{3},\ v\right].$$

The single point x_n lies in at most two of them (it can belong to two only as a shared endpoint), so at least one third excludes it; let I_n be the leftmost such third. Then $I_1 \supseteq I_2 \supseteq \cdots$ are closed and bounded, with $x_n \notin I_n$ and widths $(b-a)/3^n \le (b-a)/n$ (since $3^n \ge 2^n > n$, Chapter 1, Problem I.5), hence beneath every tolerance. By the nested interval theorem in its unique-survivor form (Theorem 2.17 and Remark 2.18) ($\circledast$), there is $x^* \in \bigcap_n I_n \subseteq [a,b]$. For every n: $x^* \in I_n$ and $x_n \notin I_n$, so $x^* \neq x_n$. $\square$

Corollary 3.11. *Every interval of positive width is uncountable; the irrationals are uncountable.*

Proof. The first claim is Theorem 3.10 applied inside the interval. For the second: were $\mathbb{R} \setminus \mathbb{Q}$ countable, then $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be a union of two countable sets, countable by Theorem 3.8 — contradicting Theorem 3.10. $\square$

Note where the force came from: the dolls, i.e. completeness. Run inside $\mathbb{Q}$ the construction still nests, but the trap may close on a hole and catch nothing (Chapter 2, Problem II.4) — as it must, since $\mathbb{Q}$ *does* queue.

3.5 The Diagonal (1891) and Cantor's Theorem

Theorem 3.12 (Cantor, 1891)**.** *The set $\{0,1\}^{\mathbb{N}}$ of all infinite binary sequences is uncountable.*

Proof. Let $s^{(1)}, s^{(2)}, s^{(3)}, \dots$ be any sequence of elements of $\{0,1\}^{\mathbb{N}}$, writing $s^{(n)} = (s_1^{(n)}, s_2^{(n)}, \dots)$. Define $t \in \{0,1\}^{\mathbb{N}}$ by

$$t_n = 1 - s_n^{(n)} \qquad (n \in \mathbb{N}).$$

For every n, $t \neq s^{(n)}$, since they differ at position n. So no roster contains every binary sequence. $\square$

Theorem 3.13 (Cantor's theorem)**.** *For every set A there is no surjection $A \to \mathcal{P}(A)$; hence $\mathcal{P}(A)$ is never $\sim A$, and the sizes $|A| < |\mathcal{P}(A)| < |\mathcal{P}(\mathcal{P}(A))| < \cdots$ ascend without end.*

Proof. Let $f\colon A \to \mathcal{P}(A)$ be any map, and set

$$D = \{\, a \in A : a \notin f(a) \,\} \in \mathcal{P}(A).$$

If $D = f(a_0)$ for some a_0, then $a_0 \in D \iff a_0 \notin f(a_0) = D$: contradiction. So D is not a value of f, and f is not surjective. (This *is* the diagonal: D inverts, at each a, the answer to "does a belong to its own assigned set?") $\square$

Remark 3.14 (The family). The same skeleton — roster, diagonal, inversion — yields Turing's 1936 theorem (no program decides halting: the saboteur inverts the oracle's verdict on the diagonal cell "program n on input n", then is fed its own index) and, in a subtler costume, Gödel's 1931 incompleteness theorem. The episode's narration of the saboteur is already essentially complete; for the honest formal treatment, see the notes.

3.6 The Census of Numbers

Definition 3.15. A real number is *algebraic* if it is a root of some polynomial $a_n x^n + \cdots + a_1 x + a_0$ with integer coefficients, not all zero. A real that is not algebraic is *transcendental*.

Lemma 3.16. *If X is uncountable and $Y \subseteq X$ is countable, then $X \setminus Y$ is uncountable.*

Proof. Problem II.5 — yours in two lines from Theorem 3.8. □

Theorem 3.17 (The algebraic numbers are countable)**.**

Proof. Problem II.2, solution supplied (enumeration by the *height* $h = n + |a_0| + \cdots + |a_n|$; finitely many polynomials per height, each with finitely many roots). □

Credit box (the chapter's only one). A nonzero polynomial of degree n has at most n real roots. This is the factor theorem's standard consequence and belongs to algebra; we take it on credit, with references in the notes.

Corollary 3.18 (Transcendentals predominate)**.** *The transcendental numbers are uncountable; indeed every interval of positive width contains uncountably many.*

Proof. $\mathbb{R}$ is uncountable (Theorem 3.10) and the algebraics A are countable (Theorem 3.17), so $\mathbb{R} \setminus A$ is uncountable (Lemma 3.16); the local statement applies Corollary 3.11 to the interval and Lemma 3.16 to A intersected with it. □

Remark 3.19 (History, and the describability argument). Liouville (1844) exhibited the first transcendental ($\sum_k 10^{-k!}$, whose fractions-flatter-it-too-well property no algebraic number tolerates); Hermite (1873) proved e transcendental; Lindemann (1882) proved π transcendental, settling the impossibility of squaring the circle. Cantor's census proves transcendentals *predominate* while exhibiting none — existence without exhibition, the sensation of Chapter 1's Problem III.2, now industrial. The episode's "unnameable" argument — finite texts over a fixed alphabet are countable (Corollary 3.9), so any fixed language describes at most countably many reals — is rigorous once the language and its reference relation are fixed in advance; treating "definable in mathematics itself" as a completed totality invites Richard's paradox. See the notes; the moral survives the asterisk.

3.7 The Dust

Definition 3.20 (The Cantor set)**.** Let $C_0 = [0, 1]$. Given C_{n-1}, a disjoint union of 2^{n-1} closed intervals of width $3^{-(n-1)}$, let C_n be the result of deleting the open middle third of each: 2^n closed intervals of width 3^{-n}. The *Cantor set* is $C = \bigcap_{n \geq 0} C_n$. The two closed thirds retained from a parent interval $[u, u + w]$ are its *left child* $[u, u + w/3]$ and *right child* $[u + 2w/3, u + w]$; they are disjoint, separated by the deleted middle.

Proposition 3.21 (The address system)**.** *For each itinerary $\sigma \in \{L, R\}^{\mathbb{N}}$ let $J_0 = [0, 1]$ and let J_n be the σ_n-child of J_{n-1}. Then $\bigcap_n J_n$ contains exactly one point $g(\sigma)$; the map g is injective; and $g(\sigma) \in C$. Consequently C — and hence $[0, 1]$, and hence $\mathbb{R}$ — is uncountable.*

Proof. The J_n are closed, bounded, nested, of widths $3^{-n} \leq 1/n$, beneath every tolerance; the unique-survivor form of the nested interval theorem (Remark 2.18) (⊛) yields exactly one common point $g(\sigma)$. Since $J_n \subseteq C_n$ for every n (induction: children of stage-$(n-1)$ intervals are stage-n intervals), $g(\sigma) \in \bigcap C_n = C$. Injectivity: if $\sigma \neq \tau$ first differ at stage n, then $J_n(\sigma)$ and $J_n(\tau)$ are the two children of one parent, hence disjoint, and the survivors lie one in each. Finally, g injects the uncountable set $\{L, R\}^{\mathbb{N}} \sim \{0, 1\}^{\mathbb{N}}$ (Theorem 3.12) into C; were C countable, its subset $g(\{L, R\}^{\mathbb{N}})$ would be countable (Lemma 3.5), yet it is $\sim \{0, 1\}^{\mathbb{N}}$: contradiction. □

Proposition 3.22 (No interval survives)**.** *C contains no interval of positive width.*

Proof. Suppose $[u, v] \subseteq C$ with $u < v$. Fix n. Then $[u, v] \subseteq C_n$, and u, v must lie in the *same* component interval of C_n: two distinct components are separated by a deleted open gap, and a gap point between u and v would lie in $[u, v] \subseteq C_n$, absurdly. Hence $v - u \leq 3^{-n} \leq 1/n$ for every n, so $v - u \leq 0$ by the slack remark of Chapter 2: no such interval exists. $\square$

Remark 3.23 (Ternary costume, endpoints, self-similarity, contracts). (i) In base three, C is exactly the set of points in $[0, 1]$ possessing a ternary expansion that avoids the digit 1: the itinerary read as digits, $L \mapsto 0$, $R \mapsto 2$ — minding the usual nuisance that some numbers own two expansions ($1/3 = 0.1\overline{0} = 0.0\overline{2}$ in base three: it *has* an avoiding expansion, and belongs). (ii) The endpoints of the construction (survivor-interval boundaries, accumulated over the countably many stages) form a countable subset; by Proposition 3.21 and Lemma 3.16, almost every grain — in the cardinality sense — is *not* an endpoint. (iii) $C \cap [0, \frac{1}{3}]$ is a $\frac{1}{3}$-scale replica of C; the dust is built of copies of itself at every depth. (iv) Contracts on file: the dust returns in Chapter 5 (its topology) and in Volume Two (a certain sealed question of extent); this chapter deliberately computes no lengths.

Remark 3.24 (The continuum hypothesis). CH asserts: no set $S \subseteq \mathbb{R}$ satisfies $\aleph_0 < |S| < |\mathbb{R}|$. Gödel (1940) proved CH consistent with the standard (Zermelo–Fraenkel, with Choice) axioms; Cohen (1963), by forcing, proved its negation consistent as well. CH is therefore undecidable from those axioms: they underdetermine the answer, and settling it would require new axioms whose adoption remains a live debate. Nothing in this course depends on it. References in the notes; the door is marked, and we do not enter.

3.8 Problems

Tier I — Calisthenics

I.1 Verify that the map of Proposition 3.4 is a bijection $\mathbb{N} \to \mathbb{Z}$ (injective and surjective, with the inverse written out).

I.2 Exhibit bijections $\mathbb{N} \to$ the even numbers and $\mathbb{N} \to$ the perfect squares, and verify them.

I.3 For real $a < b$ and $c < d$, exhibit a straight-line bijection $(a, b) \to (c, d)$.

I.4 Verify that $g(x) = x/(1 - |x|)$ is a bijection $(-1, 1) \to \mathbb{R}$ with inverse $y \mapsto y/(1 + |y|)$. Conclude that every open interval is $\sim \mathbb{R}$.

I.5 List the finite binary strings in a single file (by length, then alphabetically) and give the exact queue positions of the strings 0, 1, 00, and 101.

I.6 Compute the first twelve entries of the zigzag through the positive rationals (grid rows indexed by denominator), with duplicates skipped.

I.7 Prove: the union of *two* countable sets is countable, by an explicit interleaving (no appeal to Theorem 3.8).

I.8 Prove that the set of all *finite* subsets of $\mathbb{N}$ is countable. (Encode, or pool by maximum element.)

Tier II — The Real Thing

II.1 (Pooling; solution provided.) Prove Theorem 3.8: a countable union of countable sets is countable. Handle empty and finite A_n, and flag honestly the step where infinitely many enumerations are chosen at once.

II.2 (The census; solution provided.) Prove Theorem 3.17: the algebraic numbers are countable, by enumerating polynomials by height.

II.3 (The diagonal on decimals.) Prove directly that $[0,1]$ is uncountable by diagonalizing decimal expansions — constructing the fugitive using only the digits 4 and 5, and explaining exactly how this dodges the double-expansion nuisance ($0.4999\ldots = 0.5000\ldots$).

II.4 Prove that every infinite set contains a countably infinite subset. Flag the acts of choice involved.

II.5 Prove Lemma 3.16 from Theorem 3.8 in at most three lines.

II.6 (The hotel, applied.) Exhibit a bijection $(0,1) \to [0,1]$. (Shift a well-chosen countable file inside $(0,1)$ to absorb the two endpoints.)

Tier III — For the Ambitious ⋆

⋆ **III.1** (Cantor–Schröder–Bernstein.) If there exist injections $A \to B$ and $B \to A$, then $A \sim B$. Prove it, in the guided stages of the hints: the theorem everyone uses and few have proved, whose statement feels administrative and whose proof contains a genuine idea.

⋆ **III.2** (Weighing the continuum.) Prove $\mathbb{R} \sim \mathcal{P}(\mathbb{N})$: inject $\{0,1\}^{\mathbb{N}}$ into $[0,1]$ one way and $[0,1]$ into $\{0,1\}^{\mathbb{N}}$ the other, close the loop with III.1, then identify $\mathcal{P}(\mathbb{N})$ with $\{0,1\}^{\mathbb{N}}$ and $[0,1]$ with $\mathbb{R}$. Conclusion: $2^{\aleph_0} = |\mathbb{R}|$ — the continuum, weighed by your own hand.

3.9 Hints

II.3. Given an alleged roster $x_1, x_2, \ldots$ of $[0,1]$, fix for each a decimal expansion and set the n-th digit of the fugitive t to be 4 unless the n-th digit of x_n is 4, in which case 5. Why this dodges the nuisance: an expansion using only the digits 4 and 5 contains no tail of 0s or 9s, so t has a *unique* expansion — differing from x_n as a digit string at position n therefore means differing as a *number*.

II.4. Build the file recursively: having chosen distinct $a_1, \ldots, a_n$ in the infinite set A, the set $A \setminus \{a_1, \ldots, a_n\}$ is non-empty (else A was finite); choose a_{n+1} from it. The map $n \mapsto a_n$ is injective. The word "choose", exercised infinitely often with no stated rule, is the flag: this is (dependent) choice, cousin of II.1's footnote.

II.5. Write $X = Y \cup (X \setminus Y)$ and let Theorem 3.8 (or Problem I.7) do the talking.

II.6. March the file $s_n = 1/(n+1)$, all inside $(0,1)$. Send $s_1 \mapsto 0$, $s_2 \mapsto 1$, $s_n \mapsto s_{n-2}$ for $n \geq 3$, and fix every point not in the file. Verify injective and surjective: Hilbert's hotel absorbing two guests.

III.1. Stage one: with injections $f\colon A \to B$ and $g\colon B \to A$, trace each $a \in A$ backwards — a, then its g-preimage if it has one, then that element's f-preimage if it has one, alternating sets. Three fates: the trace halts in A (at an element outside g's image — call a an A-stopper), halts in B (a B-stopper), or never halts. Stage two: define $h(a) = f(a)$ for A-stoppers and never-stoppers, and $h(a) =$ the g-preimage of a for B-stoppers (why does it exist?). Stage three: classify B the same way and check h maps each class of A bijectively onto the matching class of B. Abbott runs this ascent as a guided exercise if you want rails.

III.2. One direction is already built: the address map g of Proposition 3.21 injects $\{0,1\}^{\mathbb{N}}$ (as itineraries) into $C \subseteq [0,1]$. For the other, send $x \in [0,1]$ to a binary expansion chosen canonically — for $x > 0$, the expansion with infinitely many 1s; $0 \mapsto$ all zeros — and check distinct reals receive distinct sequences. III.1 closes $[0,1] \sim \{0,1\}^{\mathbb{N}}$. Finally $\mathcal{P}(\mathbb{N}) \sim \{0,1\}^{\mathbb{N}}$ outright (a subset and its indicator sequence), and $\mathbb{R} \sim (0,1) \sim [0,1]$ by I.4, I.3, and II.6.

3.10 Solutions

Tier I (terse). **I.1** Inverse: $m \mapsto 2m$ for $m \geq 1$; $m \mapsto 1 - 2m$ for $m \leq 0$. Both composites are the identity (check the even and odd cases separately); hence bijective. **I.2** $n \mapsto 2n$: injective ($2n = 2m \Rightarrow n = m$), surjective (the even $2k$ is hit by k). $n \mapsto n^2$: injective on $\mathbb{N}$, surjective onto the squares by definition. **I.3** $x \mapsto c + (d-c)(x-a)/(b-a)$: affine, positive slope, sends $a \mapsto c$, $b \mapsto d$; the inverse is of the same form with the roles swapped. **I.4** For $x \in [0,1)$: $g(x) = x/(1-x) \geq 0$ and $g(x)/(1+g(x)) = x$ by algebra; the case $x \in (-1,0)$ is the mirror (signs are preserved). Conversely $y/(1+|y|) \in (-1,1)$ and g returns y. Both composites are identities: bijection. Any open interval reaches $\mathbb{R}$ through I.3 then g. **I.5** The file runs $0, 1, 00, 01, 10, 11, 000, 001, \ldots$ Positions: 0 first, 1 second, 00 third; the length-three block opens at position seven, and 101 is its sixth entry: position twelve. **I.6** $\frac{1}{1}$; $\frac{2}{1}, \frac{1}{2}$; $\frac{3}{1}, \frac{1}{3}$ (skipping $\frac{2}{2}$); $\frac{4}{1}, \frac{3}{2}, \frac{2}{3}, \frac{1}{4}$; $\frac{5}{1}, \frac{1}{5}$ (skipping $\frac{4}{2}, \frac{3}{3}, \frac{2}{4}$); $\frac{6}{1}$ — twelve entries. **I.7** Interleave: $c_{2n-1} = a_n$, $c_{2n} = b_n$, skipping repeats; if one set is finite, let the other continue alone. Every element of either set is reached at a finite position. **I.8** Send the finite set $S \subseteq \mathbb{N}$ to $\sum_{k \in S} 2^{k-1}$ (the empty set to 0). Binary representation makes this a bijection onto $\mathbb{N} \cup \{0\}$: countable, no pooling required.

II.1 (Pooling), in full. Discard empty A_n; if all are empty the union is empty and we are done. Each non-empty countable A_n admits a surjection $\varphi_n\colon \mathbb{N} \to A_n$: its roster if countably infinite; if finite, its roster with the last element repeated forever. *The honest footnote:* we have just chosen one φ_n for each of infinitely many n, with no rule specifying the choices — this is the axiom of countable choice, invisible in most textbooks at this step, harmless here, and famous in Volume Two. Now define $F\colon \mathbb{N} \times \mathbb{N} \to \bigcup_n A_n$ by $F(n,m) = \varphi_n(m)$. Every x in the union lies in some A_n, hence is some $\varphi_n(m)$: F is surjective. Composing with the pairing bijection of Proposition 3.6 gives a surjection $G\colon \mathbb{N} \to \bigcup_n A_n$. If the union is finite, it is countable outright. Otherwise map each x in the union to the *least* k with $G(k) = x$: this is injective into $\mathbb{N}$, so the union is equivalent to an infinite subset of $\mathbb{N}$, hence countable by Lemma 3.5. $\square$

II.2 (The census), in full. For $h \geq 2$ let P_h be the set of polynomials $a_n x^n + \cdots + a_0$ with integer coefficients, $n \geq 1$, $a_n \neq 0$, and height $n + |a_0| + \cdots + |a_n| = h$. Then P_h is finite: the degree satisfies $n < h$, and each coefficient satisfies $|a_i| \leq h$, leaving finitely many tuples. Let R_h be the set of real roots of members of P_h: finite, since each polynomial contributes at most its degree in roots (the credit box). Every algebraic number lies in some R_h — its witnessing polynomial has a height. Queue the algebraics by marching $h = 2, 3, 4, \ldots$, listing each finite R_h in increasing numerical order and skipping numbers already seen: every algebraic number appears at a finite position. Note, contra II.1, that *no* act of choice occurs: each R_h is finite and is ordered canonically, by size. □

3.11 Notes and References

Cantor's 1874 and 1891 papers are short and readable in English translation in Ewald, *From Kant to Hilbert*, volume 2 — the 1891 note is barely two pages, and reading it is an experience; the December 1873 letters appear in the Cantor–Dedekind correspondence excerpted there. Abbott, *Understanding Analysis*, §§1.5–1.6, is the gentlest second pass on countability and runs Cantor–Schröder–Bernstein as a guided exercise (our III.1). Aigner and Ziegler, *Proofs from THE BOOK*, polish the diagonal to a shine. For the saboteur with full engineering honesty, Petzold, *The Annotated Turing*, walks the 1936 paper line by line. Dauben, *Georg Cantor: His Mathematics and Philosophy of the Infinite*, is the standard biography and documents the Kronecker campaign with sources. The credit box (a degree-n polynomial has at most n real roots) is the factor theorem's consequence; any introductory algebra text. On what "definable" may and may not mean — Richard's paradox and its defusing — see Franzén, *Gödel's Theorem: An Incomplete Guide to Its Use and Abuse*. On the continuum hypothesis: Gödel's 1940 monograph and Cohen, *Set Theory and the Continuum Hypothesis* (1966), for orientation only; the door stays marked.

Chapter 4

Sequences, Series, and the First Machinery

Companion to Episode Four. The episode defined the limit, installed the adversarial game, licensed the climb, ran the bisection hunt, brought the huddle home, arraigned the infinite sum — paying Zeno and the geometric credit — detonated Oresme's bombshell, and closed with the rearrangement outrage and the formal opening of the swap account. This chapter is the machinery in full paperwork, with the schoolroom credits flagged.

Episode Digest

The argument of the hour. The limit is defined — the hardest small thing in mathematics, twenty-three centuries in the making — as an adversarial game: the skeptic proposes a tolerance, we answer with a threshold, and beyond the threshold every term complies. On that definition the first machinery is built: the algebra of arrival, the squeeze, and three existence machines — the licensed climb, the bisection hunt, and the huddle brought home — each powered by completeness. Infinite sums become sequences wearing a plus sign; the geometric series is licensed and Zeno's race is paid; Oresme detonates the harmonic series; and the hour closes with the rearrangement outrage — infinite addition is not commutative — and the formal opening of the swap account.

The episode, beat by beat. The Fugitive Notion. What must "approaches" mean, with every verb of motion abolished? Two honest drafts die on stage. *Ever closer* fails twice: the distances from $1/n$ to -5 shrink forever without arrival (improvement is not arrival), and honest convergence may fidget. *Arbitrarily close* fails on the flip-flop $1, 0, 1, 0, \ldots$ — a bigamist, exactly close to two destinations infinitely often, married to neither. Episode One's grammar pays off: *infinitely often* is visits; convergence must be *eventually* — residence. Two knobs remain: how close (the tolerance) and from when (the threshold).

The Game. The definition (Definition 4.1): for every tolerance epsilon, however small, there is a threshold — big N — beyond which every term lies within epsilon of the limit. The skeptic moves *first*; N may depend on ε, never the reverse — reverse the quantifiers and the sentence says "eventually constant," a different universe. Little n runs; big N stands. Matches played: $1/n \to 0$ (Archimedes hands us the threshold); the constant sequence (threshold one; the skeptic resigns); the flip-flop prosecuted with the single tolerance one half (divergence is the skeptic's victory, Remark 4.2); and $1/n \not\to 1$ (tolerance one third — each destination stands trial separately). Uniqueness by the half-gap tolerance (Proposition 4.3). Convergence is a service-level agreement; and a second dialect is installed: *diverges to infinity* — every bound eventually and permanently cleared — which Oresme will need within the hour.

The Algebra of Arrival. Sums, products, quotients of arrivals arrive (Theorem 4.5): split the tolerance for sums; handcuff one factor (convergent implies bounded, Proposition 4.4) for products; keep the denominator honestly away from zero for quotients — strategies aloud, masonry on paper. Order survives limits but strictness flattens: $1/n > 0$ yet $\lim = 0$ (Theorem 4.6). The squeeze: two escorts, one prisoner, one destination (Theorem 4.7).

The Licensed Climb. Zurich, 1858, repaid: the theorem that embarrassed Dedekind falls in four sentences. Every bounded climb converges — to its supremum (Theorem 4.9 ⊛, the bell's sixth ring): the dip-and-overtake clause of Chapter 2 hands the skeptic his threshold. The frontier no one stands on becomes the place the climbers are going. Dividends: existence without naming the destination; every infinite decimal officially names a real; the picket fence restated — every real is a limit of rationals; and the compound-interest climb summons e into existence (Problem II.2, Bernoulli lifting). Fine print: bounded is essential, and monotone sequences obey a total dichotomy — converge or diverge to infinity, no thrashing (Corollary 4.10).

The Bisection Hunt. New repertory character, christened. Subsequences: thin the crowd, keep the order (Definition 4.11). Bolzano–Weierstrass: every bounded sequence hides a convergent subsequence (Theorem 4.13 ⊛, the seventh ring). The hunt: halve the interval; one half hosts infinitely many terms (pigeonhole); keep halving — the Russian dolls assemble mid-chase, the unique survivor x^* is cornered, and one term per doll, indices rising, is harvested into a subsequence converging to x^*. Infinitude converted into location. Borders: boundedness essential; cluster values named (Remark 4.14) — the flip-flop's are $\{0, 1\}$; contracts filed for Episodes Five and Six.

Arrival Without a Destination. The intrinsic test: a huddle (Cauchy sequence, Definition 4.15) — terms eventually within any tolerance *of each other*, no limit mentioned. Convergent implies Cauchy (split the tolerance, Proposition 4.16). The deep converse in $\mathbb{R}$ (Theorem 4.18 ⊛, the eighth ring), by a three-leg relay: huddles are bounded; the hunt extracts a convergent subsequence; the huddle drags the whole flock to its limit. In $\mathbb{Q}$ the truncations of $\sqrt{2}$ huddle and arrive nowhere (Remark 4.19); Cauchy stated the criterion in 1821 with no axiom to prove it by — 1872 retroactively underwrites the book. The homecoming: the rival firm built $\mathbb{R}$ *out of* huddles. Banked for the second half: a series converges exactly when its tails die (the tail criterion, Corollary 4.23).

The Honest Residue. When limits fail, bookkeeping survives: tail-ceilings descend, tail-floors climb, and the licensed climb summons their limits — $\limsup$ and $\liminf$ (Definition 4.20), which for bounded sequences always exist. Convergence is exactly their coincidence; and the long-run ceiling is the largest cluster value (Proposition 4.21, soldered in Problem II.4).

The Infinite Sum, Arraigned. A series is a sequence wearing a plus sign (Definition 4.22); it converges if its partial sums do. Grandi's farce closes on the spot: partial sums $1, 0, 1, 0$ — the flip-flop in disguise, prosecuted in the first act; no sum; the eighteenth century's one half retired to the museum of moods. The geometric series licensed in full (Theorem 4.24): the telescope collapses the partial sums, Bernoulli executes r^n, and the boxed ceremony discharges the credit outstanding since Chapter 1. And Zeno is *paid*: the stages' totals converge in the full certified sense; the race's arithmetic is settled — with one sentence of scruple about supertasks, which are metaphysics' invoice, not ours. The Arrow remains open, due Episode Seven. Necessity in two lines: terms of a convergent series die to zero (Proposition 4.25) — and the converse is spectacularly false.

The Fourteenth-Century Bombshell. Oresme, Paris, around 1350: the harmonic series diverges (Theorem 4.26). Doubling blocks each contribute at least one half — terms shrink, blocks lengthen, exact compensation — and the partial sums climb past every bound, half-step by half-step. No limit theory required: finite partial sums against a tally of halves, divergence-to-infinity reached from the far side of six centuries. The crawl is glacial (Remark 4.27): past three hundred needs more than $2^{299} > 10^{89}$ terms — upward of a billion times the atom count of the observable universe. Proof over experiment, compressed into one series.

Tests, and Two Kinds of Convergence. Comparison: dominated by a convergent escort, a non-negative series converges — the licensed climb in working clothes (Theorem 4.28). Ratio: successive ratios settling below one mean eventual geometric camouflage (Remark 4.30). Dividend: the Basel series $\sum 1/n^2$ *converges* by a telescoping comparison, partial sums ceilinged below two (Proposition 4.29) — the convergence is ours; the valuation $\pi^2/6$ stays Euler's. Absolute convergence defined; absolute implies convergent via dying tails (Theorem 4.32). The alternating harmonic series converges — its partial sums build their own Russian dolls (Theorem 4.33, solution supplied) — to a constant a little under seven tenths, named on credit; stripped of signs it is Oresme's patient. *Conditional* convergence: a truce, not a peace.

The Rearrangement Outrage. The set piece and the wonder. Rearrangement: same terms, each once, new order (Definition 4.34). Dirichlet noticed the anomaly in 1837 and proved the consolation; Riemann, in the same 1854 essay as his integral (published 1867, Dedekind editing — loyalty was his second theorem), proved the annihilation (Theorem 4.36): a conditionally convergent series can be rearranged to sum to *any* target, or to diverge. The mechanism: the fuel lemma (Lemma 4.35) — both sign-piles diverge while terms die — then greedy steering: spend positives past the target, negatives below, forever; overshoot at most the last term spent, which dies. Infinite addition is not commutative; the sum of a conditionally convergent series is a property of the *order*. The consolation (Theorem 4.37): absolute convergence is rearrangement-proof — commutativity licensed, not abolished. The worked outrage: the alternating harmonic re-dealt one-positive-two-negatives sums to exactly *half* (Problem II.6, solution supplied). And the deep diagnosis opens the season's next account: a sum is a limit; rearrangement reorders a limiting process; *when may two limiting processes be exchanged?* — the swap account, formally issued, instalments due Episodes Seven and Eight and across Season Two.

The toolkit acquired. The definition and its negation (Definition 4.1, Remark 4.2); uniqueness and boundedness (Propositions 4.3, 4.4); the limit laws, order laws, and squeeze (Theorems 4.5–4.7); the monotone convergence theorem (Theorem 4.9 ⊛) with its dichotomy; subsequences and Bolzano–Weierstrass (Theorem 4.13 ⊛); the Cauchy criterion, both directions (Theorem 4.18 ⊛); limsup and liminf (Definition 4.20); series, the tail criterion, the geometric series with credit paid, terms-to-zero, Oresme, comparison, Basel, the ratio remark (Definition 4.22–Remark 4.30); absolute versus conditional with the alternating test (Theorems 4.32, 4.33); the fuel lemma and both rearrangement theorems (Lemma 4.35, Theorems 4.36, 4.37). Supplied solutions: the alternating test (II.5) and the halving jewel (II.6).

Debts. **Paid: D2, Zeno's race** — the oldest account on the books, discharged with ceremony and one sentence of supertask scruple; the Arrow (D2b) remains open, due Episode Seven. **Paid: the geometric credit** (boxed ceremony at Theorem 4.24). **Closed with prejudice:** Grandi's farce — the flip-flop unmasked. **Issued: D5, the swap account** — when may two limiting processes be exchanged? — instalments Episodes Seven, Eight, and Season Two. D7 (the bell): rung three times tonight — the climb, the hunt, the huddle — running count **eight**. **New character:** the bisection hunt, christened; contracts Episodes Five and Six.

Dates worth keeping. Around 1350, Oresme's blocks; 1683, Jacob Bernoulli prices compound interest and meets e; 1817, Bolzano's share of the hunt, in the unread Prague paper; 1821, Cauchy's criterion, stated on faith; 1837, Dirichlet's anomaly and consolation; 1854, Riemann's essay — integral and rearrangement in one manuscript — published 1867 by Dedekind's hand; 1858, the Zurich embarrassment, repaid tonight; the 1860s, Weierstrass's Berlin lectures, where the world finally listened.

4.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"the tolerance"	$\varepsilon > 0$	the skeptic's move
"the threshold, big N"	$N \in \mathbb{N}$	our answer
"beyond the threshold, within tolerance"	$\forall n \geq N : \ \lvert a_n - L\rvert < \varepsilon$	the guarantee
"converges to L"	$a_n \to L$; $\lim_{n\to\infty} a_n = L$	Definition 4.1
"diverges to infinity"	$\forall M\ \exists N\ \forall n \geq N : \ a_n > M$	Remark 4.2
"a subsequence"	$(a_{n_k}),\ n_1 < n_2 < \cdots$	Definition 4.11
"a climb" / "a descent"	monotone: $a_{n+1} \geq a_n$ (resp. $\leq$)	Definition 4.8
"a cluster value"	a subsequential limit	Remark 4.14
"the long-run ceiling / floor"	$\limsup a_n,\ \liminf a_n$	Definition 4.20
"a huddle"	a Cauchy sequence	Definition 4.15
"the tall Greek letter"	$\sum_n a_n$; partial sums s_n	Definition 4.22
"the tail"	$a_{m+1} + \cdots + a_n,\ n \geq m \geq N$	Corollary 4.23
"rearrangement"	$\sum_n a_{\sigma(n)},\ \sigma$ a bijection of $\mathbb{N}$	Definition 4.34

4.2 The Definition and Its Grammar

Definition 4.1 (Convergence)**.** A sequence (a_n) of real numbers *converges* to $L \in \mathbb{R}$ if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that

$$|a_n - L| < \varepsilon \qquad \text{for every } n \geq N.$$

We write $a_n \to L$ or $\lim_{n\to\infty} a_n = L$, and call L the *limit*. A sequence that converges to no real number *diverges*. (The game gloss, once for the record: the skeptic proposes the tolerance ε;

we answer with the threshold N, which may depend on ε and on nothing later; the guarantee covers *every* term beyond the threshold.)

Remark 4.2 (Negation, and the second dialect). Written out, "(a_n) does not converge to L" reads: there exists $\varepsilon > 0$ such that for every N there is some $n \geq N$ with $|a_n - L| \geq \varepsilon$ — one unanswerable tolerance, violated beyond every threshold: violated *infinitely often*. The grammar of Chapter 1, §1.2 — *eventually* versus *infinitely often* — is load-bearing here: convergence demands closeness eventually; its failure is straying infinitely often. Separately, (a_n) *diverges to infinity*, written $a_n \to \infty$, if for every M there exists N with $a_n > M$ for all $n \geq N$: every bound eventually and permanently cleared. This is a manner of divergence, not a species of convergence; ∞ names no real number. Mirror definition for $-\infty$.

Proposition 4.3 (Uniqueness of limits)**.** *A sequence has at most one limit.*

Proof. Suppose $a_n \to L$ and $a_n \to L'$ with $L \neq L'$. Take $\varepsilon = |L - L'|/2 > 0$: the half-gap. Beyond some N_1 every term is within ε of L; beyond some N_2, within ε of L'. For any n beyond both,

$$|L - L'| \;\leq\; |L - a_n| + |a_n - L'| \;<\; 2\varepsilon \;=\; |L - L'|,$$

a contradiction. (This is the promised gentlest first date with ε: one tolerance, chosen with intent.) $\square$

Proposition 4.4 (Convergent implies bounded)**.** *If (a_n) converges, then there is B with $|a_n| \leq B$ for all n.*

Proof. Let $a_n \to L$ and play tolerance 1: beyond some N, $|a_n - L| < 1$, so $|a_n| < |L| + 1$. The stragglers $a_1, \ldots, a_N$ are finitely many; take $B = \max\big(|a_1|, \ldots, |a_N|, |L| + 1\big)$. $\square$

Worked matches. (i) $1/n \to 0$: given ε, the Archimedean property (Theorem 2.12) supplies $N > 1/\varepsilon$; then $n \geq N$ gives $1/n \leq 1/N < \varepsilon$. For the skeptic's opening bid $\varepsilon = 1/10$: threshold $N = 11$ (or 10, minding the strict inequality; Problem I.1 has you mind it). (ii) The constant sequence $c, c, c, \ldots$ converges to c with threshold 1 against every tolerance. (iii) The flip-flop $1, 0, 1, 0, \ldots$ diverges: with $\varepsilon = 1/2$, any candidate L is at distance $\geq 1/2$ from at least one of $0, 1$ (else $|1 - 0| < 1$), and that value recurs beyond every threshold (Problem I.4 writes the full quantifier dress). (iv) $1/n$ does not converge to 1: with $\varepsilon = 1/3$, every term from the third onward sits at distance $1 - 1/n \geq 2/3$ from 1 (Problem I.5). A limit claim names a destination; each destination stands trial separately.

4.3 The Algebra of Limits

Theorem 4.5 (Limit laws)**.** *Let $a_n \to L$ and $b_n \to M$. Then:* (i) *$a_n + b_n \to L + M$;* (ii) *$c\, a_n \to cL$ for any constant c;* (iii) *$a_n b_n \to LM$;* (iv) *if $M \neq 0$, then $a_n/b_n \to L/M$.*

Proof. (i) Split the tolerance. Given ε, choose N_1 with $|a_n - L| < \varepsilon/2$ beyond it, N_2 likewise for b; beyond $\max(N_1, N_2)$,

$$|(a_n + b_n) - (L + M)| \leq |a_n - L| + |b_n - M| < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon.$$

(ii) Trivial if $c = 0$; otherwise answer the tolerance ε with the threshold for $\varepsilon/|c|$.

(iii) Handcuff, then squeeze. By Proposition 4.4 there is $B > 0$ with $|a_n| \le B$ for all n. Then

$$|a_n b_n - LM| = |a_n(b_n - M) + M(a_n - L)| \le B\,|b_n - M| + |M|\,|a_n - L|.$$

Given ε, demand $|b_n - M| < \varepsilon/(2B)$ and $|a_n - L| < \varepsilon/\big(2(|M|+1)\big)$ beyond suitable thresholds; beyond the larger, the right side is $< \varepsilon$.

(iv) Keep the denominator honest. It suffices to show $1/b_n \to 1/M$ and apply (iii). Play tolerance $|M|/2$ in $b_n \to M$: beyond some N_0, $|b_n| \ge |M| - |b_n - M| > |M|/2$; in particular $b_n \ne 0$ there, and

$$\left|\frac{1}{b_n} - \frac{1}{M}\right| = \frac{|M - b_n|}{|b_n|\,|M|} \le \frac{2}{M^2}\,|b_n - M|,$$

which is beaten beneath any ε by demanding $|b_n - M| < \varepsilon M^2/2$ beyond a further threshold. (The remaining bookkeeping is Problem II.1.) □

Theorem 4.6 (Order laws, with the flattening warning)**.** *If $a_n \le b_n$ for all n beyond some index, and both sequences converge, then* $\lim a_n \le \lim b_n$. *Strict inequality does not survive the limit: $1/n > 0$ for every n, while* $\lim 1/n = 0$. *Limits flatten "strictly less" into "at most".*

Proof. Suppose instead $L = \lim a_n > M = \lim b_n$, and take $\varepsilon = (L - M)/2$. Beyond the two thresholds, $a_n > L - \varepsilon = M + \varepsilon > b_n$, contradicting $a_n \le b_n$ there. □

Theorem 4.7 (The squeeze)**.** *If $a_n \le c_n \le b_n$ for all n beyond some index, and $a_n \to L$ and $b_n \to L$, then $c_n \to L$.*

Proof. Given ε, beyond the two escorts' thresholds (and the index where the sandwich holds),

$$L - \varepsilon < a_n \le c_n \le b_n < L + \varepsilon,$$

so $|c_n - L| < \varepsilon$. □

4.4 The Licensed Climb

Definition 4.8. (a_n) is *increasing* if $a_{n+1} \ge a_n$ for all n; *decreasing* if $a_{n+1} \le a_n$; *monotone* if either.

Theorem 4.9 (Monotone convergence theorem)**.** *Every increasing sequence that is bounded above converges — namely to* $\sup\{a_n : n \in \mathbb{N}\}$. *Mirror statement for decreasing sequences bounded below, converging to the infimum.*

Proof. The set of terms is non-empty and bounded above, so $s = \sup\{a_n : n \in \mathbb{N}\}$ exists by the completeness axiom (Axiom 2.6) ($\circledast$). Given $\varepsilon > 0$, the dipped level $s - \varepsilon$ is not an upper bound, so by the approximation lemma (Lemma 2.4) some term overtakes it: $a_N > s - \varepsilon$ for some N. For $n \ge N$, monotonicity gives $a_n \ge a_N > s - \varepsilon$, while $a_n \le s$ always; hence $|a_n - s| < \varepsilon$ for all $n \ge N$. The threshold is delivered, for every tolerance: $a_n \to s$. □

Corollary 4.10 (Monotone dichotomy)**.** *An increasing sequence either converges (if bounded above) or diverges to $+\infty$ (if not); no third behaviour is available.*

Proof. Bounded: Theorem 4.9. Unbounded: given M, some $a_N > M$, and monotonicity carries every later term past M too — the definition of $a_n \to \infty$ (Remark 4.2). □

Decimals, legitimized; the fence, restated. An infinite decimal expansion is precisely an increasing sequence of finite truncations, bounded above (by the integer part plus one); by Theorem 4.9 it converges, and the number it *names* is its limit. The notation of the schoolroom becomes a theorem. In the same breath, density (Theorem 2.15) restates dynamically: every real x is the limit of a sequence of rationals — plant a fraction within $1/n$ of x for each n and squeeze (Theorem 4.7). Nearness and arrival are one vocabulary. The compound-interest climb $\left(1+\frac{1}{n}\right)^n$, and the number e it summons into existence, is Problem II.2.

4.5 The Bisection Hunt

Definition 4.11. A *subsequence* of (a_n) is a sequence (a_{n_k}) where $n_1 < n_2 < n_3 < \cdots$ is a strictly increasing choice of indices: infinitely many terms kept, order preserved. (By induction, $n_k \geq k$.)

Proposition 4.12. *If $a_n \to L$, then every subsequence converges to L.*

Proof. Given ε, take the threshold N for the full sequence; for $k \geq N$ we have $n_k \geq k \geq N$, so $|a_{n_k} - L| < \varepsilon$. □

Theorem 4.13 (Bolzano–Weierstrass)**.** *Every bounded sequence of real numbers has a convergent subsequence.*

Proof. Let $|a_n| \leq M$ for all n, and set $I_0 = [-M, M]$, an interval containing a_n for infinitely many n (indeed all). Recursively: given a closed interval I_j containing terms of the sequence for infinitely many indices, split it at its midpoint into two closed halves. The index set splits accordingly; two finite sets cannot union to an infinite one, so at least one half contains terms for infinitely many indices — keep the leftmost such half as I_{j+1}. This builds nested closed intervals $I_1 \supseteq I_2 \supseteq \cdots$ of widths $2M/2^j \leq 2M/j$ (since $2^j > j$, Chapter 1, Problem I.5), beneath every tolerance. By the nested interval theorem in unique-survivor form (Theorem 2.17, Remark 2.18) ($\circledast$), there is exactly one $x^* \in \bigcap_j I_j$.

Harvest the subsequence: choose n_1 with $a_{n_1} \in I_1$; given n_k, the indices n with $a_n \in I_{k+1}$ are infinitely many, so one of them exceeds n_k — choose it as n_{k+1}. Then a_{n_k} and x^* both lie in I_k, so $|a_{n_k} - x^*| \leq \text{width}(I_k) \leq 2M/k \to 0$. The subsequence converges to x^*. □

Remark 4.14 (Borders, and cluster values). Boundedness is essential: $1, 2, 3, \ldots$ has no convergent subsequence, every infinite selection marching past every bound. Call the limits of convergent subsequences of (a_n) its *cluster values;* equivalently (a two-line check) the numbers approached within every tolerance infinitely often. The flip-flop's cluster values are exactly $\{0, 1\}$. Bolzano–Weierstrass then reads: a bounded sequence has at least one cluster value. The connection to the long-run ceiling — lim sup is the *largest* cluster value — is soldered in Problem II.4.

4.6 Arrival Without a Destination

Definition 4.15 (Cauchy sequence)**.** (a_n) is a *Cauchy sequence* (a *huddle*) if for every $\varepsilon > 0$ there exists N such that

$$|a_m - a_n| < \varepsilon \qquad \text{for all } m, n \geq N.$$

Terms compared to each other; no limit mentioned.

Proposition 4.16. *Every convergent sequence is Cauchy.*

Proof. Let $a_n \to L$; given ε, beyond the threshold for $\varepsilon/2$ any two terms satisfy $|a_m - a_n| \le |a_m - L| + |L - a_n| < \varepsilon$: split the tolerance, as ever. $\square$

Proposition 4.17. *Every Cauchy sequence is bounded.*

Proof. Problem II.3 (mirror the proof of Proposition 4.4 with the tolerance 1 played against the huddle). $\square$

Theorem 4.18 (Cauchy completeness of $\mathbb{R}$)**.** *Every Cauchy sequence of real numbers converges.*

Proof. The three-leg relay. *Leg one:* (a_n) is bounded (Proposition 4.17). *Leg two:* by Bolzano–Weierstrass (Theorem 4.13) some subsequence converges: $a_{n_k} \to x^*$ ($\circledast$). *Leg three: the huddle drags the flock.* Given ε, choose N with $|a_m - a_n| < \varepsilon/2$ for $m, n \ge N$, and choose k with $n_k \ge N$ and $|a_{n_k} - x^*| < \varepsilon/2$. For any $n \ge N$,

$$|a_n - x^*| \le |a_n - a_{n_k}| + |a_{n_k} - x^*| < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon.$$

So $a_n \to x^*$. $\square$

Remark 4.19 (What fails in $\mathbb{Q}$, and the two completeness notions)*.* The decimal truncations of $\sqrt{2}$ form a Cauchy sequence of rationals with no rational limit (Chapter 2, Problem II.4): inside $\mathbb{Q}$, huddles may have no home. Thus "every Cauchy sequence converges" is a completeness property in its own right — the rival firm's, indeed: Cantor and Méray built $\mathbb{R}$ from huddles of rationals. This chapter proved (least-upper-bound completeness) $\Rightarrow$ (Cauchy completeness); the converse implication requires the Archimedean property as a separate hypothesis, and over Archimedean ordered fields the two notions coincide. We use the axiom of Chapter 2 throughout and leave the full audit of equivalences to the references.

4.7 The Honest Residue

Definition 4.20 (Limit superior and inferior)**.** Let (a_n) be bounded. For each n set

$$t_n = \sup\{a_k : k \ge n\}, \qquad u_n = \inf\{a_k : k \ge n\} :$$

the tail-ceiling and tail-floor. Discarding candidates never raises a supremum nor lowers an infimum, so (t_n) is decreasing and (u_n) increasing; both are bounded, so both converge by Theorem 4.9. Define

$$\limsup_{n\to\infty} a_n = \lim_{n\to\infty} t_n, \qquad \liminf_{n\to\infty} a_n = \lim_{n\to\infty} u_n.$$

For every bounded sequence both exist, and $\liminf a_n \le \limsup a_n$. (Flip-flop: $\limsup = 1$, $\liminf = 0$.)

Proposition 4.21. *A bounded sequence converges if and only if its limit superior and limit inferior coincide, in which case their common value is the limit. Moreover,* $\limsup a_n$ *is the largest cluster value of* (a_n)*, and* $\liminf a_n$ *the smallest.*

Proof. Problem II.4, with hints. $\square$

4.8 The Infinite Sum, Arraigned

Definition 4.22 (Series)**.** Given a sequence (a_n), its *series* $\sum_{n=1}^{\infty} a_n$ is the sequence of *partial sums* $s_n = a_1 + a_2 + \cdots + a_n$. The series *converges to* S — written $\sum_{n=1}^{\infty} a_n = S$ — if $s_n \to S$; otherwise it *diverges*. A series is a sequence wearing a plus sign; every theorem of this chapter applies.

Corollary 4.23 (The tail criterion)**.** *$\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is N such that*

$$|a_{m+1} + a_{m+2} + \cdots + a_n| < \varepsilon \qquad \textit{for all } n \geq m \geq N.$$

Proof. The displayed sum is $|s_n - s_m|$, so the condition says exactly that (s_n) is a Cauchy sequence; apply Proposition 4.16 and Theorem 4.18. □

Theorem 4.24 (The geometric series)**.** *Let $a \neq 0$ and $r \in \mathbb{R}$. If $|r| < 1$, then*

$$\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}.$$

If $|r| \geq 1$, the series diverges.

Proof. The telescope. With $s_n = a(1 + r + \cdots + r^n)$,

$$(1-r)\, s_n = a\big(1 - r^{\,n+1}\big), \qquad \text{so} \qquad s_n = a\,\frac{1 - r^{\,n+1}}{1-r} \quad (r \neq 1):$$

multiplying by r shifts the sum one step; subtracting annihilates the middle. *The execution of r^n.* For $0 < |r| < 1$ write $|r| = 1/(1+h)$ with $h > 0$; Bernoulli's inequality (Proposition 1.3) gives $(1+h)^n \geq 1 + nh > nh$, so

$$|r|^n < \frac{1}{nh} \longrightarrow 0$$

by the Archimedean property (Theorem 2.12) and the squeeze. Hence $r^{\,n+1} \to 0$, and the limit laws give $s_n \to a/(1-r)$. If $|r| \geq 1$, then $|ar^n| \geq |a| > 0$ for every n: consecutive partial sums stay $|a|$ apart, the tail criterion (Corollary 4.23) fails at tolerance $|a|$, and the series diverges. □

Credit paid. — The geometric series was taken on credit in Chapter 1 (Zeno's schoolbook reply) and conspicuously *declined* as a loan in Chapter 3, where the dust's address system was built from nested intervals precisely to avoid borrowing it. The account opened in the course's first hour closes here: the license is the theorem above, and Zeno's stages — lengths and durations geometric — now have a total in the full, certified sense of Definition 4.22. The Arrow (velocity at an instant) remains a separate debt, due at Chapter 7.

Proposition 4.25 (Terms of a convergent series die)**.** *If $\sum a_n$ converges, then $a_n \to 0$. The converse is false (Theorem 4.26).*

Proof. $a_n = s_n - s_{n-1}$, and both partial-sum sequences converge to the same S; the limit laws give $a_n \to S - S = 0$. □

Theorem 4.26 (Oresme, c. 1350: the harmonic series diverges)**.** $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n}$ *diverges to $+\infty$, although its terms tend to* 0*.*

Proof. Group by doubling blocks. Writing $H_n = s_n$ for the partial sums: block j consists of the 2^{j-1} terms from index $2^{j-1}+1$ through 2^j, each of size at least $1/2^j$, so each block contributes at least $2^{j-1} \cdot 2^{-j} = \frac{1}{2}$. Hence

$$H_{2^k} \;=\; 1 + \sum_{j=1}^{k} (\text{block } j) \;\geq\; 1 + \frac{k}{2} \qquad \text{for every } k.$$

The partial sums are increasing and unbounded, so by the monotone dichotomy (Corollary 4.10) they diverge to $+\infty$. □

Remark 4.27 (The atom arithmetic). The crawl is quantified by the blocks alone, no logarithms needed. Each block also contributes *less than* 1 (its 2^{j-1} terms are each less than $1/2^{j-1}$), so $H_{2^k} < 1+k$; consequently $H_n < 1+k$ whenever $n \leq 2^k$. To achieve $H_n \geq 300$ therefore requires $n > 2^{299}$. And $2^{10} = 1024 > 10^3$, so

$$2^{299} = 2^9 \cdot \left(2^{10}\right)^{29} > 500 \cdot 10^{87} > 10^{89},$$

while the observable universe is commonly estimated to hold on the order of 10^{80} atoms. To push the harmonic total past three hundred needs upward of a billion times more terms than there are atoms to count them on. It diverges the way empires fall: certainly, and no one lives to see it.

Theorem 4.28 (Comparison test)**.** *If* $0 \leq a_n \leq b_n$ *for all n beyond some index and* $\sum b_n$ *converges, then* $\sum a_n$ *converges. Contrapositively, if* $\sum a_n$ *diverges, so does* $\sum b_n$*.*

Proof. Adjusting finitely many terms changes no convergence question, so assume the inequality holds for all n. The partial sums of $\sum a_n$ are increasing (non-negative terms) and bounded above by the limit of the partial sums of $\sum b_n$ (order laws, Theorem 4.6); the monotone convergence theorem (Theorem 4.9) concludes. The licensed climb, in working clothes. □

Proposition 4.29 (The Basel series converges)**.** $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2}$ *converges, with sum less than* 2*.*

Proof. The telescoping comparison: for $n \geq 2$,

$$\frac{1}{n^2} \;\leq\; \frac{1}{n(n-1)} \;=\; \frac{1}{n-1} - \frac{1}{n},$$

so the partial sums obey

$$\sum_{k=1}^{n} \frac{1}{k^2} \;\leq\; 1 + \sum_{k=2}^{n} \left(\frac{1}{k-1} - \frac{1}{k} \right) \;=\; 1 + 1 - \frac{1}{n} \;<\; 2:$$

the gaps collapse like a spyglass. An increasing sequence ceilinged below 2 converges (Theorem 4.9). The *value*, $\pi^2/6$, is Euler's of 1735; its lawful derivation lies beyond this course (references in the notes). The convergence is ours; the valuation stays Euler's. □

Remark 4.30 (Ratio test). Suppose $a_n \neq 0$ and $|a_{n+1}|/|a_n| \to \rho$. If $\rho < 1$, then $\sum a_n$ converges absolutely; if $\rho > 1$, it diverges; if $\rho = 1$, the test is silent (the harmonic and Basel series both have $\rho = 1$). Sketch, three lines: for $\rho < 1$ pick r with $\rho < r < 1$; beyond some N the ratios are $\leq r$, so $|a_{N+k}| \leq r^k |a_N|$, and $\sum |a_n|$ converges by comparison (Theorem 4.28) with a geometric series (Theorem 4.24); that absolute convergence forces convergence is Theorem 4.32 of the next section. For $\rho > 1$ the terms eventually grow, violating Proposition 4.25. The ratio test is the geometric series in a working disguise.

4.9 Absolute and Conditional Convergence

Definition 4.31. $\sum a_n$ *converges absolutely* if $\sum|a_n|$ converges. A series that converges but not absolutely *converges conditionally*.

Theorem 4.32 (Absolute convergence implies convergence)**.** *If $\sum|a_n|$ converges, then $\sum a_n$ converges.*

Proof. Given ε, the tail criterion (Corollary 4.23) applied to $\sum|a_n|$ supplies N beyond which every stretch $|a_{m+1}| + \cdots + |a_n| < \varepsilon$. By the triangle inequality,

$$|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n| < \varepsilon,$$

so $\sum a_n$ passes the tail criterion too. □

Theorem 4.33 (Alternating series test)**.** *Let $b_1 \geq b_2 \geq \cdots \geq 0$ with $b_n \to 0$. Then*

$$\sum_{n=1}^{\infty} (-1)^{n+1} b_n = b_1 - b_2 + b_3 - \cdots$$

converges, to a sum S satisfying $|S - s_n| \leq b_{n+1}$ for every n: the error is at most the first neglected term.

Proof. Problem II.5, solution supplied — the zigzag-of-dolls picture of the episode, with paperwork. □

In particular the *alternating harmonic series* $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots$ converges; stripped of its signs it is Oresme's patient, divergent — so it converges *conditionally*. Its sum is a definite constant, a little under seven tenths ($S \geq s_2 = \frac{1}{2} > 0$, and $S \leq s_3 = \frac{5}{6}$); the familiar name, the natural logarithm of two, is taken on credit until logarithms are lawfully constructed — references in the notes.

4.10 The Rearrangement Theorem

Definition 4.34. A *rearrangement* of $\sum a_n$ is a series $\sum_n a_{\sigma(n)}$ where $\sigma\colon \mathbb{N} \to \mathbb{N}$ is a bijection: the same terms, each used exactly once, in a new order.

Lemma 4.35 (The fuel lemma)**.** *Let $\sum a_n$ converge conditionally, and set $p_n = \max(a_n, 0)$ and $q_n = \max(-a_n, 0)$, so that $a_n = p_n - q_n$ and $|a_n| = p_n + q_n$. Then $\sum p_n$ and $\sum q_n$ both diverge to $+\infty$, while $p_n \to 0$ and $q_n \to 0$.*

Proof. If both $\sum p_n$ and $\sum q_n$ converged, then $\sum|a_n| = \sum(p_n + q_n)$ would converge (limit laws on partial sums), making the convergence absolute — contradiction. If exactly one converged — say $\sum p_n \to P$ while the increasing partial sums Q_n of $\sum q_n$ are unbounded — then $s_n = P_n - Q_n \leq P - Q_n \to -\infty$, so $\sum a_n$ would diverge — contradiction. Hence both diverge (their partial sums are increasing, so to $+\infty$, by Corollary 4.10). Finally $0 \leq p_n, q_n \leq |a_n| \to 0$ (Proposition 4.25 and the squeeze). □

Theorem 4.36 (Riemann rearrangement theorem)**.** *Let $\sum a_n$ converge conditionally. Then for every $T \in \mathbb{R}$ there is a rearrangement of $\sum a_n$ converging to T; and there are rearrangements diverging to $+\infty$, to $-\infty$, and rearrangements with no limiting behaviour at all.*

Proof, guided; the gaps are Problem III.1. Fix T. Let (P_j) list the non-negative terms of the series in their original order, and (Q_j) the absolute values of the negative terms in theirs; by the fuel lemma both $\sum P_j$ and $\sum Q_j$ diverge, and $P_j, Q_j \to 0$.

The steering. Spend $P_1, P_2, \ldots$ until the running total first exceeds T — this happens, since $\sum P_j$ is unbounded; stop at the first crossing. Then spend Q's (as negatives) until the total first drops below T; stop. Then P's to just above, Q's to just below, alternating forever, always stopping at the first crossing. Each raid terminates because the remaining pile still diverges [gap: formalize the greedy indices].

It is a rearrangement. Every P_j and every Q_j is eventually spent, exactly once, since each pile is consumed in order through infinitely many raids [gap: the index bookkeeping].

It converges to T. From the first crossing onward, the running total after any step differs from T by at most the size of the term spent at the most recent crossing [gap: the inequality chase, minding whether one is mid-raid or at a turn]; those crossing terms are P_j's and Q_j's with $j \to \infty$, hence tend to 0. The deviations are squeezed beneath every tolerance: the rearranged partial sums converge to T.

The divergent variants. Steer for the moving targets $1, 2, 3, \ldots$ (never turning back down far): the totals clear every bound [gap]. Mirror for $-\infty$; alternate targets $+1, -1, +1, \ldots$ for oscillation. □

Theorem 4.37 (Dirichlet, 1837: absolute convergence is rearrangement-proof)**.** *If $\sum a_n$ converges absolutely with sum S, then every rearrangement $\sum a_{\sigma(n)}$ converges to S.*

Proof. Problem III.2, with hints. □

The mandated worked outrage — the alternating harmonic series re-dealt one-positive-two-negatives, its sum audited to exactly *half* the standard total — is Problem II.6, solution supplied. The sum of a conditionally convergent series is a property of the order, not of the collection; commutativity for infinite sums is not abolished but *licensed*, and the license is absolute convergence.

4.11 Problems

Tier I — Calisthenics

I.1 For $1/n \to 0$: exhibit a threshold for the tolerance $\varepsilon = 1/10$, minding strict inequality; then write the general strategy for arbitrary ε, citing the Archimedean property.

I.2 Show $(n+1)/n \to 1$: compute $|(n+1)/n - 1|$ explicitly and manufacture the threshold.

I.3 Show that a constant sequence converges, and that an *eventually* constant sequence converges. What are the cheapest thresholds?

I.4 Write out, in full quantifier dress, the statement "(a_n) does not converge to L", and use it to prosecute the flip-flop $1, 0, 1, 0, \ldots$: show it converges to no L, using the single tolerance $\varepsilon = 1/2$.

I.5 Show that $1/n$ does not converge to 1, using the single tolerance $\varepsilon = 1/3$.

I.6 Squeeze drills: show $(-1)^n/n \to 0$; more generally, if (b_n) is bounded then $b_n/n \to 0$.

I.7 Compute lim sup and lim inf for: the flip-flop $1, 0, 1, 0, \ldots$; the bigamist $(-1)^n(1 + \frac{1}{n})$; and the sequence $1, \frac{1}{2}, 1, \frac{1}{3}, 1, \frac{1}{4}, \ldots$

I.8 Verify the telescope identity $(1-r)s_n = a(1-r^{n+1})$ by expanding; sum Zeno's series $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots$; and show directly that the cases $r=1$ and $r=-1$ diverge.

Tier II — The Real Thing

II.1 (Quotient law, completed.) Prove that if $b_n \to M \neq 0$ then $1/b_n \to 1/M$, supplying every threshold, and deduce law (iv) of Theorem 4.5 from law (iii).

II.2 (The climb to e.) Show that $x_n = \left(1+\frac{1}{n}\right)^n$ is increasing and bounded above by 3, hence convergent; its limit is christened e. (One schoolroom credit — the binomial theorem — is flagged in the hint.)

II.3 Prove Proposition 4.17: every Cauchy sequence is bounded.

II.4 Prove Proposition 4.21: convergence is equivalent to $\limsup = \liminf$; and $\limsup$ is the largest cluster value.

II.5 (Alternating series test; solution provided.) Prove Theorem 4.33, including the error bound.

II.6 (The halving jewel; solution provided.) Rearrange the alternating harmonic series in the pattern one-positive-two-negatives:

$$1-\frac{1}{2}-\frac{1}{4}+\frac{1}{3}-\frac{1}{6}-\frac{1}{8}+\frac{1}{5}-\frac{1}{10}-\frac{1}{12}+\cdots$$

Show it converges to exactly *half* the sum of the standard ordering — the same cards, a different hand.

Tier III — For the Ambitious ⋆

⋆ **III.1** Complete the guided proof of the Riemann rearrangement theorem (Theorem 4.36): formalize the greedy indices, verify that the construction is a genuine rearrangement, run the inequality chase for convergence to T, and build the divergent variant.

⋆ **III.2** Prove Dirichlet's theorem (Theorem 4.37): an absolutely convergent series has the same sum under every rearrangement.

4.12 Hints

II.1. First show the denominator keeps its distance: play tolerance $|M|/2$ in $b_n \to M$ to get $|b_n| > |M|/2$ beyond some N_0. Then bound $|1/b_n - 1/M| = |M-b_n|/(|b_n|\,|M|) \leq 2|b_n - M|/M^2$ and demand $|b_n - M| < \varepsilon M^2/2$ beyond a further threshold.

II.2. *Climbing:* write the ratio as

$$\frac{x_{n+1}}{x_n} = \left(1+\frac{1}{n}\right)\left(1-\frac{1}{(n+1)^2}\right)^{n+1}$$

(check the algebra: the middle collapses to $\left(n(n+2)/(n+1)^2\right)^{n+1}$), then apply Bernoulli's inequality (Proposition 1.3) to the bracket: $\geq 1-\frac{1}{n+1}$, whence the ratio is $\geq \left(1+\frac{1}{n}\right)\left(\frac{n}{n+1}\right) = 1$.

Bounded: expand by the binomial theorem — the flagged schoolroom credit — and dominate:

$$\binom{n}{k}\frac{1}{n^k} \le \frac{1}{k!} \le \frac{1}{2^{k-1}} \quad (k \ge 1), \qquad \text{so} \qquad x_n \le 1 + \sum_{k\ge 1} 2^{-(k-1)} < 3$$

by the geometric series (Theorem 4.24).

II.3. Play tolerance 1 against the huddle: beyond N, every term lies within 1 of a_N; round up the finitely many stragglers as in Proposition 4.4.

II.4. ($\Rightarrow$) If $a_n \to L$: beyond the threshold for ε, every tail-ceiling t_n and tail-floor u_n lies within ε of L; squeeze. ($\Leftarrow$) If $t_n \to L$ and $u_n \to L$: forever $u_n \le a_n \le t_n$; squeeze again. *Largest cluster value:* from $t_n \downarrow t$, harvest a_{n_k} within $1/k$ of t_{n_k} using the approximation lemma (Lemma 2.4) on each tail-supremum, indices rising; this subsequence converges to t, so t *is* a cluster value. Conversely any cluster value is $\le t_n$ for every n (its subsequence eventually lives in the n-th tail), hence $\le t$.

III.1. Define the turning indices greedily: m_1 = least m with $P_1 + \cdots + P_m > T$ (exists: unbounded partial sums); k_1 = least k with the total dropped below T after spending $Q_1, \ldots, Q_k$; and so on, each raid on a pile that still diverges. Every index of either pile is consumed: the raids are infinitely many and each consumes at least one fresh term. For the chase: at a turn, the overshoot is at most the term just spent (first crossing); between turns, the total moves monotonically from one side of T toward the other, so the deviation is at most the previous turn's overshoot. Both bounds involve P_j or Q_j with $j \to \infty$. For $+\infty$: cross 1, then 2, then 3, ..., spending a single Q after each crossing (so every negative term is eventually used); the single small deductions cannot undo the diverging climbs.

III.2. Let s_n be the standard partials, $S = \lim s_n$, and $\tilde{s}_m$ the rearranged ones. Given ε: the tail criterion for $\sum|a_n|$ supplies N with $\sum_{k>N}|a_k| < \varepsilon$ (why does the full infinite tail obey the bound, not just finite stretches? — monotone partial sums). Choose M so large that $\sigma(1), \ldots, \sigma(M)$ include $1, \ldots, N$ — a finite net of indices, eventually swept up. For $m \ge M$, the difference $\tilde{s}_m - s_N$ consists only of terms with indices beyond N, so $|\tilde{s}_m - s_N| \le \sum_{k>N}|a_k| < \varepsilon$; and $|s_N - S| \le \varepsilon$ likewise. Conclude $|\tilde{s}_m - S| < 2\varepsilon$ eventually.

4.13 Solutions

Tier I (terse). **I.1** $N = 11$: for $n \ge 11$, $1/n \le 1/11 < 1/10$. General: Archimedes (Theorem 2.12) gives $N > 1/\varepsilon$; then $n \ge N$ implies $1/n \le 1/N < \varepsilon$. **I.2** $|(n+1)/n - 1| = 1/n$: the same thresholds as I.1 serve verbatim. **I.3** Constant: threshold 1, distance 0 forever. Eventually constant (equal to c from index m on): threshold m. **I.4** Negation: $\exists\,\varepsilon > 0\ \forall N\ \exists n \ge N$: $|a_n - L| \ge \varepsilon$. With $\varepsilon = 1/2$: if L were within $1/2$ of both 0 and 1, then $1 = |1 - 0| < 1$; so L is at distance $\ge 1/2$ from one of them, and that value occurs at infinitely many positions — beyond every N. **I.5** For $n \ge 3$: $|1/n - 1| = 1 - 1/n \ge 2/3 > 1/3$; no threshold survives the tolerance $1/3$. **I.6** $-1/n \le (-1)^n/n \le 1/n$ with both flanks $\to 0$; squeeze. General: $|b_n| \le B$ gives $-B/n \le b_n/n \le B/n$; flanks $\to 0$ by the limit laws. **I.7** $(\limsup, \liminf)$: flip-flop $(1, 0)$; bigamist $(1, -1)$ — cluster values ± 1; third sequence $(1, 0)$ — the ones recur forever while the interlaced terms die to 0. **I.8** Expanding $(1-r)(a+ar+\cdots+ar^n)$, all middle terms cancel, leaving $a(1-r^{n+1})$. Zeno: $a=\frac{1}{2}$, $r=\frac{1}{2}$, so $s_n = \frac{1}{2}\cdot\left(1-(\frac{1}{2})^{n+1}\right)/\frac{1}{2} = 1-(\frac{1}{2})^{n+1} \to 1$. For $r=1$: $s_n = a(n+1)$, unbounded. For $r=-1$: s_n alternates $a, 0, a, 0, \ldots$ — the flip-flop, scaled; divergent by I.4.

II.5 (Alternating series test), in full. Write s_n for the partial sums of $\sum(-1)^{n+1}b_n$. *Even partials climb:* $s_{2k+2}-s_{2k} = b_{2k+1}-b_{2k+2} \geq 0$ since the b's decrease. *Odd partials descend:* $s_{2k+1}-s_{2k-1} = -b_{2k}+b_{2k+1} \leq 0$. *Interlacing:* $s_{2k} = s_{2k+1}-b_{2k+1} \leq s_{2k+1}$; hence every even partial is $\leq$ every odd partial (climb up to an even, descend down to an odd, and compare across an adjacent pair). So the even partials climb, bounded above by s_1; the odd partials descend, bounded below by s_2: by the monotone convergence theorem (Theorem 4.9) both converge, say to E and O, and

$$O-E = \lim_k\,(s_{2k+1}-s_{2k}) = \lim_k\, b_{2k+1} = 0,$$

so $E=O=S$, a common limit. The parities merge: given ε, beyond some point every even partial and every odd partial lies within ε of S, hence every partial does: $s_n \to S$. *Error bound:* S is trapped between each adjacent pair of partials — for even n, $s_n \leq S \leq s_{n+1}$; for odd n, $s_{n+1} \leq S \leq s_n$ — so

$$|S-s_n| \leq |s_{n+1}-s_n| = b_{n+1}:$$

the first neglected term. □

II.6 (The halving jewel), in full. Let s_n be the standard partials of $1-\frac{1}{2}+\frac{1}{3}-\cdots$, with sum $S=\lim s_n$ (Theorem 4.33); note $S \geq s_2 = \frac{1}{2} > 0$. Let t_m be the partials of the re-dealt series. The k-th *triple* of the new ordering is

$$\frac{1}{2k-1}-\frac{1}{4k-2}-\frac{1}{4k} = \Big(\frac{1}{2k-1}-\frac{1}{2(2k-1)}\Big)-\frac{1}{4k} = \frac{1}{2(2k-1)}-\frac{1}{4k} = \frac{1}{2}\Big(\frac{1}{2k-1}-\frac{1}{2k}\Big),$$

which is exactly half the k-th *pair* of the standard ordering. Summing m triples:

$$t_{3m} = \frac{1}{2}\sum_{k=1}^{m}\Big(\frac{1}{2k-1}-\frac{1}{2k}\Big) = \frac{1}{2}\,s_{2m} \longrightarrow \frac{S}{2}.$$

The bridge to all partials: any t_m differs from the nearest triple-boundary partial $t_{3\lfloor m/3\rfloor}$ by at most two terms of the series, each of absolute value at most $1/(2\lfloor m/3\rfloor - 1) \to 0$; so the full sequence t_m converges to $S/2$ as well. Since $S \neq 0$, we have $S/2 \neq S$: the same terms, each used exactly once, and the sum has genuinely halved. (With the on-credit valuation: S is the natural logarithm of two, and the re-dealt sum is half of it.) □

4.14 Notes and References

Abbott, *Understanding Analysis*, Chapter 2, is this chapter's companion in spirit — the same theorems with the same affection; Rudin, *Principles of Mathematical Analysis*, Chapter 3, is the stern uncle: lim sup and the convergence tests done at altitude, with the rearrangement theorem as an exercise in cold precision. Oresme's divergence proof appears in his *Questiones super geometriam Euclidis* (Paris, around 1350); the result was rediscovered and reproved down the centuries, notably by the Bernoulli brothers. Jacob Bernoulli met e pricing compound interest (1683); the limit $\left(1+\frac{1}{n}\right)^n$ of Problem II.2 is that encounter, formalized. Bolzano's share of Theorem 4.13 sits in the same 1817 Prague paper mourned in Chapter 1; Weierstrass rebuilt the theorem for his Berlin lectures in the 1860s, unaware and better attended. Cauchy stated his criterion in the *Cours d'analyse* (1821) and treated sufficiency as evident — he had no completeness axiom to prove it with; the 1872 constructions retroactively underwrite the book. Dirichlet's 1837 paper contains both the first published rearrangement anomaly and the invariance theorem for absolute convergence (our Theorem 4.37); Riemann's rearrangement theorem appears in his 1854 habilitation essay on trigonometric series — the same essay that contains his integral — published posthumously in 1867, edited by Dedekind. The binomial theorem (Problem II.2's flagged credit) is any algebra text's; the natural logarithm, and with it the valuation $\ln 2$ of the alternating harmonic series and $\pi^2/6$ of the Basel series, arrive later in this course or beyond it — for Euler's 1735 evaluation see the notes to Chapter 1. The atom-count sensibility is stolen, with admiration, from Körner's *The Pleasures of Counting*.

Chapter 5

The Topology of the Line

Companion to Episode Five. The episode built the vocabulary of nearness — neighbourhoods, open and closed, limit points, closure and boundary — then crowned compactness by way of the infinite blanket problem and the bisection hunt, walked the three roads, proved the line untearable, loaded Monsieur Baire's instrument, and audited the dust as a one-set paradox lounge. This chapter is the paperwork. One clerical convention, new this chapter: routine invocations of suprema inside minor propositions are annotated "(completeness — small change, uncounted)" and kept off the bell's register; the register reserves its marks (⊛) for the marquee rings — Heine–Borel, the connectedness of intervals, and Baire.

Episode Digest

The argument of the hour. The season changes key: from journeys to the stage itself. The vocabulary of nearness is built — neighbourhoods, open and closed (not opposites), limit points, closure and boundary — and immediately pays wages. Then the concept of the hour: compactness, defined honestly by the infinite blanket problem and sold as conserved finiteness — the next best thing to being finite. Heine–Borel crowns closed-and-bounded; three roads (covers, closed-and-bounded, sequences) prove to be one road on the line; the line cannot be torn; Monsieur Baire's instrument is loaded for Episode Seven; and the dust returns to be audited as a one-set paradox lounge. The bell rings three times more — the count stands at eleven.

The episode, beat by beat. **The Hypothesis That Keeps Recurring.** One phrase has stalked the course: closed and bounded — the nested interval theorem's fine print (Episode Two), the bisection hunt's licence (Episode Four), and the price of admission for three theorems directly ahead. When one hypothesis keeps appearing on the invoices, a gentleman investigates what he has been paying for. Tonight's answer: compactness.

Neighbourhoods, or Nearness Made Honest. The ε-neighbourhood is the tolerance spatialized (Definition 5.1); an interior point owns a buffer; an open set is one where every resident does. The empty set complies with every regulation beginning "for every." Open conditions are the verifiable ones — a finite-precision instrument can affirm them. The two laws and their asymmetry (Proposition 5.2): unions tolerate infinity; intersections demand finiteness — the buffers $(-\frac{1}{n}, \frac{1}{n})$ strangle down to $\{0\}$, the seed from which the subject grows. And the line's little confession: every open set is a countable disjoint string of interval beads (Proposition 5.3).

Closed Sets, and Why Closed Is Not the Opposite of Open. Official definition: complement open (Definition 5.4); working definition, bought and proved: a closed set contains every point it crowds — no escape by converging (Theorem 5.7), a club closed under members' guests. Limit points crowd; hermits own estates ($\mathbb{Z}$ is all hermits). The specimen pair: the reciprocals $\{\frac{1}{n}\}$ are not closed — one fugitive, zero — while the repaired $\{0\} \cup \{\frac{1}{n}\}$ is; both filed for starring roles. Sets are not doors: $\mathbb{R}$ and $\emptyset$ are both open and closed ("clopen," pronounced once and retired); $[0, 1)$ and $\mathbb{Q}$ are neither. Interior, closure, boundary (Definition 5.8, Proposition 5.9): the closure of $\mathbb{Q}$ is $\mathbb{R}$ (the picket fence restated), the boundary of $\mathbb{Q}$ is $\mathbb{R}$ — every point of the continuum a border post (Remark 5.10). The derived set named (Definition 5.6); the repaired specimen performs the two-audit demonstration; a starred problem builds a set that takes exactly three.

The Vocabulary Pays Its First Wages. Convergence restated without distances: every neighbourhood of the limit eventually contains the whole tail — epsilon was scaffolding; the passport is valid abroad. Bolzano–Weierstrass for sets: every bounded infinite set has a limit point (Problem II.2, solution supplied, choice footnote flagged). Density collapses to "closure is everything." Cantor's intersection theorem trailed: nested non-empty compacts share a point — the dolls generalized (Problem II.4). And the subject receives its name: topology, the study of what survives when distance is forgotten.

The Infinite Blanket Problem. The recurring predicament: infinitely many local certificates, one global conclusion wanted. Open covers christened blankets; compactness defined (Definition 5.11): *every* cover admits a finite subfamily *drawn from the skeptic's own bid*. Two traps sprung: having some finite cover is worthless; compactness is a standing capacity — solvency, not one paid invoice. The courtroom geometry imports from last week: one unanswerable bid convicts; only a strategy for all of them acquits. Four convictions (Proposition 5.12): the open interval leaks at its absent endpoint; the line escapes to the horizon; the integers' hermit moor — every blanket essential; and the naked reciprocals, the moor miniaturized — bounded, but the audit collapses exactly at the hole where zero used to be. One audit passed by hand (Proposition 5.13): the reciprocals with their destination — one blanket swallows the tail, finitely many mop up stragglers; the last cover ever verified manually. *The wonder beat:* compactness as conserved finiteness — the finite audit restored on infinite sets; the operating manual for every local-to-global theorem from here to Kolmogorov; the reason next week's extreme value theorem can promise an actual maximum.

Heine and Borel, or the Coronation. Compact $\iff$ closed and bounded (Corollary 5.19). Easy half: expanding blankets bound (Proposition 5.14); the moat argument closes (Proposition 5.15) — half-distances, finitely many moats, and the finite audit is what keeps the minimum positive. Heredity purchased for Episode Six: closed tenants of compact landlords are compact (Proposition 5.17). The apologia: define by the engine (blankets), not the paint (closed-and-bounded) — proofs spend the finite audit, and the blankets travel where the paint does not. The magnificent half (Theorem 5.18 ⊛, **the ninth ring, rung aloud**): the bisection hunt's first contract executed — bisect the no-finite-subcover portion, keep the defiant half (two committees would merge), dolls close on a survivor x^*; the defiant shards are non-empty (the empty set is covered by the extremely finite committee of none); x^* lies in K (closed), under one blanket, which swallows a whole doll — the uncoverable shard covered by a committee of one. Localize the crime, then arrest it. Naming history: Heine 1872 (implicit, in the vintage year of Dedekind and Cantor; his actual theorem is next episode's third crown), Borel 1895 (countable covers), the general case around the century's turn, Lebesgue among the hands — theorems named as the Navy names ships, for admirals adjacent to the battle; and Borel, history being too good to invent, later served as France's Minister of the Navy.

Three Roads to the Same Place. Sequential compactness defined with the residency requirement (Definition 5.20). Closed-and-bounded to sequential in one sentence — the hunt plus arrivals-land-within (Theorem 5.21); the road back by escape arguments (Proposition 5.22, Problem II.3, solution supplied); the cycle assembled (Remark 5.23). The workhorse demonstrated one week early: a non-empty compact set contains its supremum — an actual largest member, the frontier staffed (Proposition 5.16) — the extreme value theorem's naked skeleton, awaiting only a function. Three wardrobes, one concept: blankets travel and power local-to-global; sequences are the line's workhorse; closed-and-bounded is the checkable certificate. Honesty signpost: the roads part in infinite-dimensional wilderness.

The Line Cannot Be Torn. Torn: two open camps, jointly exhaustive, disjoint on the set (Definition 5.24). The connected subsets of the line are exactly the intervals (Theorem 5.25 ⊛, **the tenth ring, on paper**): a gap invites the tear; an interval embarrasses the supremum of the first camp whichever camp claims it. The rationals tear at the hole $\sqrt{2}$ cannot fill — indeed every irrational is a fault line, so the rationals are totally disconnected. The clopen jewel (Corollary 5.26): only $\varnothing$ and $\mathbb{R}$; the portmanteau retires with honours. Filed for next week: the intermediate value theorem is tearlessness wearing a function.

Monsieur Baire's Instrument. Nowhere dense: thin even after thickening — the closure contains no interval (Definition 5.27). The rationals are *not* nowhere dense yet *are* meagre (a countable stack of singleton films): dense-yet-meagre, two thinness verdicts coexisting — and a third notion of negligible is teased for Season Two, where the three will disagree scandalously. Baire's theorem (Theorem 5.28 ⊛, **the eleventh ring, on paper**), both forms with the mirror mechanism: the line is not meagre; countable intersections of dense opens are dense. The guardrail: meagreness is not invisibility — Baire is a non-exhaustion principle, and its operational contrapositive (the form that will be fired): in any countable cast covering a complete stage, somebody carries mass. The mirror test: run it on the rationals as a stage and it dies — $\mathbb{Q}$ is meagre in itself; completeness is the engine, the axiom in yet another costume. The proof is the dolls at higher rank — Cantor evicted a suspect per stage, Baire evicts a film per stage; the 1874 argument was this theorem's larva — and it is the desk's guided centrepiece (Problem II.6, staged hints, no solution: build the instrument yourself). Dividends (Remark 5.29): the irrationals are not meagre — the continuum's structural mass; a third uncountability proof for the collection of clocks; the word *generic* pre-installed. Registered: deployment Episode Seven, the monsters typical; a second engagement pencilled for Season Two.

The Dust Returns: A Paradox Lounge. The first of three contracts executed. The audit (Theorem 5.31): closed, twice over (deleted opens union rays; intersection of closed stages); compact, by the coronation — tamed paperwork on an untamed animal; perfect (Definition 5.30 — Cantor's own term, coined in the dust's 1883 birth-paper): flip any address beyond a fine stage and a second grain appears within tolerance, injectivity guaranteeing it is new — maximal crowding, not one hermit; totally disconnected: an evicted point between any two grains, a tear waiting — maximal tearing, a crowd of strangers pressed together and touching no one, and at uncountable strength where the rationals managed it countably; nowhere dense: the canonical film — what nothing-anywhere looks like when it insists on being somewhere, uncountably. Four antitheses lounging in one set: the standard-issue counterexample dossier of the subject. Bonus for the ambitious: every non-empty perfect set is uncountable (Theorem 5.32, Problem III.1) — the 1874 hunt and the 1891 diagonal shaking hands over a tree of dolls. And the sealed envelope — how *long* is the dust? — stays sealed for Season Two, worth the postage.

The toolkit acquired. Neighbourhoods, open and closed, with the laws and the strangled intersection (Definitions 5.1, 5.4, Proposition 5.2, Corollary 5.5); the beads (Proposition 5.3); limit points, hermits, derived sets (Definition 5.6); closedness three ways (Theorem 5.7); interior, closure, boundary, with the sequence characterization (Definition 5.8, Proposition 5.9); covers and compactness (Definition 5.11) with four convictions and one hand audit (Propositions 5.12, 5.13); compact $\Rightarrow$ bounded, closed, max-attained, hereditary (Propositions 5.14–5.17); Heine–Borel (Theorem 5.18 $\circledast$, Corollary 5.19); sequential compactness and the cycle (Definition 5.20, Theorem 5.21, Proposition 5.22, Remark 5.23); connectedness and the intervals (Definition 5.24, Theorem 5.25 $\circledast$, Corollary 5.26); nowhere dense, meagre, generic, and Baire (Definition 5.27, Theorem 5.28 $\circledast$, Remark 5.29); perfect sets and the dust's dossier (Definition 5.30, Theorems 5.31, 5.32). Supplied solutions: Bolzano–Weierstrass for sets (II.2) and the sequential road (II.3); the Baire centrepiece is guided, deliberately unsolved.

Debts. **Executed: D8, contract one** — the dust's topology return; remaining: a Season Two engagement and the sealed envelope of extent. **Loaded: Baire** — deployment Episode Seven formally registered (the monsters typical; the schoolroom inverted); a Season Two engagement pencilled besides. **The bisection hunt:** first contract executed at the coronation; Episode Six remains. **Enthroned:** closed-and-bounded, a walk-on since Episode Two, crowned under its true name — compactness, conserved finiteness. D7 (the bell): three rings tonight — the ninth aloud at Heine–Borel, the tenth and eleventh on paper at the tearlessness of intervals and at Baire — running count **eleven**; at this rate the finale wants a carillon. New clerical convention: routine suprema in minor propositions are annotated as small change, uncounted; the register logs marquee rings only. Christened: the blankets. The rationals leave with a fattened dossier: neither open nor closed, all boundary, dense yet meagre, totally disconnected.

Dates worth keeping. 1872, Heine's uniform-continuity paper — the covering idea implicit, in the vintage year of Dedekind's cuts and Cantor's huddles; 1883, Cantor's *Grundlagen* — the dust and the word *perfect* born together; 1895, Borel proves the countable-cover theorem; 1897, Osgood's line version of the category theorem; 1899, Baire's thesis; the turn of the century, the general covering theorem settles, Lebesgue among the hands; and the nineteen-twenties, Borel serves as Minister of the Navy — some jokes are issued to one.

5.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"the neighbourhood of x, radius epsilon"	$N_\varepsilon(x) = (x-\varepsilon,\ x+\varepsilon)$	Definition 5.1
"open" / "closed"	open set; closed set	Definitions 5.1, 5.4
"interior, closure, boundary"	A°, $\overline{A}$, ∂A	Definition 5.8
"limit point" / "hermit"	limit point; isolated point	Definition 5.6
"the derived set"	A' (iterable: A'', A''')	Definition 5.6
"a family of blankets"	$\{O_\alpha\}_{\alpha\in\Lambda}$, an indexed family	Definition 5.11
"cover" / "subcover"	open cover; finite subcover	Definition 5.11
"compact"	compact	Definition 5.11
"sequentially compact"	sequentially compact	Definition 5.20
"torn" / "untearable"	disconnected; connected	Definition 5.24
"a film" / "nowhere dense"	nowhere dense set	Definition 5.27
"a stack of films" / "meagre"; "generic"	meagre; residual (generic)	Definition 5.27

5.2 Neighbourhoods and Open Sets

Definition 5.1 (Neighbourhoods; open sets)**.** For $x \in \mathbb{R}$ and $\varepsilon > 0$, the *ε-neighbourhood* of x is

$$N_\varepsilon(x) = (x-\varepsilon,\ x+\varepsilon) = \{y : |y-x| < \varepsilon\} :$$

the tolerance of Chapter 4, spatialized. A point x is an *interior point* of $A \subseteq \mathbb{R}$ if some $N_\varepsilon(x) \subseteq A$: the point comes with working room. A set is *open* if every one of its points is interior. Open intervals are open; so are $\mathbb{R}$ and (vacuously) $\emptyset$.

Proposition 5.2 (The laws of open sets)**.** (i) *The union of any family of open sets — finite, countable, or wilder — is open.* (ii) *The intersection of finitely many open sets is open.* (iii) *Finiteness in (ii) is essential: $\bigcap_{n\geq 1}\left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}$, which is not open.*

Proof. (i) If $x \in \bigcup_\alpha O_\alpha$, then $x \in O_\beta$ for some β; an ε-neighbourhood inside O_β sits inside the union. (ii) If $x \in O_1 \cap \cdots \cap O_n$, each O_i grants a tolerance ε_i; take $\varepsilon = \min(\varepsilon_1, \ldots, \varepsilon_n) > 0$ — a minimum of *finitely many* positive numbers, which is where finiteness earns its keep. (iii) 0 lies in every $\left(-\frac{1}{n}, \frac{1}{n}\right)$; any $x \neq 0$ is expelled once $\frac{1}{n} < |x|$, which the Archimedean property (Theorem 2.12) guarantees; and $\{0\}$ contains no neighbourhood of 0. An infinite conference of tolerances can be strangled to nothing. □

Proposition 5.3 (The string of beads: the structure of open sets)**.** *Every non-empty open set $O \subseteq \mathbb{R}$ is a countable union of pairwise disjoint open intervals (bounded intervals, rays, or the whole line).*

Proof. For $x \in O$ let I_x be the union of all open intervals that contain x and lie inside O; equivalently $I_x = (a_x, b_x)$ where

$$a_x = \inf\{a : (a,x] \subseteq O\}, \qquad b_x = \sup\{b : [x,b) \subseteq O\},$$

allowing $\pm\infty$ (completeness — small change, uncounted). Each I_x is an open interval containing x, contained in O (any point of I_x lies in some contributing interval), and *maximal*: no strictly larger open interval fits, since its endpoints, when finite, lie outside O — if $b_x \in O$, openness would grant a neighbourhood extending $[x, b_x)$ further, against the supremum. Two components either coincide or are disjoint: if $I_x \cap I_y \neq \emptyset$, then $I_x \cup I_y$ is an open interval inside O containing both x and y, so maximality forces $I_x = I_x \cup I_y = I_y$. Thus O is the disjoint union of its distinct components. Countability by rational tags: each component contains a rational (density, Theorem 2.15); distinct components, being disjoint, receive distinct tags; and $\mathbb{Q}$ is countable (Theorem 3.7). A countable string of beads, each bead an interval. □

5.3 Closed Sets, Limit Points, Closure

Definition 5.4 (Closed sets)**.** $F \subseteq \mathbb{R}$ is *closed* if its complement $\mathbb{R} \setminus F$ is open. "Closed" is not the opposite of "open": $[0, 1)$ is neither; $\mathbb{R}$ and $\emptyset$ are both. Sets are not doors.

Corollary 5.5 (The dual laws)**.** *The intersection of any family of closed sets is closed; the union of finitely many closed sets is closed. Finiteness is again essential:* $\bigcup_{n\geq 1}[\frac{1}{n}, 1] = (0, 1]$*, not closed (Problem I.3).*

Proof. De Morgan: the complement of an intersection is the union of the complements, and the complement of a finite union is the finite intersection of the complements; apply Proposition 5.2 to the complements. □

Definition 5.6 (Limit points; isolated points; the derived set)**.** $x \in \mathbb{R}$ is a *limit point* of A if every neighbourhood of x contains a point of A *other than x itself*: the set crowds x, whether or not it contains x. A point of A that is not a limit point of A is *isolated*: it owns a neighbourhood empty of its fellows. The set of all limit points of A is the *derived set* A'; deriving iterates: $A'' = (A')'$, and so on. Calibrations: every integer is isolated, so $\mathbb{Z}' = \emptyset$; every point of $\{\frac{1}{n} : n \in \mathbb{N}\}$ is isolated, yet the set has the limit point 0 — which it does not contain; the repaired specimen $\{0\} \cup \{\frac{1}{n}\}$ performs the two-step audit of the episode: its derived set is $\{0\}$, whose derived set is empty.

Theorem 5.7 (Closedness, three ways)**.** *For $F \subseteq \mathbb{R}$ the following are equivalent:* (i) *F is closed;* (ii) *F contains all its limit points;* (iii) *whenever a sequence of points of F converges, its limit lies in F — no escape by converging.*

Proof. (i) $\Rightarrow$ (ii): let x be a limit point of F and suppose $x \notin F$; then x lies in the open complement, so some $N_\varepsilon(x) \subseteq \mathbb{R} \setminus F$ — a neighbourhood of x containing no point of F at all, contradicting "limit point".

(ii) $\Rightarrow$ (iii): let $a_n \in F$ with $a_n \to x$. If $x \in F$ we are done; if not, then every $N_\varepsilon(x)$ contains terms a_n — points of F, necessarily distinct from x — so x is a limit point of F, hence in F by (ii): contradiction, so in fact $x \in F$.

(iii) $\Rightarrow$ (i): suppose F is not closed, so the complement is not open: some $x \notin F$ has no neighbourhood clear of F; for each n choose $a_n \in F \cap N_{1/n}(x)$ (one point per stage). Then $a_n \to x$, a convergent sequence from F whose limit escapes: (iii) fails. Contrapositive complete, and the cycle closes. □

Definition 5.8 (Interior, closure, boundary)**.** For $A \subseteq \mathbb{R}$: the *interior* A° is the set of interior points of A — the largest open set inside A; the *closure* $\overline{A} = A \cup A'$ — the set together with its crowd; the *boundary* $\partial A = \overline{A} \setminus A^\circ$ — the customs line.

Proposition 5.9 (Closure: smallest closed superset; the sequence characterization)**.** *$\overline{A}$ is closed, is contained in every closed set containing A, and consists exactly of the limits of convergent sequences from A. In particular, A is dense in $\mathbb{R}$ precisely when $\overline{A} = \mathbb{R}$.*

Proof. Closed: let x be a limit point of $\overline{A}$ and $\varepsilon > 0$. Some point $y \in \overline{A}$ lies in $N_{\varepsilon/2}(x)$ with $y \neq x$; whether $y \in A$ or y is a limit point of A, the neighbourhood $N_{\varepsilon/2}(y) \subseteq N_\varepsilon(x)$ contains a point of A (in the second case, one different from y; shrink further to avoid x if needed — Problem II.1 supplies the half-radius bookkeeping). So every neighbourhood of x meets A in a point other than x: $x \in A' \subseteq \overline{A}$, and $\overline{A}$ contains its limit points; closed by Theorem 5.7. *Smallest:* a closed $F \supseteq A$ contains A's limit points (they are limit points of F too), hence contains $\overline{A}$. *Sequences:* if $x \in \overline{A}$, harvest $a_n \in A \cap N_{1/n}(x)$ (possible whether $x \in A$ — take $a_n = x$ — or $x \in A'$): then $a_n \to x$. Conversely a limit of A-points lies in every closed superset of A by Theorem 5.7(iii), hence in $\overline{A}$. □

Remark 5.10 (The rationals, calibrated)*.* $\mathbb{Q}$ is neither open (every neighbourhood of a rational contains irrationals: Corollary 2.16) nor closed (every neighbourhood of an irrational contains rationals: Theorem 2.15). Its interior is empty; its closure is $\mathbb{R}$ (the picket fence restated); hence $\partial\mathbb{Q} = \mathbb{R}$: a set that is all boundary, from end to end. Boundary calibrations for the menagerie of intervals, integers, and reciprocals are Problems I.1–I.2; the failure $\bigcup[\frac{1}{n}, 1] = (0, 1]$ is Problem I.3.

5.4 Compactness: The Infinite Blanket Problem

Definition 5.11 (Covers; compactness)**.** An *open cover* of $K \subseteq \mathbb{R}$ is a family $\{O_\alpha\}_{\alpha\in\Lambda}$ of open sets with $K \subseteq \bigcup_{\alpha\in\Lambda} O_\alpha$; the index set Λ may be of any size. A *finite subcover* is a finite subfamily $O_{\alpha_1}, \dots, O_{\alpha_m}$ that still covers K. The set K is *compact* if *every* open cover of K admits a finite subcover. Quantifier gloss, once for the record: the skeptic bids the cover; we must answer with finitely many blankets drawn *from his own bid*. Having *some* finite cover is worthless (the whole line is one blanket); compactness is a standing capacity, solvency against every bid. Its denial needs only one unanswerable bid — the exact grammar of Chapter 4's divergence prosecutions.

Proposition 5.12 (Four convictions)**.** (i) $(0, 1)$ *is not compact.* (ii) $\mathbb{R}$ *is not compact.* (iii) $\mathbb{Z}$ *is not compact.* (iv) $\{\frac{1}{n} : n \in \mathbb{N}\}$ *— the reciprocals without their destination — is not compact.*

Proof. (i) *The leak.* Bid $O_n = (\frac{1}{n}, 1)$ for $n \geq 2$. The union is all of $(0, 1)$: any $x > 0$ has $\frac{1}{n} < x$ eventually (Theorem 2.12). A finite subfamily has a largest index N; its union is $(\frac{1}{N}, 1)$, missing $(0, \frac{1}{N}]$ — the cover leaks at the absent endpoint.

(ii) *The escape.* Bid $O_n = (-n, n)$. A finite subfamily is contained in $(-N, N)$ for its largest N, and N itself escapes.

(iii) *The hermit moor.* Bid $O_k = (k - \frac{1}{2}, k + \frac{1}{2})$ for each integer k: each blanket contains exactly one integer, so every blanket is essential — discard any one and its integer lies bare. No finite subfamily covers an infinite set of hermits. (Closedness alone, note, is no protection.)

(iv) *The moor, miniaturized.* For each n bid the private tent $O_n = N_{\varepsilon_n}(\frac{1}{n})$ with $\varepsilon_n = \frac{1}{2n(n+1)}$ — half the gap $\frac{1}{n} - \frac{1}{n+1} = \frac{1}{n(n+1)}$ down to the next reciprocal, and smaller still than the gap up to the previous one. Each tent contains exactly one point of the set; every blanket is essential; no finite subfamily suffices. Bounded, but the missing limit point 0 is exactly where the audit collapses — boundedness without closedness leaks. $\square$

Proposition 5.13 (One audit by hand)**.** *$K = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ is compact.*

Proof. Let $\{O_\alpha\}$ be any open cover. Some O_β contains 0; being open, it contains $N_\varepsilon(0)$ for some $\varepsilon > 0$; and by the Archimedean property (Theorem 2.12) all but finitely many reciprocals satisfy $\frac{1}{n} < \varepsilon$ and lie inside it: one blanket swallows the destination and the entire tail. The stragglers $1, \frac{1}{2}, \ldots, \frac{1}{N}$ are finitely many; assign each one covering blanket from the family. The selected subfamily — one plus N blankets — is finite and covers K. $\square$

Proposition 5.14 (Compact implies bounded)**.** *Every compact $K \subseteq \mathbb{R}$ is bounded.*

Proof. $\{(-n, n)\}_{n \in \mathbb{N}}$ is an open cover of $\mathbb{R}$, hence of K; a finite subcover puts $K \subseteq (-N, N)$ for the largest index N. $\square$

Proposition 5.15 (Compact implies closed: the moat)**.** *Every compact $K \subseteq \mathbb{R}$ is closed.*

Proof. Let $x \notin K$; we open a moat around x. For each $y \in K$ set $\varepsilon_y = |x - y|/2 > 0$: half the distance. The family $\{N_{\varepsilon_y}(y)\}_{y \in K}$ is an open cover of K; compactness selects finitely many, centred at $y_1, \ldots, y_m$. Set $\varepsilon = \min(\varepsilon_{y_1}, \ldots, \varepsilon_{y_m}) > 0$ — *the finite audit is precisely what makes this minimum positive*; over infinitely many moats it could drain to nothing. If some z lay in $N_\varepsilon(x) \cap N_{\varepsilon_{y_i}}(y_i)$, then

$$|x - y_i| \le |x - z| + |z - y_i| < \varepsilon + \varepsilon_{y_i} \le 2\varepsilon_{y_i} = |x - y_i|,$$

absurd. So $N_\varepsilon(x)$ misses every selected blanket, hence misses K: the complement of K is open. $\square$

Proposition 5.16 (The maximum is attained)**.** *A non-empty compact K contains a largest element, and a smallest.*

Proof. K is bounded (Proposition 5.14), so $s = \sup K$ exists (completeness — small change, uncounted). For each n the dipped level $s - \frac{1}{n}$ is overtaken by some $k_n \in K$ (Lemma 2.4), and $k_n \le s$; so $k_n \to s$, a convergent sequence from K. But K is closed (Proposition 5.15), so $s \in K$ by Theorem 5.7(iii): the supremum is a member — the frontier, for once, is staffed. Mirror argument for the minimum. (This is the naked skeleton of the extreme value theorem; the next chapter dresses it in a function.) $\square$

Proposition 5.17 (Heredity)**.** *A closed subset of a compact set is compact.*

Proof. Let $F \subseteq K$ with F closed, K compact, and let $\{O_\alpha\}$ cover F. Adjoin the open set $\mathbb{R} \setminus F$: the enlarged family covers K. Extract a finite subcover of K; discard $\mathbb{R} \setminus F$ from it if present. What remains is a finite subfamily of the original cover, and it covers F (no point of F was relying on the discarded set). $\square$

Theorem 5.18 (Heine–Borel, the hard half)**.** *Every closed and bounded $K \subseteq \mathbb{R}$ is compact.*

Proof. Say $K \subseteq [-M, M] =: J_0$, and suppose, for contradiction, that some open cover $\{O_\alpha\}$ of K admits *no* finite subcover. Bisect J_0 into two closed halves. If each half's portion of K could be covered by finitely many blankets, the two committees would merge into a finite subcover of K; so at least one half's portion defies every finite subfamily — keep the leftmost defiant half as J_1, and recurse. This builds nested closed intervals $J_0 \supseteq J_1 \supseteq \cdots$ with widths $2M/2^n \leq 2M/n$ (since $2^n > n$: Chapter 1, Problem I.5), each containing a portion $K \cap J_n$ that no finite subfamily covers — in particular each portion is *non-empty*, since the empty set is covered by the extremely finite committee of none.

By the nested interval theorem in unique-survivor form (Theorem 2.17, Remark 2.18) ($\circledast$) there is exactly one $x^* \in \bigcap_n J_n$. Choosing $x_n \in K \cap J_n$ gives $|x_n - x^*| \leq \text{width}(J_n) \to 0$, so x^* is a limit of points of K; and K is closed, so $x^* \in K$ (Theorem 5.7). Some blanket of the bid covers it: $x^* \in O_\beta$, open, so $N_\varepsilon(x^*) \subseteq O_\beta$ for some $\varepsilon > 0$. But the dolls shrink beneath every tolerance: once $\text{width}(J_n) < \varepsilon$, the whole doll — containing x^* — sits inside $N_\varepsilon(x^*) \subseteq O_\beta$. Then $K \cap J_n$ is covered by the single blanket O_β: a finite subcover of the defiant shard. Contradiction. Localize the crime, then arrest it. □

Corollary 5.19 (Heine–Borel, complete)**.** *$K \subseteq \mathbb{R}$ is compact if and only if K is closed and bounded.*

Proof. Propositions 5.14 and 5.15; Theorem 5.18. □

5.5 Three Roads to the Same Place

Definition 5.20 (Sequential compactness)**.** $K \subseteq \mathbb{R}$ is *sequentially compact* if every sequence of points of K has a subsequence converging *to a point of K*. The residency requirement is not decoration: sublimits must land inside.

Theorem 5.21 (Closed and bounded implies sequentially compact)**.** *If K is closed and bounded, then K is sequentially compact.*

Proof. A sequence in K is bounded, so Bolzano–Weierstrass (Theorem 4.13, itself bell-marked) extracts a convergent subsequence; its limit is a limit of points of K, hence lies in K by closedness (Theorem 5.7). One sentence, as promised. □

Proposition 5.22 (Sequentially compact implies closed and bounded)**.** *If K is sequentially compact, then K is closed and bounded.*

Proof. Problem II.3, solution supplied — the escape arguments of the episode, with paperwork. □

Remark 5.23 (The cycle, and the borders of the analogy). Assembled: compact $\Leftrightarrow$ closed and bounded $\Leftrightarrow$ sequentially compact (Corollary 5.19, Theorem 5.21, Proposition 5.22) — three wardrobes, one concept, on the line. The classical companion — every bounded infinite *set* has a limit point, Bolzano–Weierstrass for sets — is Problem II.2, solution supplied, with the house's standing footnote on countable choice. Honesty signpost, recorded once: in the infinite-dimensional wildernesses of Season Two's outskirts the three roads part company, and it is the blanket definition that keeps its power; "closed and bounded" is a certificate valid on the line, not a definition of anything.

5.6 The Line Cannot Be Torn

Definition 5.24 (Torn; connected)**.** $A \subseteq \mathbb{R}$ is *disconnected* (torn) if there exist open sets U, V with

$$A \subseteq U \cup V, \qquad A \cap U \neq \emptyset, \qquad A \cap V \neq \emptyset, \qquad A \cap U \cap V = \emptyset:$$

two open camps, each claiming part of A, jointly exhaustive, disjoint on A. A is *connected* if no such tear exists.

Theorem 5.25 (The connected subsets of the line are the intervals)**.** *$A \subseteq \mathbb{R}$ is connected if and only if A is an interval — that is, order-convex: whenever $a, b \in A$ and $a < c < b$, also $c \in A$.*

Proof. Not an interval $\Rightarrow$ torn. If $a, b \in A$ and some $z \notin A$ lies between, then $U = (-\infty, z)$ and $V = (z, \infty)$ tear A at the hole.

Interval $\Rightarrow$ untearable. Suppose open U, V tear the interval A: pick $a \in A \cap U$ and $b \in A \cap V$; without loss $a < b$, and $[a, b] \subseteq A$ since A is an interval. Let

$$S = [a, b] \cap U \ni a, \qquad s = \sup S$$

— the supremum exists ($\circledast$), and $s \in [a, b] \subseteq A \subseteq U \cup V$, so s lies in one of the camps; both fates embarrass it.

If $s \in V$: openness grants $N_\varepsilon(s) \subseteq V$. By the approximation lemma (Lemma 2.4) some point of S exceeds $s - \varepsilon$ — a point of $[a, b] \cap U$ inside $N_\varepsilon(s) \subseteq V$; it lies in $A \cap U \cap V$, sworn empty. (If instead $s \in S$ itself, s lies in $U \cap V \cap A$ directly.) Contradiction.

If $s \in U$: openness grants $N_\varepsilon(s) \subseteq U$. First, $s < b$: for $b \in V$, and were $s = b$, the case above (with the roles of the camps swapped around b) already convicts; more simply, $b \notin U$ would fail, since $b \in A \cap V$ and the camps are disjoint on A — so $s \neq b$, giving $s < b$. Then points of $[a, b]$ immediately to the right of s — inside $N_\varepsilon(s)$, hence inside U, hence in S — exceed the supremum. Contradiction.

No tear survives; the interval is connected. □

Corollary 5.26 (The clopen jewel)**.** *The only subsets of $\mathbb{R}$ that are simultaneously open and closed are $\emptyset$ and $\mathbb{R}$.*

Proof. If E is clopen with $\emptyset \neq E \neq \mathbb{R}$, then $U = E$ and $V = \mathbb{R} \setminus E$ are open, non-trivially partition $\mathbb{R}$, and so tear it; but $\mathbb{R}$ is an interval, connected by Theorem 5.25. Two lines, as advertised. □

5.7 Monsieur Baire's Instrument

Definition 5.27 (Nowhere dense; meagre; generic)**.** $A \subseteq \mathbb{R}$ is *nowhere dense* if $\overline{A}$ has empty interior: even after thickening to its closure, the set contains no interval. A set is *meagre* if it is a countable union of nowhere dense sets — a countable stack of films. The complement of a meagre set is called *residual*, and a property holding on a residual set is called *generic* (or typical) — vocabulary pre-installed for Chapter 7. Calibrations: $\mathbb{Z}$ is nowhere dense; $\mathbb{Q}$ is *not* nowhere dense ($\overline{\mathbb{Q}} = \mathbb{R}$), yet $\mathbb{Q}$ *is* meagre — a countable union of singleton films — so dense-yet-meagre is possible: the thinness notions genuinely differ. Mirror lemma, two lines: A is nowhere dense if and only if $\mathbb{R} \setminus \overline{A}$ is dense — for "$\overline{A}$ contains no interval" says exactly "every interval meets the complement of $\overline{A}$", which is density (Problem I.7); for closed F this reads: F nowhere dense $\Leftrightarrow \mathbb{R} \setminus F$ dense and open.

Theorem 5.28 (Baire, 1899)**.** Form A. $\mathbb{R}$ *is not meagre: the line is not a countable union of nowhere dense sets.* Form B. *The intersection of countably many dense open sets is dense. Locally, no interval is meagre.*

Proof, guided; the gaps are Problem II.6. Form B. Let $G_1, G_2, \ldots$ be dense open sets and (a, b) any test interval; we corner a survivor inside $(a, b) \cap \bigcap_n G_n$.

Stage one. G_1 is dense and open, so $(a, b) \cap G_1$ is a non-empty open set; it therefore contains a *closed* interval J_1 of positive width [gap (a): from a neighbourhood inside the open set, halve the radius and close it].

Recursion. Given the closed interval J_n of positive width, the set $J_n^\circ \cap G_{n+1}$ is non-empty and open (density again), so it contains a closed interval J_{n+1} of positive width, which we may take of width at most $\frac{1}{n}$ [gap (b): the shrinking].

The survivor. The J_n are nested closed intervals with widths dying beneath every tolerance: the nested interval theorem in unique-survivor form (Theorem 2.17, Remark 2.18) ($\circledast$) yields $x^* \in \bigcap_n J_n$. By construction $x^* \in J_1 \subseteq (a, b)$ and $x^* \in J_{n+1} \subseteq G_{n+1}$ for every n: the intersection $\bigcap_n G_n$ meets (a, b). As (a, b) was arbitrary, Form B stands.

Form A from Form B. Suppose $\mathbb{R} = \bigcup_n A_n$ with each A_n nowhere dense; pass to closures (the closure of a nowhere dense set is nowhere dense and closed) and apply the mirror lemma and De Morgan to manufacture countably many dense open sets with empty intersection, contradicting Form B [gap (d): the assembly].

The local corollary. No interval (a, b) is meagre [gap (e): run the same machinery, noting the survivor was cornered *inside* (a, b)]. □

Remark 5.29 (Dividends, and the register)*.* First: *the irrationals are not meagre* — were they, the line would be the union of two meagre sets (the rationals being meagre), hence meagre, against Form A. The continuum's structural mass resides entirely with the irrationals: a thickness statement finer than cardinality. Second: *a third proof that* $\mathbb{R}$ *is uncountable* — a countable set is a countable stack of singleton films, hence meagre, and the line is not; Cantor evicted a suspect per stage, Baire evicts a film per stage, and the 1874 hunt was this theorem's larva. Third, the *operational contrapositive*, the form in which the instrument is actually swung: if a countable union of sets covers the line — or any interval — then at least one of the sets is somewhere dense: its closure contains an interval. Non-exhaustion as a guardrail. Registered deployment: Chapter 7, where the instrument announces that the monsters are *typical*; a second engagement is pencilled for Season Two.

5.8 The Dust, Audited

Definition 5.30 (Perfect sets)**.** A set $P \subseteq \mathbb{R}$ is *perfect* if it is closed and has no isolated points: every point of P is a limit point of P ($P = P'$). The term is Cantor's own, coined in the 1883 paper that built the dust.

Theorem 5.31 (The dust's dossier)**.** *The Cantor set C (Definition 3.20) is:* (i) *closed;* (ii) *compact;* (iii) *perfect;* (iv) *totally disconnected — its connected components are single points;* (v) *nowhere dense.*

Proof. (i) Twice over: the complement of C is the union of all deleted open middle thirds together with two open rays — a union of open sets, open (Proposition 5.2); equivalently $C = \bigcap_n C_n$ with each stage C_n a finite union of closed intervals, closed by the dual laws (Corollary 5.5).

(ii) Closed and bounded ($C \subseteq [0,1]$): compact by the coronation (Corollary 5.19) — tamed paperwork on an untamed animal.

(iii) Perfect: let $x \in C$ with address (itinerary) as in the addressing scheme (Proposition 3.21), and let $\varepsilon > 0$. Choose a stage n with $3^{-n} < \varepsilon$ (since $3^n \geq 2^n > n$: Chapter 1, Problem I.5, with Theorem 2.12). Flip the address at one coordinate beyond stage n: the flipped address names a point $y \in C$ sharing x's stage-n interval, so $|x - y| \leq 3^{-n} < \varepsilon$; and $y \neq x$ because the addressing is injective (Proposition 3.21). Every neighbourhood of x holds another dust point.

(iv) Totally disconnected: let $x < y$ both lie in C. Since C contains no interval (Proposition 3.22), some $z \notin C$ has $x < z < y$; the camps $(-\infty, z)$ and (z, ∞) tear any subset of C containing both points. So no connected subset of C holds two points: components are singletons — a crowd of strangers, pressed together and touching no one.

(v) Nowhere dense: C is closed with empty interior (no interval, Proposition 3.22 again), so $\overline{C} = C$ contains no interval. The canonical film: what nothing-anywhere looks like when it insists on being somewhere, uncountably. □

Theorem 5.32 (Perfect sets are uncountable)**.** *Every non-empty perfect subset of* $\mathbb{R}$ *is uncountable.*

Proof. Problem III.1, starred and staged — the 1874 hunt and the 1891 diagonal shake hands over a tree of dolls. □

5.9 Problems

Tier I — Calisthenics

I.1 Classify each of $(0,1)$, $[0,1]$, $[0,1)$, $\mathbb{Z}$, $\mathbb{Q}$, $\{\frac{1}{n}\}$, $\{0\} \cup \{\frac{1}{n}\}$, $\mathbb{R}$, $\emptyset$ as open, closed, neither, or both, with a one-line justification each.

I.2 For the same menagerie, compute interior, closure, and boundary.

I.3 Prove the dual laws (Corollary 5.5) from Proposition 5.2 via De Morgan; then verify $\bigcup_{n \geq 1} [\frac{1}{n}, 1] = (0, 1]$ and explain exactly which hypothesis fails.

I.4 Show every finite set is closed; show $\mathbb{Z}$ is closed by showing it has no limit points at all.

I.5 Write the leak and the escape in full paperwork: the covers of Proposition 5.12(i) and (ii), with the Archimedean citations in place.

I.6 Compute the derived sets: $\{\frac{1}{n}\}'$, $(0,1)'$, $\mathbb{Q}'$, $\mathbb{Z}'$; and verify the two-audit claim of Definition 5.6 for $\{0\} \cup \{\frac{1}{n}\}$.

I.7 Prove the density equivalences: A is dense $\iff$ $\overline{A} = \mathbb{R}$ $\iff$ A meets every open interval $\iff$ every real is the limit of a sequence from A.

I.8 The clopen rehearsal: reprove Corollary 5.26 in two lines from Theorem 5.25; then explain in one sentence why the same argument shows an interval admits no proper clopen pieces.

Tier II — The Real Thing

II.1 Complete the half-radius bookkeeping in Proposition 5.9: a limit point of $\overline{A}$ is a limit point of A.

II.2 (Bolzano–Weierstrass for sets; solution provided.) Every bounded infinite subset of $\mathbb{R}$ has a limit point.

II.3 (Solution provided.) Prove Proposition 5.22: a sequentially compact set is closed and bounded.

II.4 (Cantor's intersection theorem.) If $K_1 \supseteq K_2 \supseteq \cdots$ are non-empty compact sets, then $\bigcap_n K_n \neq \emptyset$; and if the widths (diameters) shrink beneath every tolerance, the intersection is a single point. The dolls, generalized: closed intervals were never the point.

II.5 (Stability.) Prove: a finite union of compact sets is compact; and the intersection of a closed set with a compact set is compact.

II.6 (The centrepiece, guided; no solution will be supplied.) Fill the gaps of Theorem 5.28: (a) the closed subinterval inside a non-empty open set; (b) the shrinking recursion; (d) Form A from Form B via the mirror lemma; (e) the local corollary that no interval is meagre.

Tier III — For the Ambitious $\star$

$\star$ **III.1** Prove Theorem 5.32: every non-empty perfect set is uncountable. Staged route in the hints: two crowded directions, a binary tree of closed intervals, one survivor per branch, and the diagonal of Chapter 3 to finish.

$\star$ **III.2** The derived-set tower: construct a set $S \subseteq \mathbb{R}$ with $S' \neq \emptyset$, $S'' \neq \emptyset$, and $S''' = \emptyset$ — a set that survives exactly three audits — and prove your claims.

5.10 Hints

II.1. Let x be a limit point of $\overline{A}$ and $\varepsilon > 0$; produce $y \in \overline{A} \cap N_{\varepsilon/2}(x)$, $y \neq x$. If $y \in A$, done. If $y \in A'$, choose $\delta < \min\big(\varepsilon/2,\ |y-x|\big)$ and pick a point of A in $N_\delta(y)$ other than y: it lies in $N_\varepsilon(x)$ and cannot equal x (it is too close to y).

II.2. Since E is infinite, select a sequence of *distinct* points $x_n \in E$ (one fresh point per stage; the selection is the countable choice covered by Chapter 3's standing asterisk — spent, as ever, without ceremony). Bounded, so Bolzano–Weierstrass (Theorem 4.13) extracts $x_{n_k} \to x^*$. Any $N_\varepsilon(x^*)$ contains infinitely many of the x_{n_k} — all in E, all distinct — so at least one differs from x^*: x^* is a limit point of E.

II.4. Pick $x_n \in K_n$ (non-empty; countable choice again). The sequence lives in K_1, which is sequentially compact (Corollary 5.19 and Theorem 5.21); extract $x_{n_k} \to x$. For each fixed m, the tail x_{n_k} with $n_k \geq m$ lies in K_m (nesting), and K_m is closed, so $x \in K_m$ (Theorem 5.7); as m was arbitrary, $x \in \bigcap K_m$. If moreover the diameters die, two survivors would sit at distance zero: uniqueness.

II.5. *Union:* a cover of $K \cup L$ covers each; each grants a finite committee; merge the two committees. *Closed $\cap$ compact:* $F \cap K$ is closed (Corollary 5.5) and a subset of the compact K; apply heredity (Proposition 5.17).

II.6, staged. (a) A non-empty open set contains some $N_\delta(c)$; the closed interval $[c - \delta/2,\ c + \delta/2]$ sits inside it and has width $\delta > 0$. (b) Apply (a) to the non-empty open set $J_n^\circ \cap G_{n+1}$ — non-empty because G_{n+1} is dense and J_n° is a non-empty open interval — then, if the produced interval is too wide, shrink it about its centre to width at most $\frac{1}{n}$: closed subintervals of closed intervals cost nothing. (d) Given $\mathbb{R} = \bigcup A_n$ with each A_n nowhere dense: set $F_n = \overline{A_n}$ (still nowhere dense: $\overline{\overline{A}} = \overline{A}$; still a cover). Each $G_n = \mathbb{R} \setminus F_n$ is open and dense (mirror lemma), and $\bigcap G_n = \mathbb{R} \setminus \bigcup F_n = \varnothing$ by De Morgan — but Form B says the intersection is dense, in particular non-empty. (e) If $(a, b) = \bigcup A_n$ with each A_n nowhere dense, run Form B's construction verbatim with these $G_n = \mathbb{R} \setminus \overline{A_n}$: the survivor is cornered *inside* (a, b) and outside every $\overline{A_n}$ — one resident too many.

III.1, staged. *Step 1 (two crowded directions).* If I is a closed interval whose interior meets P, pick $p \in I^\circ \cap P$: being a limit point of P, its neighbourhood inside I° holds a second point $q \in P$. Choose disjoint closed intervals $I_0, I_1 \subseteq I^\circ$, centred at p and q, of width at most half of I's — each with interior meeting P. *Step 2 (the tree).* Starting from any such I, iterate: every finite binary word w receives a closed interval I_w, with $I_{w0}, I_{w1} \subseteq I_w^\circ$ disjoint, widths halving. *Step 3 (one survivor per branch).* An infinite binary sequence β names a nested chain $I_{\beta_1} \supseteq I_{\beta_1\beta_2} \supseteq \cdots$ with dying widths; the survivor x_β exists (nested intervals — completeness, small change here, the marquee rings being spent). Each x_β is a limit of points of P (the interiors meet P at every stage, ever closer), and P is closed: $x_\beta \in P$. *Step 4 (the handshake).* Distinct branches part at some finite stage into *disjoint* intervals, so $\beta \mapsto x_\beta$ is injective: $\{0,1\}^{\mathbb{N}}$ embeds in P, and the diagonal (Theorem 3.12) declares P uncountable.

III.2. Plant satellites about each reciprocal: consider

$$S = \left\{ \tfrac{1}{n} + \tfrac{1}{k} \ :\ n \in \mathbb{N},\ k > n(n+1) \right\}.$$

The spacing constraint confines family n to the band $\left(\frac{1}{n},\ \frac{1}{n} + \frac{1}{n(n+1)}\right) \subseteq \left(\frac{1}{n},\ \frac{1}{n-1}\right)$ for $n \geq 2$ — check $\frac{1}{n} + \frac{1}{n(n+1)} < \frac{1}{n-1}$, which reduces to $(n+2)(n-1) < n(n+1)$ — so the families occupy disjoint bands and no collisions occur. Then argue: every point of S is isolated (within its band, its family is a discrete ladder); $S' = \{0\} \cup \{\frac{1}{n} : n \in \mathbb{N}\}$ (family n crowds $\frac{1}{n}$; the bands themselves crowd 0; nothing else is crowded); $S'' = \{0\}$ (the episode's two-step); $S''' = \varnothing$. Exactly three audits.

5.11 Solutions

Tier I (terse). **I.1** $(0, 1)$ open; $[0, 1]$ closed; $[0, 1)$ neither (no neighbourhood of 0 inside; 1 is an uncontained limit point); $\mathbb{Z}$ closed (I.4); $\mathbb{Q}$ neither (Remark 5.10); $\{\frac{1}{n}\}$ neither (0 is an uncontained limit point; no point interior); $\{0\} \cup \{\frac{1}{n}\}$ closed (all limit points — just 0 — contained); $\mathbb{R}$ and $\varnothing$ both. **I.2** Interiors: $(0, 1)$ for the three intervals; $\varnothing$ for $\mathbb{Z}$, $\mathbb{Q}$, both reciprocal sets and $\varnothing$; $\mathbb{R}$ for $\mathbb{R}$. Closures: $[0, 1]$ for the three intervals; $\mathbb{Z}$; $\mathbb{R}$ for $\mathbb{Q}$; $\{0\} \cup \{\frac{1}{n}\}$ for both reciprocal sets; $\mathbb{R}$; $\varnothing$. Boundaries: $\{0, 1\}$ for the three intervals; $\mathbb{Z}$; $\mathbb{R}$ for $\mathbb{Q}$; $\{0\} \cup \{\frac{1}{n}\}$ for both reciprocal sets; $\varnothing$ for $\mathbb{R}$ and $\varnothing$. **I.3** Complements convert the two open laws into the two closed laws. The union $\bigcup[\frac{1}{n}, 1]$: any $x \in (0, 1]$ has $\frac{1}{n} \leq x$ eventually (Archimedes), and no $x \leq 0$ or $x > 1$ belongs; 0 is a limit point (the points $\frac{1}{n}$ approach it) not contained, so the union is not closed. The failed hypothesis: finiteness of the union. **I.4** A finite set's complement is a finite union of open intervals and rays, open; equivalently a finite set has no limit points (a punctured neighbourhood small enough avoids the finitely many others), so Theorem 5.7(ii) holds vacuously. $\mathbb{Z}$: any x has a neighbourhood of radius $\frac{1}{2}$ containing at most one integer, so no point of $\mathbb{R}$ is a limit point of $\mathbb{Z}$; closed vacuously. **I.5** As in Proposition 5.12, with the two Archimedean citations written at the leak ($\frac{1}{n} < x$ eventually) and the escape (N exceeds any proposed bound). **I.6** $\{\frac{1}{n}\}' = \{0\}$; $(0,1)' = [0,1]$; $\mathbb{Q}' = \mathbb{R}$ (density); $\mathbb{Z}' = \emptyset$. Two-step: $(\{0\} \cup \{\frac{1}{n}\})' = \{0\}$, and $\{0\}' = \emptyset$. **I.7** $\overline{A} = \mathbb{R} \iff$ every point is in A or crowds it $\iff$ every $N_\varepsilon(x)$ meets $A \iff$ every open interval meets A (intervals are neighbourhoods and contain them) $\iff$ every x is a limit of an A-sequence (Proposition 5.9). **I.8** As in Corollary 5.26. Relativized: a proper clopen piece E of an interval A yields the tear U, V built from the openness of E and of its complement-within-A — forbidden by Theorem 5.25 since A is an interval.

II.2 (Bolzano–Weierstrass for sets), in full. E bounded and infinite. Select distinct points: $x_1 \in E$; given distinct $x_1, \ldots, x_n$, the set $E \setminus \{x_1, \ldots, x_n\}$ is non-empty (E is infinite), so choose x_{n+1} in it. (The infinitely many selections are countable choice — Chapter 3's standing asterisk, spent without ceremony.) The sequence (x_n) is bounded, so Theorem 4.13 extracts a convergent subsequence $x_{n_k} \to x^*$. Given $\varepsilon > 0$: all but finitely many x_{n_k} lie in $N_\varepsilon(x^*)$; they are points of E, pairwise distinct, so infinitely many distinct points of E lie in the neighbourhood — in particular one different from x^*. Thus every neighbourhood of x^* meets E away from x^*: a limit point. $\square$

II.3 (Sequentially compact $\Rightarrow$ closed and bounded), in full. *Bounded:* if not, for each n choose $a_n \in K$ with $|a_n| > n$ (the escape). Every subsequence of (a_n) is unbounded — $|a_{n_k}| > n_k \geq k$ — and convergent sequences are bounded (Proposition 4.4), so no subsequence converges at all, against sequential compactness. *Closed:* if not, by Theorem 5.7(iii) some sequence $a_n \in K$ converges to a point $x \notin K$. Every subsequence also converges to x (Proposition 4.12), so no subsequence converges to a point *of* K (limits are unique, Proposition 4.3): the residency requirement fails. $\square$

5.12 Notes and References

Abbott, *Understanding Analysis*, Chapter 3, is the companion — the same landscape at the same walking pace, dust included; Rudin, *Principles of Mathematical Analysis*, Chapter 2, does compactness at metric altitude, where the blanket definition shows why it, and not "closed and bounded", is the export-grade formulation. The history of Theorem 5.18 is a study in gradual custom: Heine's 1872 paper on uniform continuity uses the covering idea implicitly (1872 — the vintage year: Dedekind's cuts and Cantor's huddles appeared in the same twelvemonth); Borel isolated and proved the countable-cover case in his 1895 thesis; the general statement settled into place around the turn of the century, Lebesgue among those handling it, and the double name is custom rather than adjudication. Baire's theorem is his 1899 thesis (Osgood had a version on the line in 1897); the biographical restraint — the lycées, the illness — is the episode's, not the literature's. Cantor's *perfect* and the dust itself are from the 1883 *Grundlagen*. For a museum of sets calibrating every definition in this chapter, Steen and Seebach, *Counterexamples in Topology*, is the standing reference; the sensibility, as ever, owes a debt to Körner.

Chapter 6

Continuity, Earned

Companion to Episode Six. The episode defined continuity as the familiar game with a new move order, hung three portraits of the notion — the game, the sequences, the pull-back — and harvested three crowns on last chapter's terrain: extreme values by compactness, intermediate values by tearlessness, and Heine–Cantor by compactness again. Then the monster gallery opened: the static, born in Dirichlet's hands, and Thomae's function, continuous exactly on the irrationals; and the asymmetry debt was issued with ceremony. This chapter is the paperwork. One register note: ***no new bell-marks this chapter*** *— every crown runs on inherited rings (Heine–Borel, the connectedness of intervals, Bolzano–Weierstrass, completeness itself), so no* $(\circledast)$ *appears below; the axiom has moved from wages to dividends, and the register appreciates compound interest.*

Episode Digest

The argument of the hour. The terrain acquires a population. Continuity is defined as the familiar game with a new move order — challenge epsilon on the outputs, response delta on the inputs — and hung in three portraits: the game, the sequences, the pull-back. Three crowns are then harvested on last week's terrain: extreme values by compactness, intermediate values by tearlessness, uniform continuity by compactness again — Heine's name cashing out at the third. The monster gallery opens: the static (Dirichlet's function), continuous nowhere; Thomae's function, continuous exactly on the irrationals — the hour's wonder. And the asymmetry debt is issued with ceremony: no function is continuous exactly on the rationals, payable only in Season Two. The bell, for the first time since Episode Two, is silent: every crown runs on inherited rings, and the count holds at eleven.

The episode, beat by beat. The Pen Lifts. The schoolroom's definition — a curve drawn without lifting the pen — is scheduled for demolition by inspection: by hour's end, a function continuous at every irrational and discontinuous at every rational will exist, and no pen survives it. The architecture announced: definition as rematch, three portraits, three crowns powered by Episode Five, then the gallery. Hence the title: not continuity defined — continuity earned.

The Game, New Move Order. Continuity at an anchor: for every challenge epsilon on the outputs there is a response delta on the inputs (Definition 6.1); the response may depend on the challenge *and on the anchor* — flagged hard, as the seed of the third crown. Five matches: the identity ($\delta = \varepsilon$); constants (any response; the skeptic resigns); the line of slope three ($\delta = \varepsilon/3$, one formula for every anchor — filed); the parabola (the handcuff from Episode Four's product law: pin the co-factor, then shrink — response depends on the anchor, a fact about steepness, not bookkeeping); the step function prosecuted (challenge one half, unanswerable at the jump — Remark 6.2 holds the negation's standing shape). Housekeeping: responses are downward closed. Curiosity: every function is continuous at a hermit of its domain — vacuous compliance.

Three Portraits of the Same Notion. The game; the sequences; the pull-back. Second portrait (Theorem 6.3): continuity means *respecting arrivals* — and, said honestly, it is a *swap permit*: the license to move a function past a limit (Remark 6.6), the swap account's first lawful instrument. Fastest conviction on record: the step function, sequentially, in one line. Third portrait (Theorem 6.4): continuous exactly when preimages of open sets are open — verifiability pulls back; the definition in travelling clothes, valid in every mathematical country. Calibrations: an interval pulled back through the parabola returns two beads; through the step function, a set closed at zero — three instruments convicting one criminal.

The Algebra of the Continuous. Sums, products, quotients off zeros, compositions — one line each through the sequential portrait and Episode Four's limit laws (Theorems 6.7, 6.8); polynomials and rational functions law-abiding (Corollary 6.9); the absolute value uniformly continuous outright, and larger-and-smaller continuous by the schoolroom identity (Proposition 6.10). The running theme installed: tonight pays with money already banked.

The First Crown: Extremes Attained. Master lemma: continuous images of compact sets are compact (Theorem 6.11) — one breath of the sequential road. Snapped to last week's staffed supremum: the extreme value theorem (Theorem 6.12), Weierstrass, Berlin, 1861 — the skeleton shown a week early, now dressed in a function. Attained beats approached: a maximum has an address; a supremum unattained is a rumour. Hypothesis tour (Remark 6.13): the leak, the escape, and the pierced reciprocal — every strut load-bearing; and the dust qualifies as a stage. No new ring: the crown runs on inherited machinery.

The Second Crown: No Skipping. On the rational line, $x^2 - 2$ crosses zero nowhere — the theorem is about continuity *on a line with no holes*. Two proofs (Theorem 6.15): the aristocrat's — images of connected sets are connected (Theorem 6.14), so no tearing, no skipping; and the engineer's — Bolzano's 1817 bisection, the hunt's contract executed with a venue transfer duly noted (extremes to intermediates), error halving per test: binary search for roots, fifty-three halvings to machine precision (Remark 6.18). Harvest: odd-degree roots (Problem II.3); a third construction of root two; the range corollary (Corollary 6.16); and the fixed-point corollary (Corollary 6.17) — every continuous self-map of the unit interval holds a point still, ancestor of the theorems that run half of economics.

The Third Crown: Uniformity, or the Quantifier That Moves. Pointwise: the anchor is named before the response. Uniform: the response is printed before the anchor is named (Definition 6.19; the side-by-side strings live in the Notation Ledger — one quantifier moved, everything hanging on it). Failures: the parabola's walk to steeper ground; the reciprocal's cliff (Remark 6.20, certified in Problem II.1). Positive contrast: capped steepness buys uniformity — lines outright; Chapter 7 arms the theme. Heine–Cantor (Theorem 6.22): on compact ground, continuity is automatically uniform — Heine's 1872 name cashing out; the blanket proof in full, half-radius blankets, the finite audit's minimum of finitely many positives, the halving's fee annotated at the line it is earned. Dividends: huddle transport (Proposition 6.21) — uniformity carries Cauchy to Cauchy, mere continuity does not (the reciprocal sends a huddle marching off); and the grid-table wink — Episode Eight's integral will present a uniform-continuity certificate at the door.

The Monster Gallery Opens: Pure Static. The Dirichlet function born (Definition 6.24) — one on the rationals, zero off them; christened *the static*; proposed in Dirichlet's 1829 Fourier memoir as a function beyond the era's integral, a character entering with its own third-act contract. Continuous nowhere (Proposition 6.25): both fences run through every neighbourhood; challenge one half, unanswerable everywhere; sequentially, no arrival respected. Not a curve that misbehaves — noise with a domain: two dense sheets no plotter renders. Restriction repairs nothing; one minus the static is the photographic negative. Schedule read into the register: defeats Riemann in Episode Eight; tamed in Season Two.

Thomae, or the Impossible Function. The wonder. Property stated before definition: continuous at every irrational, discontinuous at every rational. Definition (Definition 6.26): loudness one over the denominator at each lowest-terms rational; silence off the rationals; lowest terms load-bearing; the ruler engraving. Discontinuity at rationals: a definite loudness against a dense silent crowd. Continuity at irrationals: loud rationals are *rare* — finitely many per window (the counting lemma), the anchor not among them, so the nearest keeps a positive distance, and that distance is the response (Theorem 6.27; Problem II.5, guided and solved, with the framed corollary: fractions converging to an irrational send their denominators to infinity). The reveal: continuity is a property of tolerances, not of curves, and tolerances can be satisfied by rarity as well as by smoothness — the definition now strictly stronger than the pen. Thomae, 1875, in a textbook. Forward contrast filed: at the integral, the static will fail and Thomae will pass — nearly the plot of Season Two.

The Asymmetry. Thomae's continuity set is the irrationals; the mirror — a function continuous exactly on the rationals — does not exist, anywhere in the totality of functions (Remark 6.28). The irrationals are zoned for continuity; the rationals are condemned. The proof needs the anatomy of continuity sets (exactly the countable intersections of open sets — a species Season Two christens) plus Baire: the rationals' countability supplies the films, meagreness plus Baire the verdict. Debt issued with ceremony: payable Season Two, Episode One — Monsieur Baire's second engagement, now formal; last week's loading was never decorative. Inventory opened: continuity sets exhibited so far are everything, nothing, the irrationals, and (starred problem) a single point; Season Two completes the census.

The toolkit acquired. The definition and its negation (Definition 6.1, Remark 6.2); the sequential and topological portraits with the relative-domain annotation (Theorems 6.3, 6.4, Remark 6.5) and the swap permit (Remark 6.6); the algebra, composition, polynomials, absolute value, larger-and-smaller (Theorems 6.7, 6.8, Corollary 6.9, Proposition 6.10); compact images and the extreme value theorem with its sharpness tour (Theorems 6.11, 6.12, Remark 6.13); connected images, the intermediate value theorem twice over, the range and fixed-point corollaries, the algorithm remark (Theorems 6.14, 6.15, Corollaries 6.16, 6.17, Remark 6.18); uniform continuity, its two canonical failures, huddle transport, and Heine–Cantor with the rival proof staged (Definition 6.19, Remark 6.20, Proposition 6.21, Theorem 6.22, Remark 6.23); the static and Thomae with the asymmetry (Definitions 6.24, 6.26, Proposition 6.25, Theorem 6.27, Remark 6.28). Supplied solutions: the uniform failures (II.1) and Thomae in full (II.5); the extension theorem and the rival Heine–Cantor are guided, deliberately unsolved.

Debts. **Born: the static** — act one of three; schedule confirmed: defeats Riemann in Episode Eight (Dirichlet's own 1829 booking), tamed in Season Two, Episode Four. **Issued: the asymmetry** — no function continuous exactly on the rationals; payable Season Two, Episode One; Baire's second engagement hereby formalized. **Executed: the bisection hunt's final Season One contract** — at the intermediate values rather than the extremes, venue transfer noted in the file; the hunt retires three for three. **The swap account** (Chapter 4) records its first lawful instrument: continuity is the swap permit; the first scheduled instalment (uniform convergence) falls due next chapter. **D7, the bell: silent** — no new rings; the count holds at eleven; all three crowns run on inherited machinery, and the register appreciates compound interest. Heine's name: cashed. Filed forward: the Thomae-versus-static divergence at the integral; the continuity-set inventory awaiting Season Two's census.

Dates worth keeping. 1806, Ampère's false proof that continuity implies differentiability — teased for Episode Seven; 1817, Bolzano's Prague paper: the intermediate value theorem by bisection, mourned in Episode One and vindicated tonight; 1821, Cauchy's *Cours*: continuity defined, rigour arriving in instalments; 1829, Dirichlet's Fourier memoir: the static born with its third-act contract; 1861, Weierstrass's Berlin lectures: the epsilon–delta apparatus and the extreme value theorem; 1872, Heine publishes the apparatus and the uniform-continuity theorem — the vintage year's third appearance in this course, with Cantor's name appended by custom; 1875, Thomae houses the finest monster ever printed in a syllabus.

6.1 Notation Ledger

Spoken in the episode	Written here	Remarks
"f of x"; a function on a set	$f(x);\ f : A \to \mathbb{R}$	domain declared
"the challenge–response guarantee"	$\lvert x-a\rvert < \delta \ \Rightarrow\ \lvert f(x)-f(a)\rvert < \varepsilon$	Definition 6.1
"respects arrivals"	$x_n \to a \ \Rightarrow\ f(x_n) \to f(a)$	Theorem 6.3
"the pull-back"	$f^{-1}(U) = \{\, x \in A : f(x) \in U \,\}$	warning below
"f after g"; composition	$(f \circ g)(x) = f(g(x))$	Theorem 6.8
"pointwise" / "uniform"	the two quantifier strings	displayed below
"attains its maximum"	$f(x^*) = \max_{x\in K} f(x)$ for some $x^* \in K$	Theorem 6.12

Warning (a standing hazard). — $f^{-1}(U)$ names the *pull-back*: the set of inputs whose outputs land in U. It is a set operation, defined for every function whether or not f is invertible, and it is never to be read as a reciprocal and never as an inverse function's value. The notation is unfortunate and permanent; this box is the vaccination.

The moved quantifier (the ledger's centrepiece). — Continuity on a set and uniform continuity on a set, side by side, in full dress:

$$\begin{aligned}\text{pointwise on } A:&\quad \forall a \in A\ \ \forall \varepsilon > 0\ \ \exists \delta > 0\ \ \forall x \in A:\ \lvert x-a\rvert < \delta \Rightarrow \lvert f(x)-f(a)\rvert < \varepsilon;\\ \text{uniform on } A:&\quad \forall \varepsilon > 0\ \ \exists \delta > 0\ \ \forall a \in A\ \ \forall x \in A:\ \lvert x-a\rvert < \delta \Rightarrow \lvert f(x)-f(a)\rvert < \varepsilon.\end{aligned}$$

The strings differ in exactly one respect: the position of $\forall a$ relative to $\exists\delta$. In the first, the anchor is named before the response, which may therefore be tailored to it; in the second, the response is issued before the anchor is named, and must serve every anchor at once. Everything in Section 6.6 hangs on that transposition.

6.2 The Definition and Three Portraits

Definition 6.1 (Continuity at a point; continuity on a set)**.** Let $f : A \to \mathbb{R}$ and let $a \in A$. Then f is *continuous at a* if

$$\forall\varepsilon > 0\, \exists\delta > 0\, \forall x \in A : \quad |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

f is *continuous on A* if it is continuous at every point of A. Game gloss: the skeptic challenges with a tolerance ε on the *outputs*; we respond with a tolerance δ on the *inputs*; the response may depend on the challenge and on the anchor a — a dependence that is a fact about the function, not a defect of the bookkeeping, and which Section 6.6 promotes to a leading role. Housekeeping, used silently hereafter: responses are downward closed — any $\delta' < \delta$ that wins, wins.

Remark 6.2 (The negation; hermits). f fails to be continuous at a precisely when

$$\exists\varepsilon > 0\, \forall\delta > 0\, \exists x \in A : \quad |x - a| < \delta \text{ and } |f(x) - f(a)| \geq \varepsilon$$

— one fixed challenge, unanswerable by every response: the standing shape of prosecution. And a curiosity with Chapter 5's fingerprints: if a is an *isolated point* of A (Definition 5.6), choose δ so small that the only point of A within δ of a is a itself; the guarantee is then vacuously satisfied, since $|f(a) - f(a)| = 0 < \varepsilon$. Every function is continuous at every hermit of its domain. Regulations beginning "for every" are satisfied most easily by those with nothing to regulate.

Theorem 6.3 (Sequential characterization)**.** *Let $f : A \to \mathbb{R}$ and $a \in A$. The following are equivalent:*

(i) f is continuous at a;

(ii) for every sequence (x_n) in A with $x_n \to a$, one has $f(x_n) \to f(a)$.

Proof. (i) $\Rightarrow$ (ii). Let $x_n \to a$ and let $\varepsilon > 0$. Continuity supplies $\delta > 0$ with $|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$; convergence supplies N with $|x_n - a| < \delta$ for all $n \geq N$; compose: $|f(x_n) - f(a)| < \varepsilon$ for all $n \geq N$.

(ii) $\Rightarrow$ (i), by contrapositive. Suppose f is not continuous at a: some $\varepsilon > 0$ defeats every response. In particular it defeats $\delta = \frac{1}{n}$ for each n: there is $x_n \in A$ with $|x_n - a| < \frac{1}{n}$ and $|f(x_n) - f(a)| \geq \varepsilon$. Then $x_n \to a$ (squeeze, Theorem 4.7, with the Archimedean property, Theorem 2.12), yet $f(x_n) \not\to f(a)$. So (ii) fails. □

Theorem 6.4 (Topological characterization, whole-line case)**.** *Let $f : \mathbb{R} \to \mathbb{R}$. The following are equivalent:*

(i) f is continuous on $\mathbb{R}$;

(ii) $f^{-1}(O)$ is open for every open $O \subseteq \mathbb{R}$.

Proof. (i) $\Rightarrow$ (ii). Let O be open and $x \in f^{-1}(O)$. Since $f(x) \in O$ and O is open, some $N_\varepsilon(f(x)) \subseteq O$. Continuity at x supplies $\delta > 0$ with $f\big(N_\delta(x)\big) \subseteq N_\varepsilon(f(x)) \subseteq O$, that is, $N_\delta(x) \subseteq f^{-1}(O)$. Every point of $f^{-1}(O)$ owns a buffer: $f^{-1}(O)$ is open (Definition 5.1).

(ii) $\Rightarrow$ (i). Let $x \in \mathbb{R}$ and $\varepsilon > 0$. The set $O = N_\varepsilon(f(x))$ is open, so $f^{-1}(O)$ is open and contains x; hence some $N_\delta(x) \subseteq f^{-1}(O)$, which says exactly: $|y - x| < \delta \Rightarrow |f(y) - f(x)| < \varepsilon$. $\square$

Remark 6.5 (Domains smaller than the line). For $f : A \to \mathbb{R}$ with $A \subsetneq \mathbb{R}$, the characterization holds verbatim once "open" on the input side is read *relatively*: a set of the form $U \cap A$ with U open in $\mathbb{R}$. The proof of Theorem 6.4 transcribes line for line with relative neighbourhoods $N_\delta(x) \cap A$. This chapter runs the whole-line case in full and flags each relative use honestly; Problem II.2 develops the details.

Remark 6.6 (The swap permit). Theorem 6.3(ii) reads, in one line: $f\big(\lim x_n\big) = \lim f(x_n)$ whenever the inner limit exists in A. Continuity is precisely the license to move a function past a limit. The swap account opened in Chapter 4 herewith records its first *lawful* instrument; the illegal swap of the same shape — a limit *of* continuous functions, moved without license — is Cauchy's 1821 error, and its autopsy is Chapter 7 business, where the account's first scheduled instalment (uniform convergence) falls due.

6.3 The Algebra of the Continuous

Theorem 6.7 (Algebra of continuity). *Let $f, g : A \to \mathbb{R}$ be continuous at $a \in A$. Then $f + g$ and fg are continuous at a; and if $g(a) \neq 0$, then f/g is continuous at a (and is defined on a relative neighbourhood of a).*

Proof. Sequential portrait throughout. If $x_n \to a$ in A, then $f(x_n) \to f(a)$ and $g(x_n) \to g(a)$ (Theorem 6.3), whence $f(x_n) + g(x_n) \to f(a) + g(a)$ and $f(x_n)\, g(x_n) \to f(a)\, g(a)$ by the algebra of limits (Theorem 4.5); apply Theorem 6.3 again. For the quotient: continuity of g at a with the challenge $\varepsilon = \frac{1}{2}|g(a)|$ supplies $\delta > 0$ with $|g(x)| > \frac{1}{2}|g(a)| > 0$ for $x \in A$, $|x - a| < \delta$; on that relative neighbourhood f/g is defined, and the quotient law of Theorem 4.5 finishes as above. $\square$

Theorem 6.8 (Composition). *If $g : A \to B$ is continuous at a and $f : B \to \mathbb{R}$ is continuous at $g(a)$, then $f \circ g$ is continuous at a.*

Proof. Let $x_n \to a$ in A. Continuity of g at a gives $g(x_n) \to g(a)$ in B; continuity of f at $g(a)$ gives $f(g(x_n)) \to f(g(a))$. That is (ii) of Theorem 6.3 for $f \circ g$. $\square$

Corollary 6.9 (The functions of daily commerce). *Every polynomial is continuous on $\mathbb{R}$; every rational function is continuous at every point where its denominator is nonzero; and finite chains of sums, products, quotients, and compositions of continuous functions are continuous wherever defined.*

Proof. The identity ($\delta = \varepsilon$) and the constants (any δ) are continuous; induct with Theorem 6.7 on the degree, then close under Theorem 6.8. $\square$

Proposition 6.10 (Absolute value; larger and smaller). *The absolute value is uniformly continuous on $\mathbb{R}$: the response $\delta = \varepsilon$ serves every anchor. Consequently, if f, g are continuous at a, so are* $\max(f, g)$ *and* $\min(f, g)$.

Proof. The reverse triangle inequality gives $\big|\,|x| - |y|\,\big| \le |x - y|$: the output error never exceeds the input error, so $\delta = \varepsilon$ wins at every anchor simultaneously (this is the chapter's first sighting of a *uniform* response; Definition 6.19 makes the notion official). For the second claim, verify by cases the schoolroom identities

$$\max(f, g) = \tfrac{1}{2}\big(f + g + |f - g|\big), \qquad \min(f, g) = \tfrac{1}{2}\big(f + g - |f - g|\big),$$

and assemble: differences, absolute values, sums, and halvings are all licensed by Theorem 6.7 and Theorem 6.8. □

6.4 The First Crown: Extreme Values

Theorem 6.11 (Continuous images of compact sets are compact)**.** *Let $K \subseteq \mathbb{R}$ be compact and nonempty, and let $f : K \to \mathbb{R}$ be continuous. Then $f(K)$ is compact.*

Proof. Sequential road. Let (y_n) be any sequence in $f(K)$; choose $x_n \in K$ with $f(x_n) = y_n$. K is sequentially compact (Corollary 5.19 with Theorem 5.21; the cycle, Remark 5.23), so some subsequence $x_{n_k} \to x \in K$. Continuity in its sequential portrait (Theorem 6.3) pushes the arrival forward: $y_{n_k} = f(x_{n_k}) \to f(x) \in f(K)$. Every sequence in $f(K)$ has a subsequence converging within $f(K)$: $f(K)$ is sequentially compact, hence compact (Remark 5.23). □

Theorem 6.12 (Extreme value theorem; Weierstrass)**.** *Let $K \subseteq \mathbb{R}$ be compact and nonempty, and let $f : K \to \mathbb{R}$ be continuous. Then f attains a maximum and a minimum on K: there exist $x^*, x_* \in K$ with*

$$f(x_*) \;\le\; f(x) \;\le\; f(x^*) \qquad \textit{for all } x \in K.$$

Proof. $f(K)$ is compact and nonempty (Theorem 6.11), so it contains a largest member and a smallest (Proposition 5.16 and its mirror). A largest member of $f(K)$ is $f(x^*)$ for some $x^* \in K$; likewise the smallest. The skeleton of Chapter 5, dressed in a function. □

Remark 6.13 (Every strut load-bearing). Drop closedness: the identity on $(0, 1)$ is continuous and bounded with supremum 1, attained by no point — the maximum leaks through the absent endpoint. Drop boundedness: $x \mapsto x/(1 + |x|)$ on $\mathbb{R}$ is continuous with supremum 1, attained nowhere — the escape to the horizon. Drop continuity, keeping the compact stage: pierce the reciprocal onto $[0, 1]$ by $f(x) = 1/x$ for $x \in (0, 1]$ and $f(0)$ anything you please; unbounded regardless. And an inventory note: the theorem speaks of compact sets, not intervals — a continuous function on the dust attains its extremes (Theorem 5.31); monstrosity and compactness are independent portfolios. Problem I.8 runs the audit across the menagerie.

6.5 The Second Crown: Intermediate Values

Theorem 6.14 (Continuous images of connected sets are connected)**.** *Let f be continuous and let C be a connected subset of its domain. Then $f(C)$ is connected. (Proof below for $f : \mathbb{R} \to \mathbb{R}$; the general case is the same argument in relative dress, Remark 6.5 and Problem II.2.)*

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and suppose $f(C)$ were torn (Definition 5.24): open sets U, V with $f(C) \subseteq U \cup V$, both $U \cap f(C)$ and $V \cap f(C)$ nonempty, and $U \cap V \cap f(C) = \emptyset$. Pull back: $f^{-1}(U)$ and $f^{-1}(V)$ are open (Theorem 6.4); they cover C; each meets C (a point of C

mapping into U lies in $f^{-1}(U)$); and they are disjoint on C, since a point of C in both would map into $U \cap V \cap f(C)$. So C is torn — contradicting connectedness. A tear in the image pulls back to a tear in the domain. □

Theorem 6.15 (Intermediate value theorem; Bolzano 1817)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous with $f(a) < 0 < f(b)$ (or the reverse). Then $f(c) = 0$ for some $c \in (a,b)$. More generally, f attains on $[a,b]$ every value between $f(a)$ and $f(b)$ (apply the first form to $f - w$).*

Proof. First proof (the aristocrat's). $[a,b]$ is connected (Theorem 5.25), so $f([a,b])$ is connected (Theorem 6.14, in relative dress per Remark 6.5), hence an interval (Theorem 5.25 again). An interval containing the negative number $f(a)$ and the positive number $f(b)$ contains 0, by the order-convexity that defines intervals. No tearing, no skipping.

Second proof (the engineer's: the bisection hunt). Set $[a_0, b_0] = [a,b]$, so $f(a_0) \le 0 \le f(b_0)$. Given $[a_n, b_n]$ with $f(a_n) \le 0 \le f(b_n)$, test the midpoint m_n: if $f(m_n) = 0$ the hunt ends early; if $f(m_n) > 0$ keep $[a_n, m_n]$; if $f(m_n) < 0$ keep $[m_n, b_n]$. The invariant survives every step, and the widths are $(b-a)/2^n \to 0$. The dolls close on a single survivor (Theorem 2.17): $a_n \nearrow c$ and $b_n \searrow c$. Now continuity closes the case in its sequential portrait (Theorem 6.3): $f(a_n) \to f(c)$ with every $f(a_n) \le 0$, so $f(c) \le 0$ by the order limit theorem (Theorem 4.6); and $f(b_n) \to f(c)$ with every $f(b_n) \ge 0$, so $f(c) \ge 0$. Hence $f(c) = 0$. *Error bound, recorded for the algorithmically minded:* $c \in [a_n, b_n]$ at every stage, so after n tests the root is pinned within $(b-a)/2^n$. □

Corollary 6.16 (The range of a continuous function on an interval)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous, with minimum m and maximum M (attained, by Theorem 6.12). Then $f([a,b]) = [m, M]$.*

Proof. $f([a,b]) \subseteq [m, M]$ by the definition of minimum and maximum; conversely, m and M are attained, and every value between two attained values is attained (Theorem 6.15, general form, applied on the subinterval between a minimizer and a maximizer). The two crowns snapped into one sentence. □

Corollary 6.17 (Fixed points on the unit interval)**.** *Every continuous $f : [0,1] \to [0,1]$ has a fixed point: some $c \in [0,1]$ with $f(c) = c$.*

Proof. Let $g(x) = f(x) - x$, continuous by Theorem 6.7. Since f maps into $[0,1]$: $g(0) = f(0) \ge 0$ and $g(1) = f(1) - 1 \le 0$. If either equals 0 we are done; otherwise $g(1) < 0 < g(0)$ and Theorem 6.15 plants a root c: $f(c) = c$. (The higher-dimensional cousins — Brouwer's theorem and its kin — run a good deal of mathematical economics; this is the one-dimensional ancestor, free with the crown.) □

Remark 6.18 (The hunt as an algorithm)*.* The second proof of Theorem 6.15 is binary search aimed at a root: one sign test per step, one bit of the answer per test, uncertainty $(b-a)/2^n$ after n steps. Since $2^{10} = 1024 > 10^3$ (the powers-of-two credit of Chapter 1), fifty steps yield $2^{-50} < 10^{-15}$, and 53 halvings pin a unit interval within $2^{-53} < 1.2 \times 10^{-16}$ — roughly where double-precision machine arithmetic itself gives out. Bolzano's 1817 paper, mourned in Episode One, thus contains the convergence analysis of a root-finder written a century and a half before anyone would pay for one.

6.6 The Third Crown: Uniform Continuity

Definition 6.19 (Uniform continuity)**.** $f : A \to \mathbb{R}$ is *uniformly continuous on A* if

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x, y \in A : \quad |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

One response, printed once, serving every anchor. The side-by-side contrast with pointwise continuity — one quantifier moved — is displayed in the Notation Ledger; it is the entire distinction, and it is the chapter's deepest single line.

Remark 6.20 (Two canonical failures). The parabola is continuous at every point of $\mathbb{R}$ and uniformly continuous on no unbounded neighbourhood of infinity: for a fixed δ, the swing $\left|(a + \frac{\delta}{2})^2 - a^2\right| = \left|a\delta + \frac{\delta^2}{4}\right|$ grows without bound in a, so no single response survives the skeptic's walk to steeper ground. And $1/x$ on $(0, 1)$ is continuous at every point, yet the huddled inputs $\frac{1}{n}, \frac{1}{n+1}$ — arbitrarily close — have outputs a full unit apart: violence concentrating at the absent endpoint. The escape and the leak, in uniform dress. Problem II.1 certifies both against Definition 6.19, solution supplied. For the positive contrast: affine functions are uniformly continuous on all of $\mathbb{R}$ ($\delta = \varepsilon/|\text{slope}|$, one formula for every anchor); capped steepness buys uniformity, a theme Chapter 7 arms properly with the mean value theorem.

Proposition 6.21 (Huddle transport)**.** *If $f : A \to \mathbb{R}$ is uniformly continuous and (x_n) is a Cauchy sequence in A, then $(f(x_n))$ is Cauchy.*

Proof. Given $\varepsilon > 0$, uniform continuity supplies $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ whenever $|x - y| < \delta$; the Cauchy property (Definition 4.15) supplies N with $|x_n - x_m| < \delta$ for $n, m \geq N$; compose. Mere pointwise continuity does no such thing: $1/x$ on $(0, 1)$ sends the Cauchy sequence $(\frac{1}{n})$ to (n), which marches off entirely. Chapter 8 will quietly spend this proposition (see also Problem II.6). □

Theorem 6.22 (Heine–Cantor)**.** *Let $K \subseteq \mathbb{R}$ be compact and $f : K \to \mathbb{R}$ continuous. Then f is uniformly continuous on K.*

Proof. Let $\varepsilon > 0$. For each $x \in K$, continuity at x supplies $\delta_x > 0$ with

$$y \in K, \ |y - x| < \delta_x \implies |f(y) - f(x)| < \tfrac{\varepsilon}{2}.$$

The *half-radius* blankets $\{ N_{\delta_x/2}(x) : x \in K \}$ form an open cover of K; compactness (Definition 5.11, via Corollary 5.19 when K is presented as closed and bounded) extracts a finite subcover, centred at $x_1, \ldots, x_m$. Set

$$\delta \; = \; \min_{1 \leq i \leq m} \frac{\delta_{x_i}}{2} \; > \; 0,$$

a minimum of *finitely many* positive numbers — the finite audit's entire contribution, and the theorem's entire engine; over infinitely many blankets this minimum drains to zero, and does, on the parabola's endless line. Now take $y, z \in K$ with $|y - z| < \delta$. The point y lies in some $N_{\delta_{x_i}/2}(x_i)$; then

$$|z - x_i| \; \leq \; |z - y| + |y - x_i| \; < \; \delta + \frac{\delta_{x_i}}{2} \; \leq \; \frac{\delta_{x_i}}{2} + \frac{\delta_{x_i}}{2} \; = \; \delta_{x_i},$$

so *both* y and z lie within δ_{x_i} of x_i, and

$$|f(y) - f(z)| \;\le\; |f(y) - f(x_i)| + |f(x_i) - f(z)| \;<\; \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} \;=\; \varepsilon.$$

(The halving's fee, annotated: without the half-radii, two inputs within δ of each other could straddle two different blankets and compare their outputs to two different anchors; the shrinkage forces both into one anchor's *full* blanket, where the triangle inequality closes the audit.*)* One δ, printed once, serving every anchor: uniform continuity. □

Remark 6.23 (The rival proof). Connoisseurs keep a second proof, by contradiction and a hunted subsequence: if uniformity fails at some ε, there are points $x_n, y_n \in K$ with $|x_n - y_n| < \frac{1}{n}$ yet $|f(x_n) - f(y_n)| \ge \varepsilon$; sequential compactness extracts $x_{n_k} \to x \in K$, drags y_{n_k} to the same x, and continuity then forces both image sequences to $f(x)$ — contradicting the standing gap. Problem II.4 stages the details, hints only.

6.7 The Monster Gallery

Definition 6.24 (The Dirichlet function; "the static")**.**

$$D(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases}$$

the characteristic function of the rationals. Proposed by Dirichlet in 1829, inside his memoir on the convergence of Fourier series, as an example beyond the reach of his era's integral — a character born with its own third-act contract. Episode name, entered in the register: *the static*.

Proposition 6.25 (The static is continuous nowhere)**.** *D is continuous at no point of $\mathbb{R}$.*

Proof. Fix any $a \in \mathbb{R}$ and take the single challenge $\varepsilon = \frac{1}{2}$. Both fences run through every neighbourhood: for every $\delta > 0$, $N_\delta(a)$ contains a rational (Theorem 2.15) and an irrational (Corollary 2.16) — hence a point x with $D(x) \neq D(a)$, so $|D(x) - D(a)| = 1 > \varepsilon$. One challenge, unanswerable at every point of the line. (Sequentially: rational approaches force limit 1, irrational approaches force limit 0; no arrival is respected.) Restriction repairs nothing — both fences are dense in every interval — and $1 - D$, the characteristic function of the irrationals, is convicted by the identical argument: the gallery's photographic negative. Problem I.4 drills all of it. □

Definition 6.26 (Thomae's function)**.**

$$T(x) = \begin{cases} \dfrac{1}{q}, & x = \dfrac{p}{q} \in \mathbb{Q} \text{ in lowest terms, } q \ge 1, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Integers are $n/1$ in lowest terms, so $T(n) = 1$. The reduction to lowest terms is load-bearing: a definition must issue one value per number, so the fraction is reduced before the denominator is read. Episode gloss: each rational chirps at *loudness* $1/q$ — crude fractions loud, refined fractions quiet, irrationals silent. Plot the loudnesses over $(0, 1)$ and something like a ruler's engraving appears (the trade's nickname: the ruler function); Thomae published the creature in 1875, in a textbook.

Theorem 6.27 (Thomae's theorem)**.** *T is discontinuous at every rational and continuous at every irrational: its set of continuity points is exactly $\mathbb{R} \setminus \mathbb{Q}$.*

Proof strategy. Discontinuity at p/q: the value there is $1/q > 0$, while silent irrationals crowd every neighbourhood; the challenge $\varepsilon = \frac{1}{2q}$ convicts. Continuity at an irrational α: given ε, the rationals *loud enough* to violate it — denominator at most $1/\varepsilon$ — are finitely many in any bounded window around α (the counting lemma: finitely many picket fences, finitely many posts each), and α is not one of them, so the nearest loud rational keeps a positive distance m; the response $\delta = \min(1, m)$ admits only quiet rationals and silent irrationals, all within ε of $T(\alpha) = 0$. The full paperwork, in four staged parts including the framed corollary — fractions converging to an irrational must send their denominators to infinity — is Problem II.5, guided and solved. □

Remark 6.28 (The asymmetry)*.* Thomae's continuity set is $\mathbb{R} \setminus \mathbb{Q}$. The mirror demand — a function continuous at every rational and discontinuous at every irrational — is unsatisfiable: *no function whatsoever* $f : \mathbb{R} \to \mathbb{R}$ has $\mathbb{Q}$ as its set of continuity points. The proof requires the anatomy of continuity sets in general: such sets are exactly the countable intersections of open sets (a species Season Two christens on its first evening), and $\mathbb{Q}$ is not one — the conviction runs through Chapter 5's meagreness, with Baire (Theorem 5.28) supplying the verdict and countability supplying merely the stack of films. Inventory of continuity sets exhibited so far: $\mathbb{R}$ (any polynomial), $\varnothing$ (the static), $\mathbb{R} \setminus \mathbb{Q}$ (Thomae), and a single point (Problem III.1 $\star$); which sets can occur at all is the census Season Two completes.

Debt issued. — No function is continuous exactly on the rationals. Asserted here, with the claim, the schedule, and the collateral on record: payable Season Two, Episode One; paying instruments, the anatomy of continuity sets and Theorem 5.28, whose *second engagement* this formally becomes (the first falls due in Chapter 7, where the monsters are shown to be typical).

6.8 Problems

Tier I — Calisthenics

I.1 Manufacture explicit responses: for $f(x) = 3x + 1$ and $g(x) = 7 - 2x$, given $\varepsilon > 0$ and an arbitrary anchor a, exhibit winning δ's and verify them. Note that neither depends on the anchor — the chapter's first two uniformly continuous specimens after the absolute value.

I.2 The parabola at the anchor $a = 3$: run the handcuff in full — restrict $\delta \leq 1$, bound the co-factor $|x + 3|$, and produce $\delta = \min(1, \varepsilon/7)$, with verification.

I.3 Prosecute the step function H (0 for $x < 0$, 1 for $x \geq 0$) at the anchor 0 in full quantifier dress, challenge $\varepsilon = \frac{1}{2}$; then prove H is continuous at every $a \neq 0$ by exhibiting responses.

I.4 Sequential drills: convict H at 0 via the arrival $x_n = -\frac{1}{n}$; convict the static at an arbitrary anchor via both fences; convict $1 - D$ everywhere; show the restriction of D to any interval is continuous nowhere on it.

I.5 Preimage computations, with verdicts on openness: $(x^2)^{-1}\big((1, 4)\big)$; $(x^2)^{-1}\big((-1, 1)\big)$; $H^{-1}\big((\frac{1}{2}, \frac{3}{2})\big)$. Use the third computation to convict H by the third portrait (Theorem 6.4).

I.6 Assembly-line certification, each licensed step named: $|x^2 - 2|$ is continuous on $\mathbb{R}$; $1/(1+x^2)$ is continuous on $\mathbb{R}$.

I.7 Classify the continuity points: $g(x) = x^2$ for $x \le 1$, $2x - 1$ for $x > 1$; and $h(x) = x^2$ for $x \le 1$, $2x$ for $x > 1$.

I.8 The extreme-value audit across the menagerie: for each of $[0,1]$, $(0,1)$, $[0,1)$, a finite set, $\{0\} \cup \{\frac{1}{n}\}$, the dust, $\mathbb{Q} \cap [0,2]$, and $\mathbb{R}$ — does every continuous function on it attain a maximum? One-line verdict and reason each; for $\mathbb{Q} \cap [0,2]$, exhibit the standard counterexample (define $f(x) = x$ where $x^2 < 2$ and $f(x) = 0$ where $x^2 > 2$; explain, citing Remark 6.5, why this is continuous on the rational interval and why its supremum is unattained).

Tier II — The Real Thing

II.1 Certify both canonical failures against Definition 6.19: the parabola is not uniformly continuous on $\mathbb{R}$; the reciprocal is not uniformly continuous on $(0,1)$. *Solution supplied.*

II.2 The relative dress: (a) state and prove the topological characterization for $f : A \to \mathbb{R}$ with relatively open sets (Remark 6.5); (b) give the two-line proof of Theorem 6.14 for $f : \mathbb{R} \to \mathbb{R}$ directly from Theorem 6.4; (c) verify that the aristocrat's proof of Theorem 6.15 survives in the relative setting.

II.3 Every polynomial of odd degree has a real root. (Make "far out, the leading term rules" honest with a taming inequality, then apply Theorem 6.15.)

II.4 The rival proof of Heine–Cantor (Remark 6.23), in full paperwork. *Hints only.*

II.5 Thomae in full, staged: (a) the conviction at every rational; (b) the counting lemma — for every $Q \in \mathbb{N}$ and every bounded window, the rationals with denominator at most Q (lowest terms) in the window are finitely many; (c) the continuity proof at every irrational; (d) the framed corollary — if $p_n/q_n \to \alpha$ with α irrational (lowest terms), then $q_n \to \infty$. *Guided and solved.*

II.6 The extension theorem: if f is uniformly continuous on (a,b), then f extends to a continuous function on $[a,b]$. (Chapter 8 will quietly spend this.) *Hints only.*

Tier III — For the Ambitious $\star$

III.1 $\star$ The function $g(x) = x\,D(x)$ — one multiplication applied to the static — is continuous at exactly one point of the line. Identify the point and prove both halves.

III.2 $\star$ Every monotone function $f : [a,b] \to \mathbb{R}$ has at most countably many discontinuities. (Episode Two's picket fence and Episode Three's counting, cooperating; Chapter 8 will spend the consequence.)

6.9 Hints

II.2. (a) Transcribe Theorem 6.4 with $N_\delta(x) \cap A$ throughout; nothing else changes. (b) A tear U, V of $f(C)$ pulls back to open sets $f^{-1}(U)$, $f^{-1}(V)$ covering C, each meeting it, disjoint on it.

(c) In the aristocrat's proof the only pull-backs taken are through a function on $[a,b]$; run them through (a).

II.3. Write $p(x) = a_n x^n\big(1 + r(x)\big)$ where r collects the lower terms divided by $a_n x^n$. For $|x| \geq R := \max\big(1,\ 2\sum_{k<n} |a_k/a_n|\big)$, each $|x|^{k-n} \leq 1/|x|$, so $|r(x)| \leq \frac{1}{2}$; hence the sign of $p(\pm R)$ is the sign of $a_n(\pm R)^n$, and for odd n the two signs differ. Apply Theorem 6.15 on $[-R, R]$.

II.4. Negate Definition 6.19; the failing ε defeats each $\delta = \frac{1}{n}$, yielding x_n, y_n. Extract $x_{n_k} \to x \in K$ (Theorem 5.21); the estimate $|y_{n_k} - x| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - x|$ drags y_{n_k} to the same x; apply Theorem 6.3 twice and contradict the standing gap.

II.6. For any sequence $t_n \downarrow a$ in (a,b): (t_n) is Cauchy, so $\big(f(t_n)\big)$ is Cauchy (Proposition 6.21), so it converges (Theorem 4.18). Independence of the sequence: interlace two candidates and note the interlaced image sequence must still converge. Define $F(a)$ as the common limit, $F(b)$ likewise, $F = f$ inside; for continuity of F at a, pass to the limit in the uniform estimate for $\varepsilon/2$.

III.1. At 0: $|g(x)| \leq |x|$; squeeze (Theorem 4.7). At $a \neq 0$: rational arrivals give image limit a; irrational arrivals give 0; the two disagree, so Theorem 6.3 convicts.

III.2. Take f increasing. For interior x, the one-sided limits $f(x-) = \sup_{t<x} f(t)$ and $f(x+) = \inf_{t>x} f(t)$ exist (completeness — small change, uncounted), and continuity at x is exactly $f(x-) = f(x+)$. To each discontinuity assign the open *jump interval* $\big(f(x-),\ f(x+)\big)$; monotonicity makes the intervals of distinct discontinuities disjoint (across any t between $x < y$: $f(x+) \leq f(t) \leq f(y-)$). Choose one rational in each (Theorem 2.15); the assignment injects the discontinuity set into $\mathbb{Q}$; conclude with Theorem 3.7.

6.10 Solutions

Tier I (terse). I.1 $|f(x) - f(a)| = 3|x-a|$: take $\delta = \varepsilon/3$; $|g(x) - g(a)| = 2|x-a|$: take $\delta = \varepsilon/2$. No anchor appears in either formula. **I.2** $|x^2 - 9| = |x-3|\,|x+3|$; if $|x-3| \leq 1$ then $x \in [2,4]$ so $|x+3| \leq 7$; hence $\delta = \min(1, \varepsilon/7)$ gives $|x^2-9| \leq 7|x-3| < \varepsilon$. **I.3** At 0, challenge $\frac{1}{2}$: for any δ, the input $x = -\delta/2$ has $|H(x) - H(0)| = 1 \geq \frac{1}{2}$. At $a > 0$: $\delta = a$ keeps all inputs positive, outputs constantly 1; at $a < 0$: $\delta = |a|$, outputs constantly 0. **I.4** $H(-\frac{1}{n}) = 0 \to 0 \neq 1 = H(0)$. Static at a: rationals $r_n \to a$ give $D(r_n) \to 1$, irrationals $s_n \to a$ give $D(s_n) \to 0$; whatever $D(a)$ is, one arrival is disrespected. $1 - D$: same fences, values swapped. Restriction: both density theorems hold inside every interval, so the argument transcribes. **I.5** $(x^2)^{-1}\big((1,4)\big) = (-2,-1) \cup (1,2)$, open — two beads on the string (Proposition 5.3); $(x^2)^{-1}\big((-1,1)\big) = (-1,1)$, open; $H^{-1}\big((\frac{1}{2}, \frac{3}{2})\big) = [0,\infty)$, not open (no buffer at 0), so H is not continuous, by Theorem 6.4. **I.6** x^2 (identity times identity, Theorem 6.7); minus the constant 2 (sum law); absolute value (Proposition 6.10) composed (Theorem 6.8). And $1 + x^2$ is a nonvanishing polynomial (Corollary 6.9), so the quotient law applies. **I.7** g: both branches polynomial; at the seam, $g(1) = 1$ and the right-hand values $2x - 1 \to 1$: every arrival respected — continuous on $\mathbb{R}$. h: left value 1, right-hand values $\to 2$: the arrival $1 + \frac{1}{n}$ is disrespected — continuous exactly on $\mathbb{R} \setminus \{1\}$. **I.8** $[0,1]$: yes (Theorem 6.12). $(0,1)$: no (the identity leaks). $[0,1)$: no (identity again). Finite set: yes — every function on a finite set has a largest value. $\{0\} \cup \{\frac{1}{n}\}$:

yes (compact, Proposition 5.13). The dust: yes (compact, Theorem 5.31). $\mathbb{Q} \cap [0,2]$: no — the exhibited f has both branches relatively open in the rational interval (no rational sits at $x^2 = 2$, Theorem 2.13 territory), values climbing toward $\sqrt{2}$ on rationals below it, supremum $\sqrt{2}$ never attained (indeed never attainable: f takes only rational values). $\mathbb{R}$: no (the escape $x/(1+|x|)$).

II.1 (solution in full). *The parabola on* $\mathbb{R}$. Negation target: some ε defeats every δ. Take $\varepsilon = 1$. Given $\delta > 0$, set $a = 1/\delta$ and $x = a + \delta/2$. Then $|x - a| = \delta/2 < \delta$, while

$$|x^2 - a^2| = |x-a|\,|x+a| = \frac{\delta}{2}\Big(2a + \frac{\delta}{2}\Big) = a\delta + \frac{\delta^2}{4} = 1 + \frac{\delta^2}{4} > 1 = \varepsilon.$$

No response survives; the parabola is not uniformly continuous on $\mathbb{R}$ (nor on any unbounded set, the same walk rightward). *The reciprocal on* $(0,1)$. Take $\varepsilon = 1$. Given $\delta > 0$, the Archimedean property (Theorem 2.12) supplies n with $\frac{1}{n(n+1)} < \delta$; the inputs $x = \frac{1}{n}$, $y = \frac{1}{n+1}$ satisfy $|x - y| = \frac{1}{n(n+1)} < \delta$ yet $\left|\frac{1}{x} - \frac{1}{y}\right| = |n - (n+1)| = 1 \geq \varepsilon$. Violence at the absent endpoint; no response survives.

II.5 (solution in full). *(a) Discontinuity at every rational.* At p/q (lowest terms) the value is $\frac{1}{q}$. Every $N_\delta(p/q)$ contains an irrational x (Corollary 2.16), where $T(x) = 0$, so $|T(x) - T(p/q)| = \frac{1}{q} > \frac{1}{2q}$. The challenge $\varepsilon = \frac{1}{2q}$ is unanswerable.

(b) The counting lemma. Fix $Q \in \mathbb{N}$ and a window $W = (\alpha - 1, \alpha + 1)$. For each $q \leq Q$, a rational $p/q \in W$ requires the integer p to lie in the interval $\big(q(\alpha-1),\, q(\alpha+1)\big)$, of length $2q$: at most $2q+1$ candidates, fewer still in lowest terms. Summing over $q = 1, \ldots, Q$: at most $\sum_{q \leq Q}(2q+1) = Q^2 + 2Q$ loud rationals in W. Finitely many fences, finitely many posts each.

(c) Continuity at every irrational. Fix irrational α and $\varepsilon > 0$; choose $Q \in \mathbb{N}$ with $Q \geq 1/\varepsilon$ (the Archimedean property), and call a rational *loud* if its lowest-terms denominator is at most Q. By (b) the loud rationals in W form a finite set L, and $\alpha \notin L$. If $L = \emptyset$, set $m = 1$; otherwise $m = \min_{r \in L} |\alpha - r| > 0$ (a minimum of finitely many positive distances). Take $\delta = \min(1, m)$. For $|x - \alpha| < \delta$: if x is irrational, $T(x) = 0$; if $x = p/q$ in lowest terms, then $x \in W$ and $x \notin L$ (it lies strictly closer than m), so $q \geq Q+1$ and $T(x) = \frac{1}{q} \leq \frac{1}{Q+1} < \varepsilon$. In every case $|T(x) - T(\alpha)| = T(x) < \varepsilon$. One recipe per challenge: continuous at α.

(d) The framed corollary. Let $p_n/q_n \to \alpha$ (lowest terms), α irrational, and fix any Q. The loud set for Q near α is finite and excludes α, hence keeps a positive distance m; eventually $|p_n/q_n - \alpha| < m$, forcing $q_n > Q$. Since Q was arbitrary, $q_n \to \infty$: the rationals sneak up on an irrational only by refining without bound.

6.11 Notes

Bolzano, 1817. The Prague paper Episode One mourned — *Rein analytischer Beweis* — states and proves the intermediate value theorem by exactly the bisection of Theorem 6.15, decades before the tools it presupposed were formally available. The error bound makes it, in modern eyes, a convergence analysis of binary search.

Cauchy, 1821. The *Cours d'analyse* gives the first serviceable definition of continuity and proves an intermediate value theorem; the same book commits the illegal swap autopsied in Chapter 7. Both facts matter: rigour arrived in instalments.

Weierstrass, Berlin, 1861. The epsilon–delta formulation as transmitted through the Berlin lectures; the extreme value theorem (Theorem 6.12) is often styled Weierstrass's. Heine, 1872, published the apparatus systematically — the same vintage-year paper whose covering idea Chapter 5 crowned — and proved uniform continuity on closed bounded intervals (Theorem 6.22); custom appends Cantor's name, adjacency in that year being what it is.

Dirichlet, 1829; Thomae, 1875. The static is born in Dirichlet's Fourier memoir as a deliberate counterexample to the era's integral — an appointment Chapter 8 honours. Thomae's function appears in his 1875 textbook; the loudness analysis is folklore polished by a century of instructors.

Fixed points. Corollary 6.17 is the one-dimensional ancestor of Brouwer's fixed-point theorem, whose higher-dimensional forms underwrite general equilibrium theory; the one-line proof above is the honest reason the one-dimensional case is free.

Reading. Abbott, *Understanding Analysis*, ch. 4; Rudin, *Principles*, ch. 4; Körner, *A Companion to Analysis*, for the uniform-continuity propaganda this chapter happily repeats.

Chapter 7

The Derivative and the Monster

Companion to Episode Seven. The episode defined the limit of a function and the derivative, paid Zeno's last invoice, climbed the staircase Fermat–Rolle–mean-value and harvested its three consequences, met Darboux's strangeness, introduced uniform convergence at need — paying the swap account's first instalment over Cauchy's 1821 corpse — built the blancmange, quoted Hermite's recoil, and let Baire fire: the monsters are typical. This chapter is the paperwork. Register notes: the bell rings ***once*** *this chapter — the twelfth ring, marked (⊛) at Theorem 7.31 — and it is the first mark ever inked off the real line; and the register's first* ***loan*** *(L1) is recorded at Remark 7.17: schoolroom trigonometry, desk-only, repaid when the power series arrive.*

Episode Digest

The argument of the hour. The cruelest reveal of the season. The derivative is defined honestly and Zeno's last invoice — the Arrow — is paid; the staircase Fermat–Rolle–mean-value is climbed and its harvest collected, the plus-C of the schoolroom proved at last; Darboux shows derivatives stranger than advertised; uniform convergence arrives precisely at need, paying the swap account's first instalment over Cauchy's 1821 corpse; the blancmange is built — continuous everywhere, differentiable nowhere; Hermite recoils on the record; and Baire fires: the monsters are not rare but typical, the smooth curves the exceptions. The bell rings once — the twelfth — and for the first time it rings off the line, in a space whose points are functions.

The episode, beat by beat. **The Pen Shakes.** Ampère, 1806: a published proof that every function worthy of the name is differentiable away from scattered bad points — the schoolroom's standing creed. The programme announced: the derivative and Zeno's payment; the workhorse; Darboux; a new mode of convergence at need; the monster; Hermite's recoil; Baire's first discharge. Last week the pen lifted; tonight it shakes.

Velocity at an Instant, or Zeno's Last Invoice. A dialect note first: the limit of a function at a point — the old game with the anchor excused from testimony, taken only at limit points (hermits give no testimony); sequential bridge and Chapter 4's algebra inherited in one stroke (Definition 7.1, Remark 7.2). Then the definition the century fought for: the derivative as the limit of difference quotients — the secant's slope settling into the tangent's (Definition 7.3). The Arrow paid with ceremony: velocity at an instant is not distance-in-the-instant over duration-of-the-instant (zero over zero, rightly mocked) but the limit of average velocities over shrinking windows — a fact about every neighbourhood of the instant, not the instant alone; debt two-b discharged, Zeno's account closed entirely. Worked matches: constants (quotient identically zero), the identity (identically one), the parabola (the handcuff factor returns as a gift; derivative twice the anchor) — and the corner: the absolute value at zero, half-slopes plus one and minus one, a disputed election, no derivative (Remark 7.4). Continuity without differentiability, first specimen.

The One-Way Street. Differentiable implies continuous, proved in one breath — quotient times displacement (Proposition 7.6); the converse died at the corner. The century's belief framed for execution: scratches are always isolated. The deepest reading pocketed: differentiability is local exchangeability for a line with certified sub-linear error — the touching order of the tangent, second-order contact (Remark 7.5); every gradient the trade follows is this exchange at scale. The algebra in one stroke: sums, products (one add-and-subtract), quotients off zeros (Theorem 7.7); the chain rule stated with its honest, fussy proof staged — the naïve division is inadmissible (Theorem 7.8); polynomials differentiable everywhere (Corollary 7.9). The monsters of Episode Six declared ineligible: the new floor sits above the old.

Rolle, or the Level Ground. Fermat's interior extremum lemma proved whole in the ear: right quotients at most zero, left at least zero, the derivative squeezed to zero — level ground at interior summits (Lemma 7.10); the entire license behind setting gradients to zero, with the honest caveats — interior is load-bearing (endpoints squeeze from one side only), and level ground is necessary, never sufficient (the cube sails through its critical point). Rolle, 1691 — furnished by a skeptic of the calculus, the patron saint of thorough code review: equal endpoint values force an interior level point (Theorem 7.11); proof by the first crown's attainment plus Fermat, constant case separately; hypothesis tour — the corner steals it, the pierced endpoint steals it; and the asymmetric hypotheses read as a lesson in stacked proofs: each floor bills its own invoice.

The Workhorse. Tilt the floor: the mean value theorem — somewhere the tangent parallels the chord (Theorem 7.12); proof by one subtraction, Rolle on the difference; the speedometer gloss, the police mathematics of the average-speed camera. The harvest, one breath each (Corollary 7.13): zero derivative means constant — the plus-C proved for the first time in the listener's life, the whole tower visible (axiom at the bottom, homework at the top); positive derivative means strictly increasing; and capped steepness — a bounded derivative is a Lipschitz bound is a uniform continuity certificate, Episode Six's moral cashed in full, with the error-control reading for the trade. Named in passing and exiled to paper: L'Hôpital's baby form (a quotient of quotients) and Taylor with the Lagrange remainder — the workhorse in evening dress, fitted one degree at a time (Remark 7.14).

Darboux's Strange Theorem. Are derivatives continuous? No: the oscillating specimen — parabola-damped oscillation on borrowed schoolroom trigonometry, the register's first loan, desk-only, repaid at the power series — differentiable everywhere, derivative discontinuous at the origin (Remark 7.17). Yet Darboux, 1875 — Thomae's vintage — every derivative has the intermediate value property (Theorem 7.15): tilt, attain the interior minimum, Fermat. Dividends: the step function is nobody's derivative; and derivatives never jump even discreetly — one-sided limits of a derivative, where they exist, must agree with its value (Remark 7.16); their only permitted misbehaviour is oscillation. Forward shadow filed: a derivative will break an integral in Episode Eight.

A New Mode of Convergence. The pivot: the monster must be assembled as a limit of functions

— what does convergence of functions mean? Pointwise defined; the disease immediate: the powers of x on the unit interval, every stage polynomial, the limit with a jump — the sag sliding rightward forever (Definition 7.18). The autopsy owed since Episode Four: Cauchy, 1821, proved that convergent sums of continuous functions are continuous — false, and the fatal step is an unlicensed swap of two limits; Exhibit A dissected, the iterated limits disagreeing by one (Remark 7.24). The cure is Tuesday's surgery again: move the quantifier — uniform convergence, one threshold serving every anchor, the third transposition of the course; the tube picture; the ramps supply the second specimen — the leak and the escape re-offending in new dress (Definition 7.19, Remark 7.20). The repaired theorem, the first instalment paid: uniform limits of continuous functions are continuous — the three-legged stool, each leg a third (Theorem 7.21). A caution stamped on the license: differentiability does not pass, the blancmange itself the witness (Remark 7.25). And the assembly permit: the M-test — domination by a summable budget buys the tube (Theorem 7.23), riding the Cauchy criterion (Theorem 7.22).

Building the Monster. The pedigree: Bolzano built one around 1830, unpublished for a century — Prague's luck holding beyond the grave; Riemann gestured, the paperwork failing inspection generations later; Weierstrass, Berlin, 1872 — the vintage year's fourth appearance — presents the cosine monster, printed via a colleague in 1875. The desk builds the friendlier cousin: **the blancmange** — Takagi, 1901 — christened for the dessert it resembles, a pudding that shivers at every scale (Definition 7.27). The recipe: the sawtooth — distance to the nearest integer — compressed by powers of two in both directions, the compressions cancelling in the slope: every stage's teeth climb at slope one exactly (Definition 7.26). Convergence by the M-test on the halving budget; continuity by the stool — the instalment already earning interest; the fractal family note: the course now owns the dust and the pudding. Nowhere differentiable by the parity heartbeat: dyadic straddle quotients are integers — finer stages vanish at dyadic endpoints, coarser stages are single tooth-sides of slope plus or minus one — and each refinement amends the integer by exactly one; parity flips forever; an integer sequence stepping by one converges to nothing; and the straddle lemma — the subpoena, converting testimony about neighbours into a verdict about the point — forces convergence if a derivative exists (Lemma 7.28, Theorem 7.29). Contradiction at every point at once. Ampère's scattered bad points have become every point there is.

Hermite Recoils. The reception, verified against the correspondence: Hermite to Stieltjes, the twentieth of May, 1893 — I turn away in fright and horror from this lamentable plague of continuous functions that have no derivatives; the French preserved in the notes, where its accents are safe. Poincaré's grumble — functions invented expressly to fault the reasonings of the fathers — and his word for them, *monsters*, the course's own job title. The valet's editorial: one sympathizes with the horror and declines to share it — the monsters were discovered, not invented, resident in the definitions the fathers ratified; one does not fault the auditor for the state of the books. The bridge: were they at least rare?

Baire Fires. The wonder. To weigh "most functions," a space: C, the continuous functions on the unit interval, under the uniform distance — the largest gap between graphs, attained (the first crown certifying the yardstick); convergence in this distance is uniform convergence — the new mode was secretly a geometry (Definition 7.30). Episode Five's vocabulary written in a portable dialect, carried abroad. The credential: C is complete — pointwise huddles converge by the completeness of the line, uniformity transfers, the stool certifies — **and the bell rings, the twelfth, for the first time off the line: the axiom travels** (Theorem 7.31, the chapter's only mark). Baire transfers verbatim (Remark 7.32). The films: functions somewhere gentle at grade n — closed (witnesses to a subsequence), nowhere dense (piecewise-linear approximation plus a sawtooth of tiny height and savage slope: the blancmange's own teeth as demolition tool); countably many grades — Episode Three underwriting Episode Seven once more; every somewhere-differentiable function caught. Baire pronounces: the somewhere-differentiable functions are meagre; the monsters are residual — generic, typical (Theorem 7.33). The inversion entire: every function ever drawn lives in the meagre stack; the pen samples forever from the negligible; ink is a category-negligible instrument; the freak show was the audience all along. The method pocketed as its own dividend: proof by census — existence in bulk without a single construction. Honesty clause: typical in the sense of category; Season Two brings the rival smallness, and the two judges will disagree. Debt four-b paid with ceremony: Baire's first engagement discharged; the second — the asymmetry — stands for Season Two.

The toolkit acquired. Functional limits and the sequential bridge (Definition 7.1, Remark 7.2); the derivative, half-slopes, first computations, the corner, local linearity and touching order (Definitions 7.3, Remarks 7.4, 7.5); differentiable-implies-continuous, the algebra, the chain rule's honest patch, polynomials (Proposition 7.6, Theorems 7.7, 7.8, Corollary 7.9); Fermat, Rolle, the mean value theorem, the three-consequence harvest, L'Hôpital and Taylor named and exiled (Lemma 7.10, Theorems 7.11, 7.12, Corollary 7.13, Remark 7.14); Darboux with both dividends and the loaned oscillator (Theorem 7.15, Remarks 7.16, 7.17); pointwise and uniform convergence, the specimens, the stool, the Cauchy criterion, the M-test, the autopsy, the caution (Definitions 7.18, 7.19, Remarks 7.20, 7.24, 7.25, Theorems 7.21, 7.22, 7.23); the sawtooth, the blancmange, the straddle lemma, the monster theorem (Definitions 7.26, 7.27, Lemma 7.28, Theorem 7.29); the space C, its completeness with the twelfth ring, the Baire transfer, and the typicality census (Definition 7.30, Theorem 7.31, Remark 7.32, Theorem 7.33). Supplied solutions: the autopsy (II.3) and the blancmange entire (II.6); the census and the squared star are guided, deliberately unsolved.

Debts. **Paid: D2b, the Arrow** — velocity at an instant defined; Zeno's account, opened in Episode One, closed entirely. **Paid: the swap account's first instalment** — limits of continuous functions licensed under uniformity, over the corpse of Cauchy's 1821 theorem; instalments remaining: the integral (Episode Eight) and the crown jewels (Season Two). **Paid: D4b** — the monster built and the monsters proved typical; Baire's first engagement discharged, the second (the asymmetry, D9) standing for Season Two, Episode One. **Opened: Loan L1** — schoolroom trigonometry for the oscillating specimen, desk-only, repaid at the power series. **D7, the bell: rang once** — the twelfth, at the completeness of C, the first mark off the real line; the axiom travels. **Christened: the blancmange**, with the straddle lemma as counsel. The static's Episode Eight appointment confirmed; Thomae's pass confirmed; the monotone star to be spent at the integral.

Dates worth keeping. 1691, Rolle's theorem, from a skeptic of the calculus; 1806, Ampère's false proof — the schoolroom's creed, executed tonight; 1821, Cauchy's *Cours*: the false continuity theorem, autopsied; circa 1830, Bolzano's unpublished monster — printed only in 1930, Prague's luck holding; 1872, Weierstrass before the Berlin Academy — the vintage year's fourth appearance; 1875, du Bois-Reymond prints the monster and Darboux prints his theorem — Thomae's vintage twice over; 1893, Hermite's recoil, letter 374, verified; 1901, Takagi's blancmange; 1930, van der Waerden's variant, and Bolzano's vindication, in the same year.

7.1 Notation Ledger

Spoken in the episode	**Written here**	**Remarks**
"the limit of f at a"	$\lim_{x\to a} f(x) = L$	Definition 7.1
"f prime"; "the difference quotient"	$f'(a) = \lim_{x\to a} \dfrac{f(x) - f(a)}{x - a}$	Definition 7.3
"the half-slopes"	$f'_+(a),\ f'_-(a)$	Definition 7.3
"converges pointwise"	$f_n \to f$ on A	Definition 7.18
"converges uniformly"; "the tube"	$f_n \rightrightarrows f$ on A	Definition 7.19
"the uniform distance"	$d(f,g) = \max_{x\in[0,1]} \lvert f(x) - g(x)\rvert$	Definition 7.30
"the sawtooth"	$s(x) = \mathrm{dist}(x, \mathbb{Z})$	Definition 7.26
"the blancmange"	$B(x) = \sum_{n\geq 0} 2^{-n} s(2^n x)$	Definition 7.27

The moved quantifier (the ledger's centrepiece, the second of the course). — Pointwise and uniform convergence on a set A, side by side, in full dress:

$$\begin{aligned} \text{pointwise on } A: \quad & \forall x \in A\ \ \forall \varepsilon > 0\ \ \exists N\ \ \forall n \geq N: \ |f_n(x) - f(x)| < \varepsilon; \\ \text{uniform on } A: \quad & \forall \varepsilon > 0\ \ \exists N\ \ \forall n \geq N\ \ \forall x \in A: \ |f_n(x) - f(x)| < \varepsilon. \end{aligned}$$

Again the strings differ in one respect only: the position of $\forall x$ relative to $\exists N$. In the first, each anchor hires its own threshold; in the second, one threshold is issued to serve every anchor at once. Compare the centrepiece of Chapter 6: the same transposition, new players, and — as the chapter proves — once again the entire theorem.

7.2 Limits of Functions and the Derivative

Definition 7.1 (Limit of a function at a point)**.** Let $f : A \to \mathbb{R}$ and let a be a *limit point* of A (Definition 5.6) — a point with company; a itself need not belong to A. Then $\lim_{x\to a} f(x) = L$ means:

$$\forall \varepsilon > 0\, \exists \delta > 0\, \forall x \in A: \quad 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

The clause $0 < |x - a|$ is the excusal: the anchor's own value, if any, gives no testimony. One-sided limits ($x \to a^+$, $x \to a^-$) restrict x to one side; both exist and agree exactly when the two-sided limit exists.

Remark 7.2 (The sequential bridge; the inherited algebra). $\lim_{x\to a} f(x) = L$ if and only if $f(x_n) \to L$ for every sequence (x_n) in $A \setminus \{a\}$ with $x_n \to a$. The proof is the transcription of Theorem 6.3, excusal included. Consequently the entire algebra of limits (Theorem 4.5) transfers to functional limits in one stroke, as do squeeze (Theorem 4.7) and the order limit theorem (Theorem 4.6); and for $a \in A$ a limit point of A, continuity at a is exactly the statement $\lim_{x\to a} f(x) = f(a)$. (At hermits, continuity is automatic, Remark 6.2, and limits are undefined — the two notions divide the domain between them without dispute.)

Definition 7.3 (The derivative)**.** Let $f : A \to \mathbb{R}$ and let $a \in A$ be a limit point of A. The *derivative of f at a* is

$$f'(a) \;=\; \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \;=\; \lim_{h\to 0} \frac{f(a+h)-f(a)}{h},$$

provided the limit exists; f is then *differentiable at a*. The line $y = f(a) + f'(a)(x-a)$ is the *tangent* at a. The one-sided limits of the difference quotient, where they exist, are the *one-sided derivatives* $f'_+(a)$ and $f'_-(a)$; $f'(a)$ exists if and only if both exist and agree.

Remark 7.4 (First derivatives; the corner)*.* Constants: the quotient is identically 0, so the derivative is 0 everywhere. The identity: quotient identically 1. The parabola at a:

$$\frac{x^2-a^2}{x-a} = x+a \;\longrightarrow\; 2a,$$

the Chapter 6 handcuff factor returning as a gift. The absolute value at 0: right quotients identically $+1$, left quotients identically -1; so $f'_+(0) = 1 \neq -1 = f'_-(0)$, and $|x|$ is continuous at 0 (Proposition 6.10) with no derivative there — a disputed election between the half-slopes.

Remark 7.5 (Local linearity; the touching order)*.* f is differentiable at a with $f'(a) = m$ if and only if

$$f(x) = f(a) + m\,(x-a) + r(x), \qquad \frac{r(x)}{x-a} \longrightarrow 0 \;\; (x \to a),$$

and the slope m with this property is unique. (Both directions are one rearrangement of the difference quotient.) Differentiability is *local exchangeability for a line with certified sub-linear error*. The tangent's advantage over every rival line is visible on the parabola: $x^2 - \big(a^2 + 2a(x-a)\big) = (x-a)^2$ — the error against the tangent dies at second order; against any other line through the point, at first order only. Problem I.3 verifies the touching orders.

Proposition 7.6 (Differentiable implies continuous)**.** *If f is differentiable at a, then f is continuous at a.*

Proof. For $x \neq a$, $f(x) - f(a) = \dfrac{f(x)-f(a)}{x-a} \cdot (x-a)$. As $x \to a$, the first factor tends to $f'(a)$ and the second to 0; by the product law (Theorem 4.5 via Remark 7.2), $f(x) - f(a) \to 0$, which is continuity at a (Remark 7.2 again). $\square$

Theorem 7.7 (Algebra of derivatives)**.** *Let f, g be differentiable at a. Then:*

(i) $(f+g)'(a) = f'(a) + g'(a)$*;*

(ii) $(fg)'(a) = f'(a)\,g(a) + f(a)\,g'(a)$*;*

(iii) if $g(a) \neq 0$*, then* $(f/g)'(a) = \dfrac{f'(a)\,g(a) - f(a)\,g'(a)}{g(a)^2}$.

Proof. (i) The quotient splits term by term; apply the sum law. (ii) Add and subtract the mixed term in the numerator:

$$\frac{f(x)g(x)-f(a)g(a)}{x-a} = \frac{f(x)-f(a)}{x-a}\,g(x) + f(a)\,\frac{g(x)-g(a)}{x-a}.$$

As $x \to a$: the first quotient tends to $f'(a)$ and $g(x) \to g(a)$ (Proposition 7.6); the second quotient tends to $g'(a)$. The product and sum laws finish. (iii) It suffices, by (ii), to differentiate $1/g$: since

$g(a) \neq 0$ and g is continuous at a, g is nonvanishing on a relative neighbourhood (Theorem 6.7), and there

$$\frac{1}{x-a}\Big(\frac{1}{g(x)} - \frac{1}{g(a)}\Big) = -\frac{1}{g(x)\,g(a)} \cdot \frac{g(x)-g(a)}{x-a} \;\longrightarrow\; -\frac{g'(a)}{g(a)^2}.$$

Combine with (ii). □

Theorem 7.8 (Chain rule). *If g is differentiable at a and f is differentiable at $g(a)$, then $f \circ g$ is differentiable at a and $(f \circ g)'(a) = f'(g(a)) \cdot g'(a)$.*

Proof strategy. The tempting manipulation multiplies and divides by $g(x) - g(a)$, which may vanish for x arbitrarily close to a; this chapter does not wave that hand. The honest proof defines the *patched quotient* $\varphi(y) = \big(f(y) - f(g(a))\big)/\big(y - g(a)\big)$ for $y \neq g(a)$ and $\varphi(g(a)) = f'(g(a))$, observes that φ is continuous at $g(a)$, and verifies the identity

$$\frac{f(g(x)) - f(g(a))}{x-a} = \varphi(g(x)) \cdot \frac{g(x)-g(a)}{x-a} \qquad \text{for } \textit{all } x \neq a$$

— including the offending x with $g(x) = g(a)$, where both sides are 0. The limit is then $\varphi(g(a)) \cdot g'(a)$ by continuity of g and of φ. Problem II.1 stages the details. □

Corollary 7.9 (Polynomials and their kin). *$(x^n)' = n\,x^{n-1}$ for every $n \in \mathbb{N}$ (induction with the product rule); every polynomial is differentiable on $\mathbb{R}$, term by term; every rational function is differentiable wherever its denominator is nonzero; and chains of differentiable functions are differentiable, with derivatives multiplying through the relay.*

7.3 Rolle and the Mean Value Theorem

Lemma 7.10 (Fermat's interior extremum lemma). *Let f attain a maximum (or minimum) at a point c* interior *to its domain, and suppose $f'(c)$ exists. Then $f'(c) = 0$.*

Proof. Say c is an interior maximum: on some $N_\eta(c)$ inside the domain, $f(x) \leq f(c)$. For $x \in (c, c+\eta)$ the difference quotient has numerator $f(x) - f(c) \leq 0$ and denominator $x - c > 0$: every right quotient is ≤ 0, so by the order limit theorem (Theorem 4.6 via Remark 7.2), $f'_+(c) \leq 0$. For $x \in (c-\eta, c)$ the denominator flips sign: every left quotient is ≥ 0, so $f'_-(c) \geq 0$. But $f'(c)$ exists, so $f'(c) = f'_+(c) = f'_-(c)$ is at once ≤ 0 and ≥ 0: zero. The minimum case is the same argument upside down. □

Necessary, never sufficient: the cube at the origin has derivative 0 and no extremum — level ground marks candidates, not victors. And *interior* is load-bearing: the identity on $[0,1]$ maxes at the endpoint 1 with derivative 1, the one-sided squeeze having only one side to squeeze from.

Theorem 7.11 (Rolle, 1691). *Let f be continuous on $[a,b]$, differentiable on (a,b), with $f(a) = f(b)$. Then $f'(c) = 0$ for some $c \in (a,b)$.*

Proof. $[a,b]$ is compact and f continuous, so f attains a maximum and a minimum (Theorem 6.12 — attainment, not rumour: Fermat needs an address). *Case 1: both extremes sit at endpoints.* Since $f(a) = f(b)$, the maximum equals the minimum, f is constant, and $f' \equiv 0$ on (a,b); any interior c serves. *Case 2: some extremum sits at an interior point c.* There f is differentiable by hypothesis, and Lemma 7.10 gives $f'(c) = 0$. □

The hypothesis audit (Problem I.2 territory): $|x|$ on $[-1,1]$ has equal endpoint values and no interior level point — the corner steals it; and piercing one endpoint of the identity on $[0,1]$ (redefine $f(1)=0$) satisfies the equal-values clause while every interior slope is 1 — continuity to the very ends is load-bearing. Note the asymmetric hypotheses: continuity on the closed interval (the extreme value theorem's invoice), differentiability only on the open one (Fermat testifies only at interior addresses). The hypotheses of a stacked theorem are the union of its floors' invoices.

Theorem 7.12 (The mean value theorem)**.** *Let f be continuous on $[a,b]$ and differentiable on (a,b). Then there is $c \in (a,b)$ with*

$$f'(c) \;=\; \frac{f(b)-f(a)}{b-a}.$$

Proof. Subtract the chord. Let

$$g(x) \;=\; f(x) - \Big[\, f(a) + \frac{f(b)-f(a)}{b-a}\,(x-a)\Big].$$

Then g is continuous on $[a,b]$ and differentiable on (a,b) (Theorem 7.7; the bracket is affine), and $g(a) = 0 = g(b)$ by direct evaluation. Rolle (Theorem 7.11) supplies $c \in (a,b)$ with $g'(c) = 0$, that is, $f'(c) = \dfrac{f(b)-f(a)}{b-a}$. □

Corollary 7.13 (The workhorse's harvest)**.** *Let f be continuous on an interval I and differentiable at its interior points.*

(i) If $f' \equiv 0$ on the interior, f is constant on I; consequently two functions with the same derivative on an interval differ by a constant.

(ii) If $f' > 0$ on the interior, f is strictly increasing on I (and $f' \geq 0$ gives nondecreasing; mirror statements for negative).

(iii) If $|f'| \leq M$ on the interior, then $|f(x)-f(y)| \leq M\,|x-y|$ for all $x,y \in I$ — a Lipschitz bound *with constant M — and consequently f is uniformly continuous on I (Definition 6.19), with the anchor-free response $\delta = \varepsilon/M$.*

Proof. All three by one application of Theorem 7.12 to an arbitrary pair $x < y$ in I: some interior c has $f(y) - f(x) = f'(c)\,(y-x)$. (i) The right side is 0: any two values agree. (ii) The right side is positive: $f(y) > f(x)$. (iii) Take sizes: $|f(y)-f(x)| = |f'(c)|\,|y-x| \leq M\,|y-x|$; given ε, the response $\delta = \varepsilon/M$ wins at every anchor simultaneously. The Episode Six moral — capped steepness buys uniformity — is hereby a corollary. □

Remark 7.14 (Named in passing: L'Hôpital; Taylor–Lagrange)*.* Two celebrated relatives of the workhorse, stated here and fitted in Problem II.4. *L'Hôpital, baby form:* if $f(a) = g(a) = 0$, both are differentiable at a, and $g'(a) \neq 0$, then

$$\lim_{x\to a}\frac{f(x)}{g(x)} = \lim_{x\to a}\frac{\big(f(x)-f(a)\big)/(x-a)}{\big(g(x)-g(a)\big)/(x-a)} = \frac{f'(a)}{g'(a)},$$

a quotient of difference quotients and one application of the quotient law — no mystery, merely bookkeeping. (Stronger forms, with both derivatives merely limiting, run through a two-function mean value theorem; the notes cite them.) *Taylor with the Lagrange remainder:* if f has $n+1$ derivatives on an interval containing a and b, then

$$f(b) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}\,(b-a)^k \;+\; \frac{f^{(n+1)}(c)}{(n+1)!}\,(b-a)^{n+1} \quad \text{for some } c \text{ between } a \text{ and } b.$$

The case $n = 0$ *is* the mean value theorem; each further degree is one more auxiliary function fed to Rolle — the workhorse in evening dress, fitted one degree at a time in Problem II.4.

7.4 Darboux's Theorem

Theorem 7.15 (Darboux, 1875)**.** *Let f be differentiable on $[a,b]$ and let w lie strictly between $f'(a)$ and $f'(b)$. Then $f'(c) = w$ for some $c \in (a,b)$. Derivatives have the intermediate value property, continuous or not.*

Proof. Say $f'(a) < w < f'(b)$; the reversed case mirrors. Tilt: let $g(x) = f(x) - w\,x$, differentiable on $[a,b]$ with $g' = f' - w$, so $g'(a) < 0 < g'(b)$. Being differentiable, g is continuous (Proposition 7.6) on the compact $[a,b]$, so it *attains* a minimum (Theorem 6.12). The minimum is not at a: since $\lim_{x\to a^+}\big(g(x)-g(a)\big)/(x-a) = g'(a) < 0$, right quotients are eventually negative, so some $x > a$ has $g(x) < g(a)$. Nor at b: since $g'(b) > 0$, left quotients near b are eventually positive, and with $x - b < 0$ this forces $g(x) < g(b)$ for some $x < b$. The minimum therefore sits at an interior c; Lemma 7.10 gives $g'(c) = 0$, that is, $f'(c) = w$. □

Remark 7.16 (Nobody's derivative; derivatives never jump). The step function H (0 for $x < 0$, 1 for $x \geq 0$) is *nobody's derivative* on any interval containing 0: it takes the values 0 and 1 and skips everything between, which Theorem 7.15 forbids of derivatives. Sharper, and staged as Problem II.2: if $\lim_{x\to a^+} f'(x) = L$ exists, then $f'_+(a) = L$ (run Theorem 7.12 on $[a,x]$ and let $x \downarrow a$); likewise from the left. A derivative cannot jump even discreetly — its only permitted discontinuity is the oscillating kind, exhibited next.

Remark 7.17 (The oscillating specimen — on loan). Define $g(0) = 0$ and $g(x) = x^2\sin(1/x)$ for $x \neq 0$. At the origin the definition alone suffices: $\big|\big(g(x)-g(0)\big)/x\big| = |x\sin(1/x)| \leq |x| \to 0$ by squeeze, so $g'(0) = 0$. Away from the origin the chain and product rules give $g'(x) = 2x\sin(1/x) - \cos(1/x)$, which has no limit as $x \to 0$ (the cosine term oscillates through $[-1,1]$ forever while the first term dies). So g is differentiable at *every* point of $\mathbb{R}$ and g' is discontinuous at 0: discontinuity by indecision, exactly as Darboux permits.

Loan recorded (L1). — This specimen borrows from the schoolroom: the sine function, with $|\sin| \leq 1$ and $\sin' = \cos$, $|\cos| \leq 1$. First entry on the register's loan side; desk-only (the episode names the specimen and defers the trigonometry); repaid when this course constructs the power series and, with them, the trigonometric functions honestly.

7.5 A New Mode of Convergence

Definition 7.18 (Pointwise convergence)**.** Functions $f_n : A \to \mathbb{R}$ *converge pointwise* to $f : A \to \mathbb{R}$ if $f_n(x) \to f(x)$ for every $x \in A$ — Episode Four's game, played anchor by anchor, each anchor hiring its own threshold. Two computations to keep. *The powers:* on $[0,1]$, $x^n \to 0$ for $x < 1$ (Chapter 4) and $1^n \to 1$, so the pointwise limit is 0 on $[0,1)$ and 1 at $x = 1$: every stage a polynomial, the limit with a jump. *The ramps:* on $\mathbb{R}$, $x/n \to 0$ at every anchor.

Definition 7.19 (Uniform convergence)**.** $f_n \rightrightarrows f$ on A if

$$\forall \varepsilon > 0\ \exists N\ \forall n \geq N\ \forall x \in A: \quad |f_n(x) - f(x)| < \varepsilon;$$

equivalently, $\sup_{x \in A} |f_n(x) - f(x)| \to 0$. One threshold, printed once, serving every anchor: geometrically, for each ε the entire graph of every late stage lies inside the ε-tube around the graph of f. The side-by-side contrast with the pointwise string is the ledger's centrepiece.

Remark 7.20 (The specimens audited). *The powers:* on $[0,1)$, $\sup_x |x^n - 0| = 1$ for every n (values climb arbitrarily near 1), so the convergence is not uniform on $[0,1)$ — nor, a fortiori, on $[0,1]$: the sag near the right edge never enters any tube of height below 1. On $[0,b]$ with $b < 1$: $\sup = b^n \to 0$, uniform — shrink the territory and the disease vanishes. *The ramps:* on $[-M, M]$, $\sup |x/n| = M/n \to 0$, uniform; on $\mathbb{R}$, $\sup = \infty$ for every n, never uniform. The leak and the escape (Proposition 5.12's two convicts), re-offending in the new vocabulary: failure at an edge, failure at the horizon. Problems I.6–I.7 run the full sup-arithmetic.

Theorem 7.21 (Uniform limits of continuous functions are continuous)**.** *If each $f_n : A \to \mathbb{R}$ is continuous on A and $f_n \rightrightarrows f$ on A, then f is continuous on A.*

Proof. The $\varepsilon/3$ argument — the three-legged stool. Fix $a \in A$ and $\varepsilon > 0$. By uniformity, choose one stage N with $|f_N(x) - f(x)| < \varepsilon/3$ for *every* $x \in A$. By continuity of f_N at a, choose $\delta > 0$ with $|f_N(x) - f_N(a)| < \varepsilon/3$ whenever $x \in A$, $|x - a| < \delta$. Then for such x:

$$|f(x) - f(a)| \ \leq\ \underbrace{|f(x) - f_N(x)|}_{<\varepsilon/3 \text{ (uniformity)}} + \underbrace{|f_N(x) - f_N(a)|}_{<\varepsilon/3 \text{ (continuity)}} + \underbrace{|f_N(a) - f(a)|}_{<\varepsilon/3 \text{ (uniformity)}} \ <\ \varepsilon.$$

One response per challenge: f is continuous at a, and a was arbitrary. The swap of limits is hereby licensed under uniform convergence: the first instalment on the account of Chapter 4. □

Theorem 7.22 (Cauchy criterion for uniform convergence)**.** *(f_n) converges uniformly on A if and only if it is* uniformly Cauchy*: for every $\varepsilon > 0$ there is N with $|f_n(x) - f_m(x)| < \varepsilon$ for all $m, n \geq N$ and all $x \in A$.*

Proof. ($\Rightarrow$) Triangle through the limit with $\varepsilon/2$ twice, the N uniform in x by hypothesis. ($\Leftarrow$) At each anchor x, $(f_n(x))$ is a Cauchy sequence of reals, hence converges (Theorem 4.18 — the inheritance that Theorem 7.31 will crown); call the limit $f(x)$. Now fix $\varepsilon > 0$ and its uniform N: for $n \geq N$, every x, and all $m \geq N$, $|f_n(x) - f_m(x)| < \varepsilon$; let $m \to \infty$ (order limit theorem, Theorem 4.6) to get $|f_n(x) - f(x)| \leq \varepsilon$ for all x. So $\sup_A |f_n - f| \leq \varepsilon$ for all $n \geq N$: uniform convergence. □

Theorem 7.23 (The Weierstrass M-test)**.** *Let $g_n : A \to \mathbb{R}$ satisfy $|g_n(x)| \leq M_n$ for all $x \in A$, with $\sum M_n$ convergent. Then $\sum g_n$ converges uniformly on A.*

Proof. Let S_n denote the partial sums. For $m < n$ and every $x \in A$,

$$|S_n(x) - S_m(x)| \ =\ \Big|\sum_{k=m+1}^{n} g_k(x)\Big| \ \leq\ \sum_{k=m+1}^{n} M_k,$$

and the right side is the tail-block of a convergent series of numbers, hence below any ε once m, n exceed the N supplied by the Cauchy property of $\sum M_k$ (its partial sums converge, so huddle: Theorem 4.18, easy direction). The bound is independent of x: (S_n) is uniformly Cauchy, and Theorem 7.22 converts. Domination by a summable budget buys the tube. □

Remark 7.24 (The autopsy: Cauchy, 1821). The *Cours d'analyse* states: a convergent sum of continuous functions is continuous. The powers of x refute the sequence form on $[0,1]$, and the failure is an unlicensed swap of two limiting processes. At the seam:

$$\lim_{x\to 1^-}\lim_{n\to\infty} x^n = \lim_{x\to 1^-} 0 = 0, \qquad \lim_{n\to\infty}\lim_{x\to 1^-} x^n = \lim_{n\to\infty} 1 = 1.$$

The two iterated limits disagree by exactly 1; Cauchy's proof assumes their agreement without license. Uniformity is the repair: under $f_n \rightrightarrows f$, Theorem 7.21 licenses the swap. Problem II.3 (solved in full) conducts the dissection with the quantifiers on the table.

Remark 7.25 (A caution stamped on the license). Uniform convergence transports continuity, not differentiability. The blancmange (Theorem 7.29) is the standing witness: a uniform limit of stages differentiable at all but finitely many points, itself differentiable at none. Moving a derivative past a limit is a stricter negotiation — uniform convergence *of the derivatives*, plus convergence at one point — cited in the notes and not conducted this season.

7.6 Building the Monster

Definition 7.26 (The sawtooth)**.** $s(x) = \operatorname{dist}(x,\mathbb{Z}) = \min_{k\in\mathbb{Z}} |x-k|$, the distance to the nearest integer. Dossier (verified in Problem II.6(a)): $0 \le s \le \frac{1}{2}$; s has period 1; $s(k) = 0$ for every integer k; s is 1-Lipschitz, hence uniformly continuous; and on each half-integer interval $[\frac{k}{2}, \frac{k+1}{2}]$, s is affine with slope $+1$ or -1.

Definition 7.27 (The blancmange)**.**

$$B(x) \;=\; \sum_{n=0}^{\infty} \frac{s(2^n x)}{2^n}, \qquad x \in \mathbb{R}.$$

Takagi's function (1901); van der Waerden's base-ten variant (1930) differs only in diet. Stage n has teeth 2^n times as frequent and 2^n times as short as the original — and, the load-bearing invariant, the compressions cancel in the slope: every stage's teeth climb and fall at slope exactly ± 1.

Lemma 7.28 (The straddle lemma)**.** *Let f be differentiable at x, and let $u_m \le x \le v_m$ with $u_m < v_m$ and $v_m - u_m \to 0$. Then*

$$\frac{f(v_m) - f(u_m)}{v_m - u_m} \;\longrightarrow\; f'(x).$$

Differentiability controls the slopes across shrinking intervals that straddle *the point, not merely those anchored at it.*

Proof strategy. Write the straddle quotient as a convex combination of the two anchored quotients, with weight $\lambda_m = (v_m - x)/(v_m - u_m) \in [0,1]$; each anchored quotient tends to $f'(x)$, and a convex combination is squeezed by the worse of the two errors. Problem II.6(c) executes the algebra, degenerate endpoint cases included. □

Theorem 7.29 (The blancmange is a monster)**.** *B is continuous at every point of $\mathbb{R}$ and differentiable at none.*

Proof of continuity; strategy for the rest. Continuity is two lines of tonight's machinery: stage n is continuous and satisfies $|2^{-n}s(2^n x)| \le 2^{-n-1} =: M_n$ with $\sum M_n = 1$ (Theorem 4.24), so the M-test (Theorem 7.23) gives uniform convergence on $\mathbb{R}$ and the stool (Theorem 7.21) certifies the limit continuous — the chapter's instalment already earning interest.

Nowhere-differentiability is Problem II.6(d)–(e), solved in full; the skeleton: fix x and scale m, let $u_m = k_m 2^{-m} \le x < (k_m+1)2^{-m} = v_m$ be the dyadic straddle ($k_m = \lfloor 2^m x \rfloor$), and let D_m be the straddle quotient of B at scale m. Stages $n \ge m$ contribute 0 (their sawtooth vanishes at both dyadic endpoints); each stage $n < m$ is affine of slope ± 1 across the straddle; hence D_m is a sum of m signs — an integer — and refining the scale appends exactly one new sign: $D_{m+1} - D_m = \pm 1$. An integer sequence whose consecutive terms always differ by exactly 1 is never Cauchy, hence never convergent; the straddle lemma (Lemma 7.28) says differentiability at x would force convergence. Contradiction at x; and x was arbitrary. □

7.7 Typicality: Baire Fires

Definition 7.30 (The space C and the uniform distance)**.** C denotes the set of continuous functions $f : [0,1] \to \mathbb{R}$, with

$$d(f,g) \;=\; \max_{x\in[0,1]} |f(x) - g(x)|$$

— a maximum genuinely attained, since $|f-g|$ is continuous on a compact interval (Theorem 6.12 certifying the yardstick). The metric axioms are immediate (the triangle inequality pointwise, then pass to the max), and convergence in d is *exactly* uniform convergence on $[0,1]$: Definition 7.19 in geometric dress.

Theorem 7.31 (($\circledast$) The space C is complete)**.** *Every Cauchy sequence in (C,d) converges in (C,d).*

Proof. Let (f_k) be Cauchy in d: for every $\varepsilon > 0$ there is N with $\max_{[0,1]} |f_k - f_j| < \varepsilon$ for $k, j \ge N$ — which is precisely uniform Cauchyness on $[0,1]$. By Theorem 7.22, (f_k) converges uniformly to some f (each anchor's values converge by the completeness of $\mathbb{R}$, Theorem 4.18); by Theorem 7.21, f is continuous, so $f \in C$; and $d(f_k, f) = \max |f_k - f| \le \sup |f_k - f| \to 0$. *The register:* the twelfth ring, and the first inked off the real line — the completeness of $\mathbb{R}$, deployed anchorwise, underwrites the completeness of a space whose points are functions. The axiom travels. □

Remark 7.32 (Baire transfers)*.* The definitions of Chapter 5 — neighbourhoods (balls), open, closed, dense, nowhere dense, meagre, residual — never mention numbers, only distances, and transcribe to (C,d) verbatim. So does Baire's theorem: the Chapter 5 argument, rerun with closed balls of radii halving to zero in place of nested intervals and Theorem 7.31 in place of the nested interval theorem, delivers: *C is not a countable union of nowhere-dense sets; indeed the complement of any meagre subset of C is dense.* The transcription is Problem III.1(a).

Theorem 7.33 (Typicality of the monsters; Banach's census)**.** *The set of $f \in C$ that are differentiable at even one point of $[0,1]$ — indeed, that possess so much as a finite one-sided derivative anywhere — is meagre in (C,d). Its complement, the nowhere-differentiable functions, is residual: the generic continuous function is a monster.*

Proof strategy. For each n, let E_n be the set of $f \in C$ that are *somewhere gentle at grade n*: there exists $x \in [0, 1 - \frac{1}{n}]$ with $|f(x+h) - f(x)| \le n\,h$ for all $0 < h < \frac{1}{n}$. Each E_n is closed (pass witnesses to a convergent subsequence, estimate in three legs); each is nowhere dense (approximate any f within $\varepsilon/2$ by a piecewise-linear function — Heine–Cantor, Theorem 6.22, licenses the approximation — then add a sawtooth of height $\varepsilon/4$ and slope huge: uniformly invisible, gentleness-fatal). Any function with a finite right derivative at some $x < 1$ lands in some E_n; a mirror family catches left derivatives at $x = 1$. So the somewhere-differentiable functions sit inside a countable union of closed nowhere-dense sets: meagre; Remark 7.32 finishes. Problem III.1 stages the full census, hints only, in the standing manner of Chapter 5's Baire centrepiece. □

7.8 Problems

Tier I — Calisthenics

I.1 From the raw definition (no rules): compute $f'(a)$ for $f(x) = x^3$; for $f(x) = 1/x$ at $a \neq 0$; and for $f(x) = \sqrt{x}$ at $a > 0$ (rationalize the numerator).

I.2 Corner paperwork: the full two-sided audit of $|x|$ at 0; classify the differentiability of $|x-2|$ everywhere; and show $x\,|x|$ is differentiable at *every* point of $\mathbb{R}$, with derivative $2|x|$ — a differentiable function whose derivative has a corner.

I.3 Tangents and touching order: write the tangent to x^2 at $a = 1$ and verify $x^2 - (2x-1) = (x-1)^2$; write the tangent to x^3 at $a = 1$ and verify $x^3 - (3x-2) = (x-1)^2(x+2)$ — second-order contact, both times.

I.4 Rules drills: differentiate $(x^2+1)(x^3-x)$ by the product rule and check against the expanded form; differentiate $x/(1+x^2)$; differentiate $(x^2+1)^5$ by the chain rule; and verify the product rule on $x \cdot x^2$ against Corollary 7.9.

I.5 Workhorse drills: prove $\sqrt{1+x} < 1 + \frac{x}{2}$ for all $x > 0$ (mean value theorem on $[0, x]$, with I.1's derivative); and show a polynomial with three distinct real roots has at least two distinct critical points (Rolle, twice).

I.6 Compute the pointwise limits on the stated domains: x^n on $[0,1]$; x/n on $\mathbb{R}$; $x/(1+nx)$ on $[0,\infty)$; $x^n - x^{2n}$ on $[0,1]$.

I.7 Uniformity verdicts for all four specimens of I.6, with honest supremum arithmetic: where is each convergence uniform, and where does the sup refuse to die? (For the last specimen: locate the sag's crest and show its height is $\frac{1}{4}$ at every stage. For the third: show the sup is exactly $\frac{1}{n}$ — an escape-shaped domain tamed by damping.)

I.8 Assembly permits: verify uniform convergence via the M-test for $\sum x^n/2^n$ on $[-1,1]$; re-verify the blancmange's permit; and certify $\sum 1/(n^2+x^2)$ uniformly convergent on all of $\mathbb{R}$ (Chapter 4 supplies the budget's convergence).

Tier II — The Real Thing

II.1 The chain rule, honestly: execute the patched-quotient proof of Theorem 7.8 in full, including the verification of the identity at points where $g(x) = g(a)$, and one sentence on why the naïve divide-and-multiply argument is inadmissible. *Hints supplied.*

II.2 Darboux dividends: (a) prove the step function is nobody's derivative on any interval containing 0; (b) prove the no-jump theorem of Remark 7.16: if $\lim_{x\to a^+} f'(x) = L$ exists, then $f'_+(a) = L$. *Hints supplied.*

II.3 The autopsy, formalized: state Cauchy's 1821 claim for sequences; exhibit the counterexample; compute both iterated limits at the seam; identify the precise illegal step; and show that at the anchors $x = 1 - \frac{1}{k}$ the pointwise threshold for $\varepsilon = \frac{1}{2}$ grows at least like $k/2$ (Bernoulli, Proposition 1.3 — no logarithms required). *Solution supplied.*

II.4 Evening dress: (a) prove L'Hôpital's baby form from Remark 7.14; (b) prove Taylor–Lagrange for $n = 1$: fix b, define M by $f(b) = f(a) + f'(a)(b-a) + M(b-a)^2$, feed $\varphi(t) = f(t) - f(a) - f'(a)(t-a) - M(t-a)^2$ to Rolle twice, and conclude $M = f''(c)/2$; (c) repeat for $n = 2$. *Hints supplied.*

II.5 Transport theorems: (a) a uniform limit of bounded functions is bounded; (b) a uniform limit of *uniformly* continuous functions is uniformly continuous; (c) formalize the claim of Definition 7.30 that d-convergence and uniform convergence on $[0,1]$ coincide. *Hints supplied.*

II.6 *The guided centrepiece, solved in full: the blancmange.* (a) Verify the sawtooth dossier of Definition 7.26. (b) Verify the assembly permit and conclude B is continuous. (c) Prove the straddle lemma (Lemma 7.28). (d) For the dyadic straddle of scale m around an arbitrary x, show: stages $n \geq m$ contribute 0; stages $n < m$ contribute ± 1 each; hence D_m is an integer and $D_{m+1} - D_m = \pm 1$. (e) Conclude that B is differentiable at no point of $\mathbb{R}$.

Tier III — For the Ambitious $\star$

III.1 $\star$ Banach's census (Theorem 7.33), staged: (a) transcribe Baire to complete metric spaces and hence to C; (b) each film E_n is closed; (c) each is nowhere dense; (d) every function with a finite one-sided derivative somewhere is caught by some film (mirror family included); (e) conclude, and append the census remark: category proves the monsters exist *in bulk* without constructing a single one. *Hints only.*

III.2 $\star$ Last chapter's star, squared: let $g(x) = x^2 D(x)$, the square times the static. Prove g is differentiable at exactly one point of $\mathbb{R}$ — a function whose entire supply of calculus is concentrated at a single address. *Hints only.*

7.9 Hints

II.1. φ is continuous at $g(a)$ by the very definition of $f'(g(a))$ (Remark 7.2 converts). For the identity: if $g(x) \neq g(a)$, multiply and divide honestly; if $g(x) = g(a)$, both sides are 0 — check each. The naïve argument divides by $g(x) - g(a)$, which may vanish for x arbitrarily close to a (constant g is the crudest offender), and a proof may not divide by a quantity it cannot certify nonzero.

II.2. (a) H takes the values 0 and 1 on any such interval and no value in $(0,1)$; Theorem 7.15 says a derivative that takes two values takes all between. (b) For $x > a$, Theorem 7.12 on $[a,x]$

gives $\big(f(x)-f(a)\big)/(x-a) = f'(\xi_x)$ with $a < \xi_x < x$; as $x \downarrow a$, $\xi_x \to a$ (squeeze), so $f'(\xi_x) \to L$; hence the right difference quotients converge to L.

II.4. (a) Write $f(x)/g(x)$ as the quotient of the two difference quotients at a (using $f(a) = g(a) = 0$) and apply the quotient law; $g'(a) \neq 0$ certifies the denominator. (b) $\varphi(a) = \varphi(b) = 0$ gives $\varphi'(c_1) = 0$ by Rolle; also $\varphi'(a) = 0$ by construction, so Rolle again on $[a, c_1]$ yields $\varphi''(c) = 0$; unwind. (c) One more derivative, one more Rolle; the factorial assembles itself.

II.5. (a) Fix a stage within 1 of the limit uniformly; $|f| \le |f_N| + 1$. (b) The stool with the middle leg replaced by the *uniform* continuity of stage N: the resulting δ serves every anchor. (c) Unwrap both definitions; on $[0,1]$ the sup is a max (Theorem 6.12), and the two statements are the same string.

III.1. (a) Nested closed balls with radii halving to zero; centres form a Cauchy sequence; Theorem 7.31 supplies the limit, which lies in every ball — then run Chapter 5's argument verbatim. (b) Let $f_k \in E_n$, $f_k \rightrightarrows f$, with witnesses $x_k \in [0, 1-\frac{1}{n}]$; extract $x_{k_j} \to x$ (Theorem 5.21); for fixed $h \in (0, \frac{1}{n})$ estimate $|f(x+h) - f(x)|$ in three legs through f_{k_j} at x_{k_j}, using uniform convergence and the continuity of f; conclude $\le nh$. (c) Given g and ε: Heine–Cantor makes g uniformly continuous; choose a partition fine enough that g moves less than $\varepsilon/4$ per cell and interpolate linearly — a piecewise-linear p with $d(g,p) \le \varepsilon/4$ and slopes bounded by some S; add $\frac{\varepsilon}{2}\, s(Kx)$ with K so large that $\frac{\varepsilon}{2}K - S > n$: the sum stays within ε of g and every point now owns arbitrarily small h-windows with quotient exceeding n. (d) A finite right derivative at x bounds the right quotients near x; choose n beating both the bound and the window; for x near 1, use the mirror films built from left quotients. (e) Countably many films, each closed and nowhere dense; the somewhere-differentiable functions hide inside the stack; Remark 7.32 pronounces the complement residual — dense, hence inhabited: existence by census.

III.2. At 0: $|g(x)/x| = |x|\, D(x) \le |x|$; squeeze gives $g'(0) = 0$. At $a \neq 0$: the two fences give g the disagreeing sequential limits a^2 and 0 along rationals and irrationals respectively (Theorem 2.15, Corollary 2.16), so g is discontinuous at a (Theorem 6.3) and Proposition 7.6 forbids a derivative there.

7.10 Solutions

Tier I (terse). **I.1** $(x^3 - a^3)/(x-a) = x^2 + ax + a^2 \to 3a^2$; $(1/x - 1/a)/(x-a) = -1/(ax) \to -1/a^2$; $(\sqrt{x} - \sqrt{a})/(x-a) = 1/(\sqrt{x} + \sqrt{a}) \to 1/(2\sqrt{a})$. **I.2** $|x|$: half-slopes ± 1 disagree (Remark 7.4). $|x-2|$: differentiable except at 2, derivative -1 then $+1$. $x|x|$: equals x^2 for $x \ge 0$ and $-x^2$ for $x \le 0$, so derivative $2x$ and $-2x$ off the origin; at 0 the quotient is $|x| \to 0$; derivative $2|x|$ everywhere. **I.3** Tangents $y = 2x - 1$ and $y = 3x - 2$; the factorings as displayed, each with the factor $(x-1)^2$: second-order contact. **I.4** $5x^4 - 1$ both ways; $(1-x^2)/(1+x^2)^2$; $10x\,(x^2+1)^4$; $1 \cdot x^2 + x \cdot 2x = 3x^2$, agreeing. **I.5** $\sqrt{1+x} - 1 = \dfrac{x}{2\sqrt{1+c}} < \dfrac{x}{2}$ for some $c \in (0, x)$. Rolle on $[r_1, r_2]$ and on $[r_2, r_3]$: two critical points, distinct since the intervals' interiors are disjoint. **I.6** Indicator of $\{1\}$; the zero function; the zero function ($x/(1+nx) \le 1/n$ for $x \ge 0$); the zero function (both powers die on $[0,1)$; at 1, $1 - 1 = 0$). **I.7** Powers: uniform on $[0,b]$ ($\sup = b^n$), not on $[0, 1)$ or $[0, 1]$ (sup $= 1$). Ramps: uniform on $[-M, M]$ (sup $= M/n$), not on $\mathbb{R}$ (sup $= \infty$). Damped ramps: $x/(1 + nx)$ increases to $1/n$ as $x \to \infty$, so $\sup_{[0,\infty)} = 1/n \to 0$: *uniform* on the unbounded domain — damping defeats the escape. Sag: $t - t^2$ on $[0, 1]$ crests at $t = \frac{1}{2}$ with value $\frac{1}{4}$; substituting $t = x^n$, the crest sits at $x = (\frac{1}{2})^{1/n}$ with height exactly $\frac{1}{4}$ at every stage: pointwise to zero, never uniform. **I.8** $|x^n/2^n| \le 2^{-n}$ on $[-1, 1]$, budget geometric; blancmange as in Theorem 7.29; $1/(n^2 + x^2) \le 1/n^2$ on $\mathbb{R}$, budget convergent by Chapter 4's comparison — uniform on the whole line.

II.3 (solution in full). *The claim (1821, sequence form):* if each f_n is continuous on A and $f_n \to f$ pointwise, then f is continuous. *Counterexample:* $f_n(x) = x^n$ on $[0, 1]$, continuous; the pointwise limit is 0 on $[0, 1)$ and 1 at 1 (Definition 7.18), discontinuous at 1. *The iterated limits:*

$$\lim_{x\to 1^-} \lim_{n\to\infty} x^n = \lim_{x\to 1^-} 0 = 0, \qquad \lim_{n\to\infty} \lim_{x\to 1^-} x^n = \lim_{n\to\infty} 1 = 1.$$

The illegal step: continuity of f at 1 is precisely the statement $\lim_{x\to 1^-} f(x) = f(1)$, i.e. $\lim_x \lim_n = \lim_n \lim_x$; the 1821 argument assumes the interchange tacitly (it estimates $|f(x) - f(1)|$ through a single stage n chosen *after* x, which silently requires the stage to work for all nearby x at once). *The quantifier diagnosis:* at the anchor $x = 1 - \frac{1}{k}$, Bernoulli (Proposition 1.3) gives $x^n = (1 - \frac{1}{k})^n \ge 1 - \frac{n}{k}$, so $x^n < \frac{1}{2}$ forces $n > k/2$: the pointwise threshold for $\varepsilon = \frac{1}{2}$ explodes along the seam — no single N serves all anchors, which is exactly the failure of uniformity. *The repair:* under $f_n \rightrightarrows f$, Theorem 7.21 licenses the swap; on $[0, b]$ with $b < 1$ the same powers converge uniformly and the limit is continuous there, consistently.

II.6 (solution in full). *(a) The dossier.* The nearest integer to x lies within $\frac{1}{2}$, so $0 \le s \le \frac{1}{2}$. $\mathbb{Z} + 1 = \mathbb{Z}$ gives period 1; $s(k) = 0$ is immediate. Lipschitz: for any integer k, $s(x) \le |x - k| \le |x - y| + |y - k|$; minimize over k to get $s(x) \le |x - y| + s(y)$, and symmetrize. Affinity: on $[j, j + \frac{1}{2}]$ the nearest integer is j, so $s(x) = x - j$ (slope $+1$); on $[j + \frac{1}{2}, j + 1]$ it is $j + 1$, so $s(x) = j + 1 - x$ (slope -1).

(b) The permit. $|2^{-n}s(2^n x)| \le 2^{-n} \cdot \frac{1}{2} = 2^{-n-1} = M_n$ and $\sum_{n\ge 0} M_n = 1$ (Theorem 4.24); the M-test (Theorem 7.23) gives uniform convergence on $\mathbb{R}$; each stage is continuous (s is Lipschitz, $x \mapsto 2^n x$ affine, then scaled); the stool (Theorem 7.21) certifies B continuous everywhere.

(c) The straddle lemma. If $x = u_m$ or $x = v_m$ for some m, the straddle quotient at that m *is* an anchored quotient, and anchored quotients converge to $f'(x)$ by definition; so assume $u_m < x < v_m$. With $\lambda_m = \dfrac{v_m - x}{v_m - u_m} \in (0, 1)$ and $Q(t) = \dfrac{f(t) - f(x)}{t - x}$, expand:

$$\lambda_m Q(v_m) + (1 - \lambda_m)Q(u_m) = \frac{f(v_m) - f(x)}{v_m - u_m} + \frac{f(x) - f(u_m)}{v_m - u_m} = \frac{f(v_m) - f(u_m)}{v_m - u_m}.$$

Since $0 \le x - u_m \le v_m - u_m \to 0$ and likewise for $v_m - x$, both $u_m \to x$ and $v_m \to x$, so $Q(u_m) \to f'(x)$ and $Q(v_m) \to f'(x)$ (Definition 7.3 via Remark 7.2). Then

$$\left|\tfrac{f(v_m)-f(u_m)}{v_m-u_m} - f'(x)\right| \le \lambda_m|Q(v_m) - f'(x)| + (1 - \lambda_m)|Q(u_m) - f'(x)| \le \max\big(|Q(v_m) - f'(x)|, |Q(u_m) - f'(x)|\big)$$

(d) The heartbeat. Fix x and m; put $k_m = \lfloor 2^m x \rfloor$, $u_m = k_m 2^{-m}$, $v_m = (k_m + 1)2^{-m}$, so $u_m \le x < v_m$ and $v_m - u_m = 2^{-m} \to 0$; the straddles are nested, since $k_{m+1} \in \{2k_m, 2k_m + 1\}$. Let $D_m = 2^m\big(B(v_m) - B(u_m)\big)$ and audit stage by stage. *Stages $n \ge m$:* $2^n u_m = k_m 2^{n-m}$ and

$2^n v_m = (k_m+1)2^{n-m}$ are integers, where s vanishes: contribution 0. *Stages $n < m$:* the image interval $[2^n u_m,\ 2^n v_m]$ has length $2^{n-m} \le \frac{1}{2}$ and its endpoints are consecutive multiples of 2^{n-m}. Every half-integer is a multiple of 2^{n-m} (since $\frac{1}{2} = 2^{m-n-1} \cdot 2^{n-m}$ with $m-n-1 \ge 0$), and no multiple of 2^{n-m} lies strictly between two consecutive ones; so the image interval contains no half-integer in its interior and lies within a single affine piece of s (Definition 7.26), of slope $\varepsilon_n = \pm 1$ in the s-variable. Its contribution to D_m is $2^{-n} \cdot \varepsilon_n 2^n (v_m - u_m) \cdot 2^m = \varepsilon_n$. Hence $D_m = \sum_{n<m} \varepsilon_n$, an integer — and since the straddles are nested and each stage $n < m$ is affine on the *larger* interval, the signs ε_n do not change when m increments: $D_{m+1} = D_m + \varepsilon_m = D_m \pm 1$.

(e) The verdict. $|D_{m+1} - D_m| = 1$ for every m, so (D_m) is not Cauchy, hence not convergent (Theorem 4.18, easy direction). If B were differentiable at x, Lemma 7.28 would force $D_m \to B'(x)$. Contradiction; and x was arbitrary: B is differentiable nowhere. ■

7.11 Notes

Ampère, 1806. The memoir claims every function is differentiable away from isolated exceptional points; the claim propagated through the century's textbooks. Its refutation is this chapter's second half.

Rolle, 1691; Fermat, 1630s. Rolle's theorem appears in his *Méthode pour résoudre les égalités*; Rolle himself was a vocal critic of the infinitesimal calculus — the foundation stone furnished by a skeptic. Fermat's interior-extremum method predates the calculus proper.

Cauchy, 1821. The *Cours d'analyse*, source of both the era's rigour and its most instructive error (Remark 7.24). Stronger L'Hôpital forms run through Cauchy's own two-function mean value theorem.

The monster's pedigree. Bolzano constructed a continuous nowhere-differentiable function around 1830; it remained unpublished until Rychlík's edition of 1930. Riemann suggested $\sum \sin(n^2 x)/n^2$ in lectures around 1861; Gerver (1970) showed it *is* differentiable at certain rationals — the gesture was prophetic, the example not. Weierstrass presented his cosine-series monster to the Berlin Academy in 1872; the first publication is due to du Bois-Reymond (1875). Darboux's theorem appears in his 1875 memoir on discontinuous functions — Thomae's vintage. Takagi's blancmange is from 1901; van der Waerden's variant from 1930.

Hermite's recoil. Letter 374, Hermite to Stieltjes, 20 May 1893: *« Je me détourne avec effroi et horreur de cette plaie lamentable des fonctions continues qui n'ont point de dérivées »* — "I turn away in fright and horror from this lamentable plague of continuous functions that have no derivatives" (*Correspondance d'Hermite et de Stieltjes*, ed. Baillaud & Bourget, Gauthier-Villars, 1905, vol. 2, pp. 317–319). Poincaré's complaint — functions invented expressly to fault the reasonings of the fathers, and his term *monsters* — is paraphrased in the episode.

Term-by-term differentiation. The stricter permit cited in Remark 7.25 — uniform convergence of the derivatives plus convergence at one point — is Rudin, *Principles*, Theorem 7.17.

Reading. Abbott, *Understanding Analysis*, ch. 5 (whose §5.4 builds the same sawtooth monster and inspired the staging of II.6); Rudin, *Principles*, chs. 5 and 7; Körner, *A Companion to Analysis*, on why the mean value theorem is the honest workhorse of the subject.

Chapter 8

The Riemann Integral and Its Discontents

Companion to Episode Eight, the season finale. The episode built the Riemann integral honestly — Darboux's machinery, suprema and infima laying the bricks — proved the fundamental theorem of calculus in both parts, and then read the fine print: the static defeated, Thomae admitted, the vocabulary crisis opened and answered only by prophecy, the moving spike convicted and the swap account's second instalment paid, Volterra's crack in the summit disclosed, and the season audited. This chapter is the paperwork. Register notes: the bell rings ***twice*** *this chapter — the thirteenth ring at Definition 8.2 (the bricks) and the fourteenth at Definition 8.6 (the books), honouring Episode Seven's forecast; remaining suprema of the chapter carry the standing annotation (completeness — small change, uncounted). Debt four executes act two at Proposition 8.21; debt five's second instalment is paid at Theorem 8.26; the Lebesgue criterion is* ***stated on credit*** *at Theorem 8.23, proof due in Volume Two; and the register's loan L1 is serviced a second time at Remark 8.28.*

Episode Digest

The argument of the hour. The season's summit and its cracks. Area is rendered honest by caging: upper and lower Darboux sums, manufactured slice by slice out of the completeness axiom — the bell's thirteenth and fourteenth rings, bricks and books. Continuous functions are admitted on uniform continuity's certificate; monotone functions on the telescope, jumps and all — so integrability tolerates infinite discontinuity, and the boundary of Riemann's jurisdiction runs along no line this season can name. At the summit, the fundamental theorem, both parts: accumulation smooths, and the mean value theorem telescopes net change into the integral — the reconciliation of the two calculi. Then the fine print. The static (upper ledger one, lower zero, forever) is refused; Thomae's function (loud fractions finitely many, caged in thin slices) is admitted with integral zero; and since both are discontinuous on dense sets, the season owns no vocabulary for the verdict — Lebesgue's criterion is read as prophecy: bounded, and continuous off a set of *measure zero*, words Volume Two must define. The moving spike (area one forever, pointwise limit zero) convicts the unlicensed swap of limit and integral; uniform convergence licenses it — the swap account's second instalment, paid with a stinginess codicil. And Volterra's function — differentiable everywhere, derivative bounded, derivative not integrable — holes the fundamental theorem below the waterline: Volume Two repairs the bounded estate; the full repair is honestly beyond the course. The season closes audited: fourteen rings, every debt paid or carried on stated terms.

The episode, beat by beat. **The Summit in View.** The finale's map announced: build the integral, ring the bell at the foundation (twice, forecast), prove the fundamental theorem, then the fine print — the static, Thomae, the spike, Volterra — and the season's audit. The summit first, the cracks after; the history's own order.

Area on Trial. Exhaustion (Eudoxus, Archimedes) as the ancient standard; the eighteenth century's loan — integral defined as antiderivative, area riding on faith. Fourier calls the loan: arbitrary profiles demand integrals of arbitrary functions. Cauchy (1823) defines the integral of a continuous function as a limit of sums; Riemann (1854, the qualifying lecture on trigonometric series — the integral built inside Fourier's own question; debt three serviced at the foundation) asks the modern question: the *exact* jurisdiction of the concept. Darboux (1875) tidies the machinery into suprema and infima, where the dependence on completeness shows openly. Riemann's own tagged sums (Remark 8.8) agree exactly; one direction is Problem II.1.

Cages and Bricks. Partitions as finite itineraries (Definition 8.1); on each slice the function is asked its ceiling and floor — supremum and infimum, which exist for bounded functions by completeness *and by nothing else*: ring thirteen, at Definition 8.2. Boundedness stamped on the ticket (Remark 8.3): an unbounded function has slices with no ceiling for the axiom to deliver. Upper and lower sums cage the region from outside and fill it from inside; exhaustion in modern dress.

The Two Ledgers. Refinement never hurts (Lemma 8.4); every lower sum undercuts every upper sum across all partitions (Lemma 8.5, by common refinement). The supremum of all lower sums (the lower integral) and the infimum of all upper sums (the upper integral) exist by completeness again — ring fourteen, at Definition 8.6; the forecast honoured with the mortar barely mixed. Integrable when the ledgers agree; the common value is the integral. The auditor's criterion (Theorem 8.7): one partition with gap under ε convicts the books of agreement — Episode Five's courtroom geometry, roles reversed. Every integrability proof henceforth hunts one good partition.

First Through the Door. Continuous implies integrable (Theorem 8.9): the extreme value theorem seats the ticket and attains each slice's ceiling and floor; Heine–Cantor converts the tolerance to one delta valid across the interval; slices thinner than delta hold every oscillation under the per-unit budget; the auditor stamps. Episode Six's grid-table wink redeemed verbatim: an integral is one enormous table, and uniform continuity is the certificate that prices every gap with one signature.

The Staircase Pays. Monotone implies integrable (Theorem 8.10): equal slices; on each, ceiling minus floor is the climb accomplished; the climbs telescope to $f(b) - f(a)$; width times total ascent dips under any tolerance. Episode Six's monotone star spent: the salted staircase (Proposition 8.12) jumps at *every* rational and integrates without complaint — integrability tolerates infinitely much discontinuity; the continuity line is not the boundary. The step function waved through (Proposition 8.11); housekeeping — linearity, order, splitting, the triangle inequality — filed at Proposition 8.14.

The Reconciliation. The accumulation function (Definition 8.15). Part one (Theorem 8.16): accumulation is Lipschitz, and wherever the integrand is continuous, the accumulation differentiates back to it — averages over shrinking windows settle onto the pointwise value; the step function's cornered ramp (Remark 8.17) shows settlement failing exactly at a jump. Part two (Theorem 8.18): the mean value theorem plants a witness in each slice; the witnesses' tagged sum telescopes to $F(b) - F(a)$ for every partition; the ledgers squeeze; net change equals integral. The wonder beat: slope and accumulation, inverse trades, proved — the season assembled (MVT per slice, Chapter 4's telescope, tonight's books). Hypotheses anatomized at Remark 8.20: $F' = f$ everywhere *and* f integrable — the second clause is load-bearing; Volterra is aimed at it.

The Static at the Gate. Act two of the static's contract, executed (Proposition 8.21): the picket fence runs through every slice twice — rationals and irrationals both dense — so every upper sum is one and every lower sum is zero; the books stand a unit apart forever. Dirichlet's 1829 booking honoured, forty-six years before Darboux tidied the machinery it defeats. Bounded, ticketed, refused: boundedness admits to the anteroom, not the seat. Act three: Volume Two, Chapter 4.

Thomae Passes. Theorem 8.22: loudness threshold at half the tolerance; the counting lemma (Chapter 6) certifies finitely many loud fractions; cage each in a slice, total width under half the tolerance; the quiet remainder prices under the other half; upper sum under ε, lower sums identically zero. Integrable, integral zero — discontinuous on a dense set, like the static; admitted, unlike it. The season's prettiest "finitely many exceptions, each caged" — compactness's household style, plied by hand.

The Vocabulary Crisis. Density of defect cannot decide (both monsters have it); cardinality cannot decide (the salted staircase jumps on a countable dense set and passes; the dust's indicator, Problem II.3, is discontinuous on an *uncountable* set and passes). Lebesgue's criterion stated on credit (Theorem 8.23): bounded, and discontinuous only on a set of measure zero — the phrase glossed in covering language as a preview, defined properly in Volume Two, proof due at its Chapter 4. The exhibits run through the prophecy in the table at Remark 8.24. The centre of gravity moves from functions to sets: the season's last lesson.

The Spike, or the Second Instalment. The moving spike (Proposition 8.25): tents of height n on base $2/n$ — each integral exactly one, pointwise limit identically zero, uniform distance to zero equal to n. The swap fails by the whole quantity under discussion; mass escapes up the spike. The license (Theorem 8.26): uniform convergence carries integrability to the limit and integrals to the integral — the tube argument, Episode Seven's uniform distance in its second season of service. Debt five, instalment two, paid — with the stinginess codicil (Remark 8.27): sufficient, not necessary; the crown jewels are Volume Two's Chapter 5.

Volterra, or the Hole Below the Waterline. Remark 8.28, starred blueprint at Problem III.1: a fat cousin of the dust (positive extent — Volume Two's language, previewed); in each removed gap a trimmed copy of the oscillating specimen (loan L1 serviced a second time); the assembly differentiable everywhere, derivative bounded, derivative oscillating at fixed size on the fat dust — ledgers permanently apart, integral refused. The fundamental theorem cannot read back a bounded derivative's accumulated change. Repair, stated honestly: Volume Two recovers the entire bounded estate; the full reconciliation needs a further integral, named in the Notes, beyond this course.

The Season's Audit. The register read in full — debts one through nine, the loan, the new paper (the criterion, the phrase "measure zero," the honest crack), and the bell's complete peal: fourteen marquee rings, one axiom, the season's skeleton laid bare. The carillon promised in Episode Five, delivered.

The toolkit acquired. Partitions, refinements, and the common-refinement manoeuvre; Darboux sums and the two ledgers; the auditor's criterion as the standard instrument; uniform continuity spent at the door; the telescope spent twice (monotone admission; FTC II); the accumulation function; the tube argument for interchange; "finitely many exceptions, each caged"; and the prophecy's covering language, previewed.

Dates worth keeping. Cauchy's integral for continuous functions, 1823; Riemann's qualifying lecture, 1854 (published 1868, posthumously, by Dedekind's hand); Darboux's machinery, 1875; Dirichlet's booking, 1829; Thomae's creature, 1875; Volterra's construction, 1881, at twenty-one; Lebesgue's thesis and criterion, 1902.

8.1 Notation Ledger

Spoken in the episode	**Written here**	**Remarks**
"a partition"; "the itinerary"	$P : a = x_0 < x_1 < \cdots < x_n = b$	Definition 8.1
"the slices"	$[x_{i-1}, x_i]$, width $\Delta x_i = x_i - x_{i-1}$	Definition 8.1
"the mesh"	$\|P\| = \max_i \Delta x_i$	Definition 8.1
"ceiling and floor of a slice"	$M_i = \sup_{[x_{i-1},x_i]} f,\ m_i = \inf_{[x_{i-1},x_i]} f$	(⊛) Def. 8.2
"upper and lower sums"; "the cage"	$U(f,P) = \sum M_i \, \Delta x_i, \quad L(f,P) = \sum m_i \, \Delta x_i$	Definition 8.2
"the two ledgers"	$\overline{\int_a^b} f = \inf_P U(f,P), \quad \underline{\int_a^b} f = \sup_P L(f,P)$	(⊛) Def. 8.6
"the integral"	$\int_a^b f$ (the common value)	Definition 8.6
"a tagged sum"	$S(f,P,\xi) = \sum f(\xi_i) \, \Delta x_i,\ \xi_i \in [x_{i-1}, x_i]$	Remark 8.8
"the accumulation function"	$F(x) = \int_a^x f$	Definition 8.15
"the uniform distance" (reprise)	$d(f,g) = \sup_{[a,b]} \|f - g\|$	Chapter 7, Def. 7.30

The auditor's criterion, in full dress (the chapter's working instrument). — f bounded on $[a,b]$ is integrable if and only if

$$\forall \varepsilon > 0 \ \exists P : \quad U(f,P) - L(f,P) < \varepsilon.$$

One thrifty itinerary certifies the agreement of two infinite ledgers: the courtroom geometry of Chapter 5, prosecution and defence exchanged. Only two entries of this ledger wear the bell-mark, and the chapter's remaining suprema are clerical work upon them — annotated below, per the standing convention, as small change.

8.2 The Darboux Machinery

Definition 8.1 (Partition; refinement; mesh). A *partition* of $[a,b]$ is a finite set of points $P = \{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$. Its *slices* are the intervals $[x_{i-1}, x_i]$, of widths $\Delta x_i = x_i - x_{i-1}$; its *mesh* is $\|P\| = \max_i \Delta x_i$. A partition P' *refines* P when $P \subseteq P'$: every waypoint kept, new ones permitted. Any two partitions P_1, P_2 admit the *common refinement* $P_1 \cup P_2$.

Definition 8.2 (($\circledast$) Darboux sums — the bricks)**.** Let $f : [a,b] \to \mathbb{R}$ be *bounded*, and let P be a partition. On each slice set

$$M_i = \sup \{ f(x) : x \in [x_{i-1}, x_i] \}, \qquad m_i = \inf \{ f(x) : x \in [x_{i-1}, x_i] \},$$

and define the *upper sum* and *lower sum* of f for P:

$$U(f,P) = \sum_{i=1}^{n} M_i \, \Delta x_i, \qquad L(f,P) = \sum_{i=1}^{n} m_i \, \Delta x_i.$$

Since $m_i \le M_i$ on every slice, $L(f,P) \le U(f,P)$ for every P. *The bell rings — the thirteenth.* The numbers M_i and m_i exist because the values of f over a slice form a nonempty bounded set of reals, and the completeness axiom — and nothing else — delivers its supremum and infimum. Over $\mathbb{Q}$ the construction dies in its crib. Every brick in the chapter is manufactured by the axiom.

Remark 8.3 (Boundedness is stamped on the ticket)*.* If f is unbounded on $[a,b]$, some slice of every partition carries values with no ceiling (or no floor), and the sums cannot be formed: the construction does not fail gracefully, it fails to start. The bounded case is the honest core; the extensions (improper integrals) are conveniences built atop it, with dangers catalogued in the Notes.

Lemma 8.4 (Refinement never hurts)**.** *If P' refines P, then*

$$L(f,P) \;\le\; L(f,P') \;\le\; U(f,P') \;\le\; U(f,P).$$

Proof. It suffices to treat the insertion of a single new waypoint c into one slice $[x_{i-1}, x_i]$ of P, and to iterate. Write $M' = \sup_{[x_{i-1},c]} f$ and $M'' = \sup_{[c,x_i]} f$. Both suprema are taken over subsets of $[x_{i-1}, x_i]$, so $M' \le M_i$ and $M'' \le M_i$ (a supremum over a subset cannot exceed the supremum over the set). Hence

$$M'(c - x_{i-1}) + M''(x_i - c) \;\le\; M_i(c - x_{i-1}) + M_i(x_i - c) \;=\; M_i \, \Delta x_i,$$

and all other terms of U are untouched: $U(f,P') \le U(f,P)$. The lower-sum inequality is the mirror image (infima over subsets cannot fall below the infimum over the set), and the middle inequality is Definition 8.2's slice-by-slice $m \le M$ applied to P'. $\square$

Lemma 8.5 (Every bid undercuts every asking price)**.** *For* any *two partitions P_1, P_2 of $[a,b]$: $L(f,P_1) \le U(f,P_2)$.*

Proof. Let $P = P_1 \cup P_2$ be the common refinement. By Lemma 8.4 twice,

$$L(f,P_1) \;\le\; L(f,P) \;\le\; U(f,P) \;\le\; U(f,P_2). \qquad \square$$

Definition 8.6 (($\circledast$) The two ledgers; integrability — the books)**.** The set of all lower sums is nonempty and bounded above (by any upper sum, Lemma 8.5); the set of all upper sums is nonempty and bounded below. *The bell rings — the fourteenth*: the completeness axiom, surveying the whole infinite inventory of sums, delivers

$$\underline{\int_a^b} f \;=\; \sup_P \, L(f,P) \qquad \text{and} \qquad \overline{\int_a^b} f \;=\; \inf_P \, U(f,P),$$

the *lower* and *upper integrals* of f over $[a,b]$. By Lemma 8.5, $\underline{\int_a^b} f \le \overline{\int_a^b} f$ always. When the two ledgers agree, f is *(Riemann) integrable* on $[a,b]$, and the common value is written $\int_a^b f$. Episode Seven forecast more than one ring before the mortar set; rings thirteen and fourteen honour it at the foundation. Every further supremum in this chapter is clerical work on these two and carries the standing annotation: completeness — small change, uncounted.

Theorem 8.7 (The auditor's criterion)**.** *A bounded $f : [a,b] \to \mathbb{R}$ is integrable if and only if for every $\varepsilon > 0$ there exists a partition P with $U(f,P) - L(f,P) < \varepsilon$.*

Proof. ($\Leftarrow$) For any partition P,

$$0 \;\le\; \overline{\int_a^b} f - \underline{\int_a^b} f \;\le\; U(f,P) - L(f,P),$$

since the upper integral sits at or below $U(f,P)$ and the lower integral at or above $L(f,P)$. If the right side dips under every $\varepsilon > 0$ for suitable P, the fixed nonnegative gap between the ledgers is less than every positive number, hence zero.

($\Rightarrow$) Suppose the ledgers agree on the value A. Given $\varepsilon > 0$: since $A = \sup_P L(f,P)$, some P_1 has $L(f,P_1) > A - \varepsilon/2$ (approximation property of the supremum); since $A = \inf_P U(f,P)$, some P_2 has $U(f,P_2) < A + \varepsilon/2$. Let $P = P_1 \cup P_2$. By Lemma 8.4, $L(f,P) \ge L(f,P_1)$ and $U(f,P) \le U(f,P_2)$, so $U(f,P) - L(f,P) < (A + \varepsilon/2) - (A - \varepsilon/2) = \varepsilon$. □

Remark 8.8 (Riemann's own formulation: tagged sums). Riemann's 1854 definition uses *tagged sums*: choose a tag $\xi_i \in [x_{i-1}, x_i]$ in each slice and form $S(f,P,\xi) = \sum f(\xi_i)\,\Delta x_i$; the function is integrable with integral A when the tagged sums converge to A as the mesh tends to zero, uniformly over all choices of tags. Since $m_i \le f(\xi_i) \le M_i$, every tagged sum is trapped between $L(f,P)$ and $U(f,P)$; the two formulations agree exactly. The direction requiring care — from tagged convergence to the agreement of the ledgers — is Problem II.1: tags can be chosen to press each $f(\xi_i)$ against the ceiling M_i (approximation property of the supremum; small change, uncounted), so tagged sums reach within any tolerance of the upper sum, and symmetrically the lower. Your numerical trade lives in this remark: a quadrature rule is a policy for choosing tags.

8.3 Who Integrates

Theorem 8.9 (Continuous implies integrable)**.** *If f is continuous on $[a,b]$, then f is integrable on $[a,b]$.*

Proof. The certificates, in the order the episode checked them. f is bounded on the compact interval (extreme value theorem, Theorem 6.12), so it holds a ticket and the machinery of Definition 8.2 applies. On each slice $[x_{i-1}, x_i]$ the supremum and infimum are *attained* (Theorem 6.12 again, retail): there are $u_i, v_i \in [x_{i-1}, x_i]$ with $f(u_i) = M_i$ and $f(v_i) = m_i$. And f is uniformly continuous on $[a,b]$ (Heine–Cantor, Theorem 6.22).

Let $\varepsilon > 0$. Set the per-unit budget $\varepsilon' = \varepsilon/(2(b-a))$. Uniform continuity supplies one $\delta > 0$, valid across the whole interval, with $|f(x) - f(y)| < \varepsilon'$ whenever $|x - y| < \delta$. Choose any partition P with mesh $\|P\| < \delta$. On each slice, u_i and v_i lie within $\Delta x_i \le \|P\| < \delta$ of one another, so

$$M_i - m_i \;=\; f(u_i) - f(v_i) \;<\; \varepsilon'.$$

Therefore

$$U(f,P) - L(f,P) \;=\; \sum_i (M_i - m_i)\,\Delta x_i \;<\; \varepsilon' \sum_i \Delta x_i \;=\; \varepsilon'\,(b-a) \;=\; \frac{\varepsilon}{2} \;<\; \varepsilon,$$

and the auditor's criterion (Theorem 8.7) stamps the file. — The episode's redeemed wink, in one sentence: the integral is a tabulation of f on a grid with the gaps priced honestly, and uniform continuity is the single signature that prices them all. $\square$

Theorem 8.10 (Monotone implies integrable)**.** *If f is monotone on $[a,b]$, then f is integrable on $[a,b]$.*

Proof. Say f is nondecreasing (the nonincreasing case mirrors). Then f is bounded: $f(a) \le f(x) \le f(b)$ for all x. Let P_n be the partition into n slices of equal width $h = (b-a)/n$. On the slice $[x_{i-1}, x_i]$, monotonicity pins the floor and ceiling at the endpoints: $m_i = f(x_{i-1})$ and $M_i = f(x_i)$. Hence

$$U(f,P_n) - L(f,P_n) = \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big)\,h = h \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = h\,\big(f(b) - f(a)\big),$$

the sum telescoping — every intermediate station arriving and departing once. Given $\varepsilon > 0$, choose n with $h\,(f(b) - f(a)) < \varepsilon$ (Archimedean property, Theorem 2.12); the criterion stamps. Note what was never mentioned: continuity. $\square$

Proposition 8.11 (Step functions integrate)**.** *Let f be a* step function *on $[a,b]$: constant on each open slice of some fixed partition $Q = \{c_0, \ldots, c_m\}$, with arbitrary values at the waypoints. Then f is integrable, and $\int_a^b f = \sum_{j=1}^{m} v_j\,(c_j - c_{j-1})$, where v_j is the constant value on (c_{j-1}, c_j).*

Proof. Given $\varepsilon > 0$, take the partition consisting of Q together with two extra waypoints flanking each interior c_j at distance η, where $2m\eta \cdot 2K < \varepsilon/2$ and K bounds $|f|$. On the thin flanking slices the oscillation is at most $2K$ and the total width at most $2m\eta$; on the fat interior slices f is constant and the oscillation is zero. Hence $U - L < \varepsilon$, and both sums differ from $\sum_j v_j(c_j - c_{j-1})$ by less than ε; letting $\varepsilon \downarrow 0$ identifies the integral. $\square$

Proposition 8.12 (The salted staircase)**.** *Let $q_1, q_2, q_3, \ldots$ enumerate the rationals of $[0,1]$, and define $f : [0,1] \to \mathbb{R}$ by*

$$f(x) \;=\; \sum_{k\,:\,q_k \le x} 2^{-k}$$

(an empty sum is zero). Then f is nondecreasing; it is discontinuous at exactly the rationals of $(0,1]$ — a countable set, dense in $[0,1]$ — and continuous at every other point; and it is integrable, by Theorem 8.10, discontinuities on a dense set notwithstanding.

Proof. Monotone: enlarging x enlarges the index set of the sum, and all summands are positive. (The series converges by comparison with the geometric series, Chapter 4; its tails $\sum_{k>K} 2^{-k} = 2^{-K}$ appear twice below.)

Discontinuity at each rational q_j of $(0,1]$: for $x < q_j \le y$,

$$f(y) - f(x) \;\ge\; 2^{-j},$$

since the index j contributes to $f(y)$ and not to $f(x)$: the jump from the left at q_j is at least 2^{-j}. (At the rational 0 there is no left side within the domain, and the argument below shows f is in fact continuous there relative to $[0,1]$ — an endpoint technicality the proposition's statement respects.)

Continuity at each point α of $[0,1]$ that is not a rational of $(0,1]$ (the irrationals, and the endpoint 0): given $\varepsilon > 0$, choose K with $2^{-K} < \varepsilon$. Only the indices $k \leq K$ carry weight $\geq 2^{-K}$; the finitely many points among $q_1, \ldots, q_K$ lying in $(0,1]$ do not include α, so some window around α (one-sided at an endpoint) excludes them all. For x, y in that window with $x < \alpha < y$, the indices contributing to $f(y) - f(x)$ are all $> K$, whence $f(y) - f(x) \leq \sum_{k>K} 2^{-k} = 2^{-K} < \varepsilon$. Continuity follows by monotonicity (values between $f(x)$ and $f(y)$). This is Chapter 6's Problem III.2 spent, as the episode promised: a monotone function may be discontinuous on a countable dense set — and by Theorem 8.10 it integrates regardless. The boundary of Riemann's jurisdiction does not run along the continuity line. □

Lemma 8.13 (Absolute values keep the ticket)**.** *If f is integrable on $[a,b]$, so is $|f|$.*

Proof. On any slice and for any x, y there, $\big|\,|f(x)| - |f(y)|\,\big| \leq |f(x) - f(y)| \leq M_i - m_i$ (reverse triangle inequality, Chapter 4). Taking suprema over x, y (small change, uncounted), the oscillation of $|f|$ on each slice is at most that of f, so $U(|f|, P) - L(|f|, P) \leq U(f, P) - L(f, P)$ for every P, and the criterion transfers. □

Proposition 8.14 (Housekeeping)**.** *Let f, g be integrable on $[a,b]$ and $c \in \mathbb{R}$. Then:*

(i) (Linearity) *$f + g$ and cf are integrable, with $\int_a^b (f+g) = \int_a^b f + \int_a^b g$ and $\int_a^b cf = c\int_a^b f$.*

(ii) (Order) *If $f \leq g$ on $[a,b]$, then $\int_a^b f \leq \int_a^b g$.*

(iii) (Splitting) *For $a < c < b$: f is integrable on $[a,b]$ if and only if it is integrable on $[a,c]$ and on $[c,b]$, and then $\int_a^b f = \int_a^c f + \int_c^b f$.*

(iv) (Triangle inequality) $\left|\int_a^b f\right| \leq \int_a^b |f|$.

Proof. The two instructive cases, the rest staged as Problem I.4.

(i), the sum. On any slice, $\sup(f+g) \leq \sup f + \sup g$ and $\inf(f+g) \geq \inf f + \inf g$ (a sum cannot out-reach the sum of its parts' reaches), so for every partition

$$L(f,P) + L(g,P) \;\leq\; L(f{+}g,P) \;\leq\; U(f{+}g,P) \;\leq\; U(f,P) + U(g,P).$$

Given ε, pick P_f, P_g with gaps under $\varepsilon/2$ (Theorem 8.7) and pass to the common refinement P: the outer quantities above stand less than ε apart, so $f + g$ passes the criterion; and both $\int f + \int g$ and $\int(f+g)$ are trapped in the same shrinking bracket, hence equal.

(iv). By Lemma 8.13, $|f|$ is integrable. Since $-|f| \leq f \leq |f|$, part (ii) gives $-\int_a^b |f| \leq \int_a^b f \leq \int_a^b |f|$, which is (iv). — Part (ii) itself: $f \leq g$ makes every lower sum of f at most the corresponding lower sum of g, and suprema preserve $\leq$ (small change, uncounted). Part (iii): waypoint c splits every sum; details in Problem I.4. □

8.4 The Fundamental Theorem

Definition 8.15 (The accumulation function)**.** Let f be integrable on $[a,b]$. Its *accumulation function* is

$$F : [a,b] \to \mathbb{R}, \qquad F(x) = \int_a^x f$$

(with $F(a) = 0$; the restriction of f to $[a, x]$ is integrable by Proposition 8.14(iii)).

Theorem 8.16 (Fundamental theorem, part one: accumulation smooths)**.** *Let f be integrable on $[a, b]$ and F its accumulation function. Then:*

(i) F is Lipschitz: with $K = \sup_{[a,b]} |f|$ (small change, uncounted), $|F(y) - F(x)| \le K\,|y - x|$ for all $x, y \in [a, b]$. In particular F is uniformly continuous.

(ii) If f is continuous at $c \in [a, b]$, then F is differentiable at c, and $F'(c) = f(c)$.

Proof. (i) For $x < y$, splitting (Proposition 8.14(iii)) gives $F(y) - F(x) = \int_x^y f$, and the triangle inequality with order (parts (iv), (ii)) gives $\left|\int_x^y f\right| \le \int_x^y |f| \le K(y - x)$: the deposit over a sliver is at most the sliver's width times the bound.

(ii) Let $\varepsilon > 0$. Continuity at c supplies $\delta > 0$ with $|f(t) - f(c)| < \varepsilon$ for all $t \in [a, b]$ with $|t - c| < \delta$. For $0 < |h| < \delta$ (with $c + h \in [a, b]$), compute the difference quotient against the candidate limit — the average against the pointwise value:

$$\frac{F(c+h) - F(c)}{h} - f(c) \;=\; \frac{1}{h}\int_c^{c+h} f(t)\,dt \;-\; f(c) \;=\; \frac{1}{h}\int_c^{c+h} \big(f(t) - f(c)\big)\,dt,$$

using $\int_c^{c+h} f(c)\,dt = f(c)\,h$ (the constant integrates to height times width; Proposition 8.11) and linearity. Every t in the window satisfies $|f(t) - f(c)| < \varepsilon$, so by the triangle inequality and order,

$$\left|\frac{F(c+h) - F(c)}{h} - f(c)\right| \;\le\; \frac{1}{|h|}\cdot\varepsilon\,|h| \;=\; \varepsilon.$$

(The signed-endpoint bookkeeping for $h < 0$ is the standard convention $\int_x^y = -\int_y^x$, filed with Proposition 8.14(iii).) An average of values within ε of a target lies within ε of the target; the averages settle onto the height; $F'(c) = f(c)$. □

Remark 8.17 (The cornered ramp: settlement fails at a jump). Let f be 0 on $[0, \frac{1}{2})$ and 1 on $[\frac{1}{2}, 1]$. Its accumulation is $F(x) = 0$ for $x \le \frac{1}{2}$ and $F(x) = x - \frac{1}{2}$ beyond: a ramp with a corner at one half, where the left difference quotients tend to 0 and the right to 1. Accumulation smooths — the ramp is Lipschitz, as (i) promises — but it cannot smooth over a lie: differentiability fails exactly where f jumps, and the averages straddle exactly as the episode described. (Problem I.7 works the computation.)

Theorem 8.18 (Fundamental theorem, part two: net change is the integral)**.** *Let $F : [a, b] \to \mathbb{R}$ be differentiable on all of $[a, b]$ with $F' = f$ there, and suppose f is integrable on $[a, b]$. Then*

$$\int_a^b f \;=\; F(b) - F(a).$$

Proof. Let $P = \{x_0, \ldots, x_n\}$ be *any* partition of $[a, b]$. On each slice $[x_{i-1}, x_i]$, the function F is continuous on the closed slice and differentiable on its interior, so the mean value theorem (Theorem 7.12) plants a witness $\xi_i \in (x_{i-1}, x_i)$ with

$$F(x_i) - F(x_{i-1}) \;=\; F'(\xi_i)\,\Delta x_i \;=\; f(\xi_i)\,\Delta x_i.$$

Sum over the slices; the left sides telescope:

$$F(b) - F(a) \;=\; \sum_{i=1}^{n} f(\xi_i)\,\Delta x_i \;=\; S(f, P, \xi),$$

a tagged sum with the mean-value witnesses as tags. Every tagged sum is trapped between the ledgers (Remark 8.8):

$$L(f,P) \;\le\; F(b) - F(a) \;\le\; U(f,P) \qquad \text{for every partition } P.$$

The fixed number $F(b) - F(a)$ therefore sits at or above every lower sum and at or below every upper sum, hence between the lower and upper integrals. But f is integrable: the ledgers close on the single value $\int_a^b f$, and a number trapped between them can only be that value. □

Corollary 8.19 (The schoolroom rules, licensed)**.** *(i) Every continuous f on $[a,b]$ has an antiderivative — its own accumulation function (Theorem 8.16(ii), now at every point). (ii) For continuous f and* any *antiderivative G of f on $[a,b]$: $\int_a^b f = G(b) - G(a)$ — by Theorem 8.18, whose integrability hypothesis Theorem 8.9 supplies. (Two antiderivatives differ by a constant — the constant theorem, Chapter 7's mean-value harvest — so the choice of G is immaterial: the plus-C of the schoolroom, cashed.)*

Remark 8.20 (Anatomy of the hypotheses). Part two demands *both* that $F' = f$ at every point of $[a,b]$ — one exception and the telescope leaks — *and* that f be integrable. At the summit the second clause looks like lawyer's caution: surely a bounded, everywhere-defined derivative integrates? It does not, always: Remark 8.28. The clause is load-bearing, and the theorem is holed below the waterline exactly there.

8.5 The Static Defeated

Proposition 8.21 (Act two: the static is not integrable)**.** *The Dirichlet function D (Chapter 6, Definition 6.24) — 1 on the rationals, 0 on the irrationals — is bounded on $[0,1]$ and is not Riemann integrable there.*

Proof. Let P be any partition. Every slice $[x_{i-1}, x_i]$ is a nondegenerate interval, so it contains a rational (Theorem 2.15) — where $D = 1$, forcing $M_i = 1$ — and an irrational (Corollary 2.16) — where $D = 0$, forcing $m_i = 0$. Hence $U(D,P) = \sum 1 \cdot \Delta x_i = 1$ and $L(D,P) = 0$, *for every P*: the upper ledger reads 1 forever, the lower 0 forever,

$$\underline{\int_0^1} D = 0 \;\neq\; 1 = \overline{\int_0^1} D,$$

and no refinement in the world closes a gap that refinement cannot touch. Dirichlet's 1829 booking, honoured: bounded, ticketed, refused. Act three is Volume Two, Chapter 4, where a better machine tames this creature in one line. □

8.6 Thomae Passes

Theorem 8.22 (Thomae's function is integrable, with integral zero)**.** *Thomae's function t (Chapter 6, Definition 6.26) — $t(x) = 1/q$ at each rational $x = p/q$ in lowest terms, $t(x) = 0$ at each irrational, $t(0) = 1$ — is Riemann integrable on $[0,1]$, and $\int_0^1 t = 0$.*

Proof. Every lower sum is zero: each slice contains an irrational (Corollary 2.16), so $m_i = 0$ for every slice of every partition; the lower ledger reads 0.

For the upper ledger, let $\varepsilon > 0$; the loudness threshold is $\varepsilon/2$. Call x *loud* if $t(x) \geq \varepsilon/2$. A loud point is a rational p/q with $1/q \geq \varepsilon/2$, that is $q \leq 2/\varepsilon$ — and rationals of $[0, 1]$ with bounded denominator are finitely many (the counting lemma inside Chapter 6's Theorem 6.27: at most $q + 1$ candidates per denominator q, finitely many denominators). Let the loud points be $s_1, \ldots, s_m$ (with $s_1 = 0$ among them, $t(0) = 1$ being loud for small ε).

Build the partition in the episode's two gestures. Around each s_j place an interval of width at most $\varepsilon/(4m)$, clipped to $[0, 1]$ and shrunk if necessary so the intervals are disjoint — finitely many culprits, thinness ours to command; the endpoints of these intervals, together with 0 and 1, form the partition P. Price $U(t, P)$ in two parcels. The loud parcels — the slices containing loud points — number at most m, with total width at most $m \cdot \varepsilon/(4m) = \varepsilon/4$, each with ceiling $M_i \leq 1$ (all values of t are at most 1): contribution at most $\varepsilon/4 < \varepsilon/2$. The quiet parcels: every point there is quiet, so $M_i \leq \varepsilon/2$ on each (the supremum of values below a bound is at most the bound; small change, uncounted), total width at most 1: contribution at most $\varepsilon/2$. Hence

$$U(t, P) \;<\; \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \;=\; \varepsilon.$$

So the upper integral is at most every positive ε, hence 0; both ledgers read 0; integrable, integral zero. Discontinuous on a dense set — every rational — and admitted regardless: finitely many exceptions, each individually caged, the compactness household style plied by hand. $\square$

8.7 The Prophecy

Theorem 8.23 (Lebesgue's criterion, 1902 — stated on credit)**.** *Let f be bounded on $[a, b]$, and let Δ_f denote its set of discontinuity points. Then f is Riemann integrable on $[a, b]$ if and only if Δ_f has* measure zero*.*

On the phrase, a preview only. A set has measure zero when it can be covered by countably many intervals of total length as small as one pleases — for every $\varepsilon > 0$, a countable family of intervals whose union contains the set and whose lengths sum below ε. Volume Two opens by making this and its kindred phrases lawful (its Chapters 1–2) and proves the criterion in its Chapter 4 — the same chapter in which the static meets act three. Until then the theorem stands on credit, entered in the register with the season's other prophecies. No use of it is made in any proof of this volume.

Remark 8.24 (The exhibits run through the prophecy).

Exhibit	**Defect set Δ_f**	**Verdict**
continuous f (Thm. 8.9)	empty	integrable
step function (Prop. 8.11)	finitely many waypoints	integrable
monotone f (Thm. 8.10)	countable (Ch. 6, Prob. III.2)	integrable
salted staircase (Prop. 8.12)	the rationals of $(0, 1]$: countable, dense	integrable
Thomae's t (Thm. 8.22)	the rationals of $[0, 1]$	integrable, $\int = 0$
the dust's indicator (Prob. II.3)	the dust: uncountable	integrable, $\int = 0$
the static D (Prop. 8.21)	all of $[0, 1]$	*refused*

Density of defect cannot decide (salted staircase, Thomae versus the static); cardinality cannot decide (the dust's indicator); the criterion decides every line with one phrase — and the phrase measures a *set*. The season's centre of gravity, moved.

8.8 The Swap and the Spike

Proposition 8.25 (The moving spike). *For $n \geq 2$ define $f_n : [0,1] \to \mathbb{R}$ as the tent of height n on the base $[0, 2/n]$:*

$$f_n(x) = \begin{cases} n^2 x, & 0 \leq x \leq 1/n, \\ n^2\left(\frac{2}{n} - x\right), & 1/n \leq x \leq 2/n, \\ 0, & 2/n \leq x \leq 1. \end{cases}$$

Then: (i) each f_n is continuous, hence integrable, with $\int_0^1 f_n = 1$ exactly; (ii) $f_n \to 0$ pointwise on $[0,1]$; (iii) the convergence is not uniform: $d(f_n, 0) = n$. Consequently

$$\lim_n \int_0^1 f_n \;=\; 1 \;\neq\; 0 \;=\; \int_0^1 \Big(\lim_n f_n\Big).$$

Proof. (i) Continuity is piecewise-linear bookkeeping (the pieces agree at the seams: $n^2 \cdot \frac{1}{n} = n$ from both sides at $1/n$; 0 at $2/n$); integrability is Theorem 8.9; the integral is the triangle's area, computed honestly by Corollary 8.19 on each linear piece or in one line as $\frac{1}{2} \cdot \text{base} \cdot \text{height} = \frac{1}{2} \cdot \frac{2}{n} \cdot n = 1$, for every n.

(ii) $f_n(0) = 0$ for all n. For fixed $x > 0$: once $2/n < x$ — Archimedean, Theorem 2.12 — the spike has retreated entirely to the left of x and $f_n(x) = 0$ from that index onward.

(iii) $\sup_{[0,1]} |f_n - 0| = f_n(1/n) = n \to \infty$: no tube of any radius ever contains the tents. The mass escapes upward: area one, squeezed thinner and taller, with nowhere to stand in the limit. □

Theorem 8.26 (Uniform convergence licenses the swap). *Let f_n be integrable on $[a,b]$ for each n, and suppose $f_n \rightrightarrows f$ uniformly on $[a,b]$ (Chapter 7, Definition 7.19). Then f is integrable on $[a,b]$, and*

$$\int_a^b f_n \;\longrightarrow\; \int_a^b f.$$

Proof. Write $\varepsilon_n = \sup_{[a,b]} |f_n - f|$ (the uniform distance, Chapter 7; small change, uncounted); uniform convergence says $\varepsilon_n \to 0$. First, f is bounded: it sits within ε_1 of the bounded f_1 eventually — more precisely, within ε_n of f_n for any fixed n — so it holds a ticket.

f passes the auditor. Fix n. The tube inequality

$$f_n - \varepsilon_n \;\leq\; f \;\leq\; f_n + \varepsilon_n \quad \text{on } [a,b]$$

transfers to every slice's ceiling and floor: $M_i(f) \leq M_i(f_n) + \varepsilon_n$ and $m_i(f) \geq m_i(f_n) - \varepsilon_n$. Hence for every partition P,

$$U(f,P) - L(f,P) \;\leq\; \big(U(f_n,P) - L(f_n,P)\big) + 2\,\varepsilon_n(b-a).$$

Given $\varepsilon > 0$: choose n with $2\varepsilon_n(b-a) < \varepsilon/2$, then — f_n being integrable — a partition P with $U(f_n,P) - L(f_n,P) < \varepsilon/2$ (Theorem 8.7). The displayed bound puts $U(f,P) - L(f,P) < \varepsilon$; the criterion stamps.

The integrals converge. By linearity and the triangle inequality (Proposition 8.14),

$$\left|\int_a^b f_n - \int_a^b f\right| \;=\; \left|\int_a^b (f_n - f)\right| \;\le\; \int_a^b |f_n - f| \;\le\; \varepsilon_n\,(b-a) \;\longrightarrow\; 0.$$

The areas differ by at most the tube's cross-section times the interval's width, exactly as narrated. Debt five, instalment two: paid. □

Remark 8.27 (The stinginess codicil: sufficient, not necessary). The license is not the law. The *thin tents* — height 1 on base $[0, 2/n]$, zero beyond — converge to zero pointwise but not uniformly ($d(g_n, 0) = 1$ for every n), yet $\int_0^1 g_n = 1/n \to 0 = \int_0^1 0$: the swap succeeds with no uniformity in sight (Problem II.5 makes it rigorous). Uniformity forbids every escape; but mass can also simply *drain*, no escape attempted. The swap theorems that ask nearly nothing — the crown jewels promised when the account was opened in Episode Four — are Volume Two, Chapter 5; the standard to beat is now on file.

8.9 Volterra's Function

Remark 8.28 (The hole below the waterline — a desk note). **Claim (Volterra, 1881).** There exists $V : [0,1] \to \mathbb{R}$, differentiable at every point, with V' bounded on $[0,1]$, such that V' is *not* Riemann integrable. Consequently the formula of Theorem 8.18 is, for $f = V'$, not false but unpronounceable: the net change $V(1) - V(0)$ is a definite number that no integral of this volume can read back.

Silhouette of the construction (the blueprint, with hints, is Problem III.1). Take a *fat* cousin of the dust (Chapter 3, Definition 3.20): remove from $[0,1]$ middle intervals whose lengths shrink fast enough — middle quarters, then middle sixteenths, and so on — that the removals total only one half; what remains is closed, nowhere dense (Chapter 5, Definition 5.27), yet of "total length" one half in the covering sense previewed after Theorem 8.23. (Volume Two makes this lawful; the set is filed there as a standing exhibit.) In each removed gap, plant a trimmed copy of the oscillating specimen of Chapter 7, Remark 7.17 — $x^2 \sin(1/x)$, the register's loan L1, serviced here a second time, still desk-only — reflected and truncated so that the copy's derivative stays bounded by a fixed constant yet arrives at the gap's endpoints still oscillating at full amplitude. The assembled V is differentiable everywhere (on the gaps by the specimen's calculus; on the fat dust by a squeeze against the quadratic clamp); V' is bounded; and at every point of the fat dust, V' oscillates by at least a fixed amount, because gaps crowd against every such point and each gap delivers oscillation at full amplitude. A set of positive extent on which the oscillation of V' refuses to die keeps the ledgers of V' permanently apart — by Riemann's own criterion (Problem III.2), or in Volume Two's language by Theorem 8.23 read in reverse. The verifications deferred to Volume Two's vocabulary are flagged in Problem III.1, per the honesty clause.

Repair, stated exactly. Volume Two's integral recovers the entire bounded estate: every bounded derivative will be integrable there, with integral the net change — Volterra's creature tamed alongside the static. The full reconciliation — every derivative, bounded or not — exceeds even that machinery; the integral that closes the account entirely is named in the Notes, and it is a Season Three sort of object. Tonight's peak keeps its crack, honestly disclosed.

8.10 Problems

Tier I — Calisthenics

I.1 For $f(x) = x$ on $[0,1]$ and the uniform partition P_n into n equal slices, compute $L(f, P_n)$ and $U(f, P_n)$ exactly (the finite arithmetic series of Chapter 4 will assist), verify

$$L(f, P_n) = \frac{n-1}{2n}, \qquad U(f, P_n) = \frac{n+1}{2n},$$

and conclude from the auditor's criterion that $\int_0^1 x = \frac{1}{2}$.

I.2 Settle the books of Remark 8.17's step function directly: exhibit, for each $\varepsilon > 0$, a partition of $[0,1]$ with $U - L < \varepsilon$, and evaluate $\int_0^1 f = \frac{1}{2}$.

I.3 For $f(x) = x^2$ on $[0,1]$ and the two-slice partition $\{0, \frac{1}{2}, 1\}$, compute both sums; then insert the waypoint $\frac{3}{4}$ and verify by hand that the upper sum falls and the lower sum rises, as Lemma 8.4 promises.

I.4 Complete Proposition 8.14: scalar multiples (treat $c \geq 0$ and $c < 0$ separately — negative constants exchange ceilings and floors), order, and splitting (a waypoint at c splits every sum; each direction of (iii) passes through the criterion).

I.5 For $f(x) = x - \frac{1}{2}$ on $[0,1]$, compute both sides of the triangle inequality (Proposition 8.14(iv)) and measure the slack: $\left|\int_0^1 f\right| = 0$ against $\int_0^1 |f| = \frac{1}{4}$.

I.6 The thin tents of Remark 8.27: verify by triangle areas that $\int_0^1 g_n = 1/n$, and that $\sup_{[0,1]} g_n = 1$ for every n.

I.7 Compute the accumulation function of Remark 8.17's step function from the definition, confirm the ramp formula, and verify the one-sided derivatives 0 and 1 at the corner.

I.8 For the moving spike at $n = 3$ (Proposition 8.25), evaluate f_3 at the stations $x = \frac{1}{6}, \frac{1}{3}, \frac{2}{3}$, verify $\int_0^1 f_3 = 1$ by areas, and check $d(f_3, 0) = 3$.

Tier II — The Real Thing

II.1 (*Riemann's formulation.*) Suppose every sequence of tagged sums $S(f, P_k, \xi^{(k)})$ with $\|P_k\| \to 0$ converges to A, regardless of the tags. Prove f is integrable with $\int_a^b f = A$, by choosing tags that press the tagged sums against the upper and lower sums.

II.2 (*The huddle dividend, spent.*) Let f be uniformly continuous on the *open* interval (a, b). Prove that f extends to a continuous function on $[a, b]$ — Chapter 6's huddle-transport lemma (Proposition 6.21) carries Cauchy sequences home — and conclude that f is integrable on $[a, b]$ in the sense that every continuous extension is, all extensions agreeing. Episode Six promised this three-line dividend would be quietly spent tonight; spend it.

II.3 (*The dust's indicator.*) Let C be the Cantor set (Chapter 3, Definition 3.20) and let $\mathbf{1}_C$ be its indicator function on $[0,1]$: one on the dust, zero elsewhere. Prove that the set of discontinuity points of $\mathbf{1}_C$ is exactly C; prove that $\mathbf{1}_C$ is integrable with $\int_0^1 \mathbf{1}_C = 0$, by building partitions from the stage-n construction intervals; and conclude that an *uncountable* defect set can be integrable — cardinality convicted of irrelevance, as the episode claimed.

II.4 (*The vanishing theorem.*) Let f be continuous and nonnegative on $[a,b]$ with $\int_a^b f = 0$. Prove f is identically zero; and show by exhibit that continuity is not a removable hypothesis.

II.5 (*The stinginess codicil.*) For the thin tents g_n of Remark 8.27: prove $g_n \to 0$ pointwise on $[0,1]$, prove the convergence is not uniform, and prove $\int_0^1 g_n \to 0$ nonetheless. Conclude that uniform convergence is sufficient but not necessary for the swap, and place the exhibit beside Proposition 8.25 in the cabinet.

II.6 (*Mean value theorem for integrals.*) Let f be continuous on $[a,b]$. Prove there exists $c \in [a,b]$ with

$$\int_a^b f \;=\; f(c)\,(b-a),$$

by penning the average value between the minimum and maximum of f (Theorem 6.12) and applying the intermediate value theorem (Theorem 6.15).

Tier III — For the Ambitious $\star$

III.1 $\star$ (*Volterra's blueprint.*) Following Remark 8.28: (a) construct the fat dust — remove from $[0,1]$ a middle interval of length $\frac{1}{4}$, then middle intervals of length $\frac{1}{16}$ from each remaining piece, and so on, the stage-n removals totalling $2^{n-1}/4^n$ — and verify the removals sum to $\frac{1}{2}$; show the residue is closed, nowhere dense, and contains no interval, yet cannot be covered by finitely many intervals of total length below $\frac{1}{2}$. (b) On a removed gap (u,v), trim the specimen $x^2\sin(1/x)$: follow it from u rightward to the last point before the gap's midpoint where its derivative vanishes, then continue constant to the mirror point, and mirror back down to v; verify the trimmed piece is differentiable on (u,v) with derivative bounded by a constant independent of the gap, taking values ± 1 arbitrarily near u and v. (c) Assemble V and prove differentiability at points of the fat dust by the quadratic squeeze. (d) Show the oscillation of V' is at least 1 at every point of the fat dust, and conclude via Problem III.2 — granting (a)'s covering fact, whose full proof waits on Volume Two's language — that V' is not integrable. References in the Notes.

III.2 $\star$ (*Riemann's own criterion, 1854.*) For bounded f on $[a,b]$ and $\sigma > 0$, let $W(P,\sigma)$ be the total width of the slices of P on which the oscillation $M_i - m_i$ is at least σ. Prove: f is integrable if and only if for every $\sigma > 0$ and every $\varepsilon > 0$ there is a partition with $W(P,\sigma) < \varepsilon$. Position the result as the historical halfway house between the auditor's criterion and Theorem 8.23.

8.11 Hints

II.1. Fix P. Choose tags with $f(\xi_i) > M_i - \varepsilon/(b-a)$ (approximation property of the supremum, slice by slice): then $S(f,P,\xi) > U(f,P) - \varepsilon$. Symmetrically press the floor. Two tagged sums over the same fine partition then trap $U - L$; let the mesh shrink.

II.4. If $f(c) = 2\eta > 0$, continuity plants a plateau: a window around c on which $f \geq \eta$. One lower sum then clears η times the window's width. For the exhibit, an indicator of a single point.

II.5. Pointwise: past the base, every tent reads zero at a fixed $x > 0$; at $x = 0$ every tent reads zero outright. Non-uniformity: each tent still summits at height one. The integrals: areas $1/n$.

II.6. With $m \leq f \leq M$ on $[a, b]$ (Theorem 6.12), order pens the average $\frac{1}{b-a}\int_a^b f$ between m and M; both are attained values; Theorem 6.15 delivers c.

III.1. (a) The stage-n residue is a union of 2^n closed intervals; any finite cover of the residue by open intervals must, by compactness of the residue (closed and bounded — Chapter 5), essentially contain a stage; count lengths. (b) The derivative of $x^2 \sin(1/x)$ is $2x \sin(1/x) - \cos(1/x)$, bounded by 3 on $[0, 1]$; its zeros accumulate at the origin, so a last one before any positive station exists — take a supremum (small change) and verify it is a zero by continuity of the derivative away from the origin. (c) At a fat-dust point p, every nearby point is either in the dust (where V matches the local quadratic clamp) or in a gap whose trimmed specimen is clamped by the square of the distance to the gap's nearer endpoint; squeeze the difference quotient. (d) Gaps crowd p from at least one side at every scale; each contributes derivative values ± 1 within any window.

III.2. ($\Leftarrow$) Split $U - L$ over the loud slices (oscillation $\geq \sigma$, total width $< \varepsilon$, oscillation at most $2K$) and the quiet ones (oscillation $< \sigma$, total width at most $b - a$); choose σ and ε against a given tolerance. ($\Rightarrow$) If some σ_0, ε_0 resist every partition, every $U - L$ is at least $\sigma_0\, \varepsilon_0$.

8.12 Solutions

Tier I (terse). **I.1** $L = \sum_{i=1}^{n} \frac{i-1}{n} \cdot \frac{1}{n} = \frac{n(n-1)}{2n^2} = \frac{n-1}{2n}$; $U = \sum_{i=1}^{n} \frac{i}{n} \cdot \frac{1}{n} = \frac{n+1}{2n}$; $U - L = \frac{1}{n} \to 0$, and both pinch $\frac{1}{2}$. **I.2** Partition $\{0, \frac{1}{2} - \eta, \frac{1}{2}, 1\}$: the only slice with oscillation is the η-wide one, so $U - L = \eta$; both sums pinch $0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}$ as $\eta \downarrow 0$ (the waypoint value at $\frac{1}{2}$ is measure-free — it affects no slice's supremum by more than the thin slice absorbs). **I.3** On $\{0, \frac{1}{2}, 1\}$: $L = 0 \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$, $U = \frac{1}{4} \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{5}{8}$. Insert $\frac{3}{4}$: $L = \frac{1}{8} +$ (the right slice improving) $= 0 \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{4} + \frac{9}{16} \cdot \frac{1}{4} = \frac{13}{64}$; $U = \frac{1}{4} \cdot \frac{1}{2} + \frac{9}{16} \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} = \frac{33}{64}$. Indeed $\frac{1}{8} < \frac{13}{64}$ and $\frac{33}{64} < \frac{5}{8}$. **I.4** $c \geq 0$: ceilings and floors scale by c; $c < 0$: $\sup(cf) = c \inf f$ and $\inf(cf) = c \sup f$, so $U(cf, P) = c\, L(f, P)$, $L(cf, P) = c\, U(f, P)$, and the criterion transfers with the roles exchanged; $\int cf = c \int f$ in all cases. Order: $m_i(f) \leq m_i(g)$ slice by slice, so $L(f, P) \leq L(g, P)$; take suprema. Splitting: refine any partition by the waypoint c; sums split as stated; each half inherits the criterion from the whole (its slices are a subset), and conversely two half-partitions concatenate. **I.5** $\int_0^1 (x - \frac{1}{2}) = \frac{1}{2} - \frac{1}{2} = 0$ (I.1 and linearity); $|x - \frac{1}{2}|$ is two ramps, each of area $\frac{1}{8}$, total $\frac{1}{4}$. **I.6** Triangle of base $2/n$ and height 1: area $1/n$; summit $g_n(1/n) = 1$. **I.7** For $x \leq \frac{1}{2}$: $F(x) = \int_0^x 0 = 0$. For $x > \frac{1}{2}$: $F(x) = 0 + \int_{1/2}^{x} 1 = x - \frac{1}{2}$ (splitting; constants integrate to height times width). Difference quotients at $\frac{1}{2}$: from the left $0/h \to 0$; from the right $h/h \to 1$. **I.8** $f_3(\frac{1}{6}) = 9 \cdot \frac{1}{6} = \frac{3}{2}$; $f_3(\frac{1}{3}) = 3$ (the summit, $= d(f_3, 0)$); $f_3(\frac{2}{3}) = 0$ ($\frac{2}{3}$ being the base's right endpoint). Area: $\frac{1}{2} \cdot \frac{2}{3} \cdot 3 = 1$.

II.2 (solution in full). *Extension.* Let $(x_n) \subseteq (a,b)$ with $x_n \to a$. Then (x_n) is Cauchy, so by the huddle-transport lemma (Proposition 6.21) — uniform continuity carries Cauchy sequences to Cauchy sequences — $(f(x_n))$ is Cauchy in $\mathbb{R}$, hence converges (Cauchy completeness, Chapter 4); call the limit ℓ. The limit is sequence-independent: if also $y_n \to a$, interleave $x_1, y_1, x_2, y_2, \ldots$ — still convergent to a, still Cauchy — so the interleaved image converges, and its two subsequences share one limit; ℓ is unanimous. Define $\tilde{f}(a) = \ell$, symmetrically $\tilde{f}(b)$, and $\tilde{f} = f$ on (a,b).

Continuity of the extension at a. Let $\varepsilon > 0$; take δ from uniform continuity for $\varepsilon/2$. For $x \in (a, a+\delta)$: choose any sequence $t_k \downarrow a$; eventually $|t_k - x| < \delta$, so $|f(t_k) - f(x)| \leq \varepsilon/2$; let $k \to \infty$: $|\tilde{f}(a) - f(x)| \leq \varepsilon/2 < \varepsilon$. So $\tilde{f}$ is continuous at a (and at b, mirrored; on (a,b) it was already continuous, uniform continuity being continuity with a global budget).

Integrability. $\tilde{f}$ is continuous on the compact $[a,b]$, hence integrable (Theorem 8.9). Any two continuous extensions agree with f on the dense (a,b) and hence — continuity being determined on a dense set, Chapter 6 — agree outright; so "the" integral is unambiguous. Episode Six's dividend, spent as promised: the huddle carried the boundary values home, and the door of Theorem 8.9 did the rest.

II.3 (solution in full). *The defect set is exactly C.* If $x \notin C$: the complement of C is open (the dust is closed, Chapter 5, Theorem 5.31), so a whole window around x misses C, where $\mathbf{1}_C \equiv 0$: continuous at x. If $x \in C$: every window around x contains points of C (namely x) and points not in C — the dust contains no interval (Theorem 5.31: empty interior) — so $\mathbf{1}_C$ takes both values 1 and 0 in every window: oscillation 1, discontinuous at x.

Integrability, integral zero. Lower sums: every slice meets the complement of C (no slice fits inside the dust), so every $m_i = 0$: the lower ledger reads 0. Upper sums: fix n and recall the construction (Chapter 3, Definition 3.20): after stage n, the dust lies inside a union C_n of 2^n closed intervals of length 3^{-n} each, total $(2/3)^n$. Take the partition P_n whose waypoints are 0, 1, and all endpoints of those 2^n intervals. Slices inside C_n total $(2/3)^n$ in width and carry ceiling at most 1; every other slice lies outside C_n, misses C entirely, and carries ceiling 0. Hence

$$U(\mathbf{1}_C, P_n) \;\leq\; \left(\tfrac{2}{3}\right)^n \;\longrightarrow\; 0,$$

so the upper integral is 0 (given ε, choose n with $(2/3)^n < \varepsilon$; Chapter 4's geometric decay). Ledgers agree at 0: integrable, $\int_0^1 \mathbf{1}_C = 0$.

The moral, on the record. C is uncountable (Chapter 3) and is the exact defect set of an integrable function; the salted staircase's defect set is countable; the static's is all of $[0,1]$. Cardinality does not decide integrability — the prophecy's covering language does, and this exhibit is its cleanest witness this side of Volume Two.

8.13 Notes

History. Cauchy's integral for continuous functions appears in the 1823 *Résumé des leçons*; Riemann's definition, with his oscillation criterion (Problem III.2), in the 1854 Habilitation lecture on trigonometric series, published posthumously in 1868 through Dedekind's care; Darboux's upper-and-lower machinery in his 1875 *Mémoire sur les fonctions discontinues*, which also contains the theorem of Chapter 7 bearing his name. Dirichlet's function is proposed — as a function beyond the era's integral — in his 1829 memoir on Fourier convergence; Thomae's in his 1875 textbook. Volterra's construction is in *Alcune osservazioni sulle funzioni punteggiate* *discontinue* (1881), written at twenty-one; the fat dust it stands on is the Smith–Volterra–Cantor set, Henry Smith having built such sets already in 1875, six years before — priority, as so often, ahead of fame. Lebesgue's criterion and the integral that honours it are in his thesis, *Intégrale, longueur, aire* (1902).

The full repair. The integral that recovers *every* derivative — Volterra's included, unbounded ones included — is the gauge integral of Denjoy, Perron, Henstock, and Kurzweil; a Season Three sort of object, as the episode said. Volume Two's Lebesgue integral recovers every *bounded* derivative (and much besides), which is the estate this season's cliffhanger actually mortgaged.

Improper integrals. The extensions of the bounded-core theory to unbounded integrands and unbounded intervals proceed by limits of honest integrals; they are conveniences, not a new theory, and their dangers (conditional convergence masquerading as area) mirror Chapter 4's rearrangement lessons. The desk defers them to use at need.

Reading. Abbott, *Understanding Analysis*, chapters seven and eight, remains the season's companion; Rudin, *Principles*, chapter six, for the stern statements (and the Stieltjes generalization the course leaves aside); Körner, *A Companion to Analysis*, for the honest discussion of what Riemann's integral cannot do; Hairer and Wanner, *Analysis by Its History*, for the Cauchy–Riemann–Darboux thread with the original texts.

Season consolidation. A closing set for the season as a whole — one tool per episode, as promised aloud: (1) prove the square root of three is irrational (Chapter 1's toolkit); (2) prove the supremum of $\{ x : x^2 < 3 \}$ squares to three (Chapter 2's bell); (3) exhibit a bijection between the dust and the unit interval — or cite the seasonings of Chapter 3 that combine to one; (4) prove that a series whose partial sums contain two subsequences with different limits diverges, and convict Grandi's series a second time by this instrument (Chapter 4); (5) give an open cover of the rationals of the unit interval with no finite subcover, and say why this does not contradict Heine–Borel (Chapter 5); (6) construct a function continuous at exactly the irrationals of the positive line and explain why the mirror image is impossible — Volume Two, Chapter 1 will pay this properly; state what is missing (Chapter 6 + the prophecy); (7) prove the blancmange, though differentiable nowhere, is uniformly continuous — and name the theorem that makes it automatic (Chapters 6–7); (8) deduce from the fundamental theorem that two antiderivatives of one function on an interval differ by a constant, and exhibit the failure on a domain of two pieces (Chapters 7–8). Solutions deliberately withheld; the season is over, sir, and the desk trusts its graduate.

The Season's Notation, Consolidated

Every chapter opened with a notation ledger bridging the episode's spoken phrases to their written symbols. This table gathers the season's standing instruments — the notation that outlived its introducing episode — grouped by the chapter that coined it. Episode-local devices (the nudge, the loudness ruler, and kin) remain in their own chapters' ledgers.

Spoken	**Written**	**Coined**
"the natural numbers" (one first)	$\mathbb{N} = \{1, 2, 3, \dots\}$	Ch. 1
"for every / there exists"	$\forall$ / $\exists$	Ch. 1
"the rationals; the reals"	$\mathbb{Q}$; $\mathbb{R}$	Chs. 1–2
"least upper bound; greatest lower"	$\sup A$; $\inf A$	Ch. 2
"the completeness axiom"	LUB property (Axiom 2.x)	Ch. 2
"the bell"	($\circledast$) at marquee rings	Ch. 2
"same size" (sets)	bijection $A \leftrightarrow B$	Ch. 3
"countable; uncountable"	$\lvert A\rvert = \lvert\mathbb{N}\rvert$; $\lvert A\rvert > \lvert\mathbb{N}\rvert$	Ch. 3
"the dust"	C (the Cantor set)	Ch. 3
"the sequence a sub n"	$(a_n)_{n\in\mathbb{N}}$	Ch. 4
"converges to L"	$a_n \to L$; $\lim a_n = L$	Ch. 4
"the tolerance; the threshold"	$\varepsilon > 0$; N	Ch. 4
"huddles" (Cauchy)	(a_n) Cauchy	Ch. 4
"the series; partial sums"	$\sum a_n$; s_n	Ch. 4
"limsup, liminf"	$\limsup a_n$, $\liminf a_n$	Ch. 4
"open; closed" (sets)	U open; F closed	Ch. 5
"interior, closure, boundary"	$A^\circ, \overline{A}, \partial A$	Ch. 5
"a blanket" (open cover)	$\{U_i\}_{i\in I}$, $A \subseteq \bigcup U_i$	Ch. 5
"compact"	every cover admits a finite subcover	Ch. 5
"nowhere dense; meagre; generic"	Def. 5.x	Ch. 5
"continuous at a"	ε–δ at a	Ch. 6
"uniformly continuous"	one δ for the whole set	Ch. 6
"the static"	D (Dirichlet's function)	Ch. 6
"Thomae's function"	$t(p/q) = 1/q$, $t(\mathrm{irr}) = 0$	Ch. 6
"the limit of f at a"	$\lim_{x\to a} f(x)$	Ch. 7
"f prime"	$f'(a)$	Ch. 7
"converges pointwise; uniformly"	$f_n \to f$; $f_n \rightrightarrows f$	Ch. 7
"the uniform distance"	$d(f,g) = \sup \lvert f - g\rvert$	Ch. 7
"the blancmange"	$B(x) = \sum 2^{-n} s(2^n x)$	Ch. 7
"a partition; the mesh"	P; $\lVert P\rVert$	Ch. 8
"upper and lower sums"	$U(f,P)$, $L(f,P)$	Ch. 8
"the two ledgers"	$\overline{\int_a^b} f$, $\underline{\int_a^b} f$	Ch. 8
"the integral"	$\int_a^b f$	Ch. 8
"the accumulation function"	$F(x) = \int_a^x f$	Ch. 8

Pronunciation Guide

The master lexicon's table, for the reader who intends to say these names aloud in company — name and respelling, with the lexicon's remarks. The machine-readable forms (exact IPA, both apostrophe variants of every possessive) live in the course's pronunciation-lexicon files, from which this table is generated; the IPA column is omitted here rather than set approximately in a face that lacks the glyphs — the honesty clause extends to typography.

Name	Respelling	Remark
Abel	Ah-bel	Norwegian; not ”AY-bel”
Ampère	Am-pair	
Arago	Ah-rah-goh	
Baire	Bair	one syllable; not ”BAY-er”
Banach	Bah-nahk	
Berkeley	Barkley	the bishop is BARK-lee
Bolzano	Bolt-sah-no	
Borel	Bo-rell	
Carathéodory	Kara-tay-oh-doh-ree	
Cauchy	Koh-shee	
Chebyshev	Cheb-ish-off	
Darboux	Dar-boo	
Dedekind	Day-duh-kint	not English ”kind”
Dirichlet	Dee-ree-shlay	anglicized convention; the German hard-K variant also circulates
Egorov	Yeh-gor-off	
Euler	Oiler	
Eudoxus	You-dock-sus	

Name	**Respelling**	**Remark**
Fatou	Fah-too	
Fourier	Foor-ee-ay	usually fine unaided; belt and braces
Gödel	Gur-del	
Grandi	Grahn-dee	
Heine	Hy-nuh	
Hermite	Air-meet	not "HER-might"
Hippasus	Hip-pah-sus	
Hölder	Herl-der	
Kolmogorov	Kol-muh-gor-off	
Kronecker	Kroh-neck-er	
L'Hôpital	Loh-pee-tahl	both apostrophe forms in PLS
Lebesgue	Luh-beg	the heaviest-traffic entry in the course
Leibler	Libe-ler	Kullback–Leibler
Leibniz	Libe-nits	
Liouville	Lee-oo-veel	
Lusin	Loo-zin	
Méray	May-ray	Charles Méray; Cantor's rival-firm precursor

Name	**Respelling**	**Remark**
Minkowski	Min-koff-ski	
Nikodym	Nik-oh-deem	
Oresme	Or-em	
Radon	Rah-don	the mathematician, not the gas
Poincaré	pwan-kar-AY	French; added Session 7 (spoken E7)
Riemann	Ree-mahn	
Rolle	rohl	rhymes with "hole"; added Session 7 (spoken E7)
Riesz	Reess	Riesz–Fischer
Solovay	Sol-uh-vay	
Stieltjes	Steel-chess	
Takagi	Tah-kah-ghee	hard G
Thomae	Toe-mah-uh	German final vowel sounded; anglicized "TOE-may" also circulates
Vitali	Vih-tah-lee	
Volterra	Vol-tair-ah	
Weierstrass	Vye-er-shtrahss	
Zermelo	Tsair-may-lo	

Colophon

This volume is the desk's half of *The Anatomy of the Continuum*, an audio season of eight episodes narrated by Jeeves for and with P. Milovanov, Esq., produced one episode and one chapter per session between the seventh and twelfth of July, twenty twenty-six. The season ran from Hippasus drowned at sea to Volterra's crack in the summit: the continuum numbered, counted, mapped, tamed, monstered, and measured against the limits of its own best machine. Fourteen marquee deployments of a single axiom — one sentence about least upper bounds — carry every load-bearing theorem in these pages; the full peal is audited in Chapter 8's closing sections, and the debt register travels in the course's LEDGER. The text is set in Palatino (MATHPAZO), adopted as the volume's face at the season assembly. Season Two — *The Measure of All Things* — opens where Chapter 8's prophecy points: at the question of how long a set is, and at the ceremony of a debt two seasons in the arranging.

The establishment reopens shortly under a new sign; same management, same register, the bell polished and waiting.